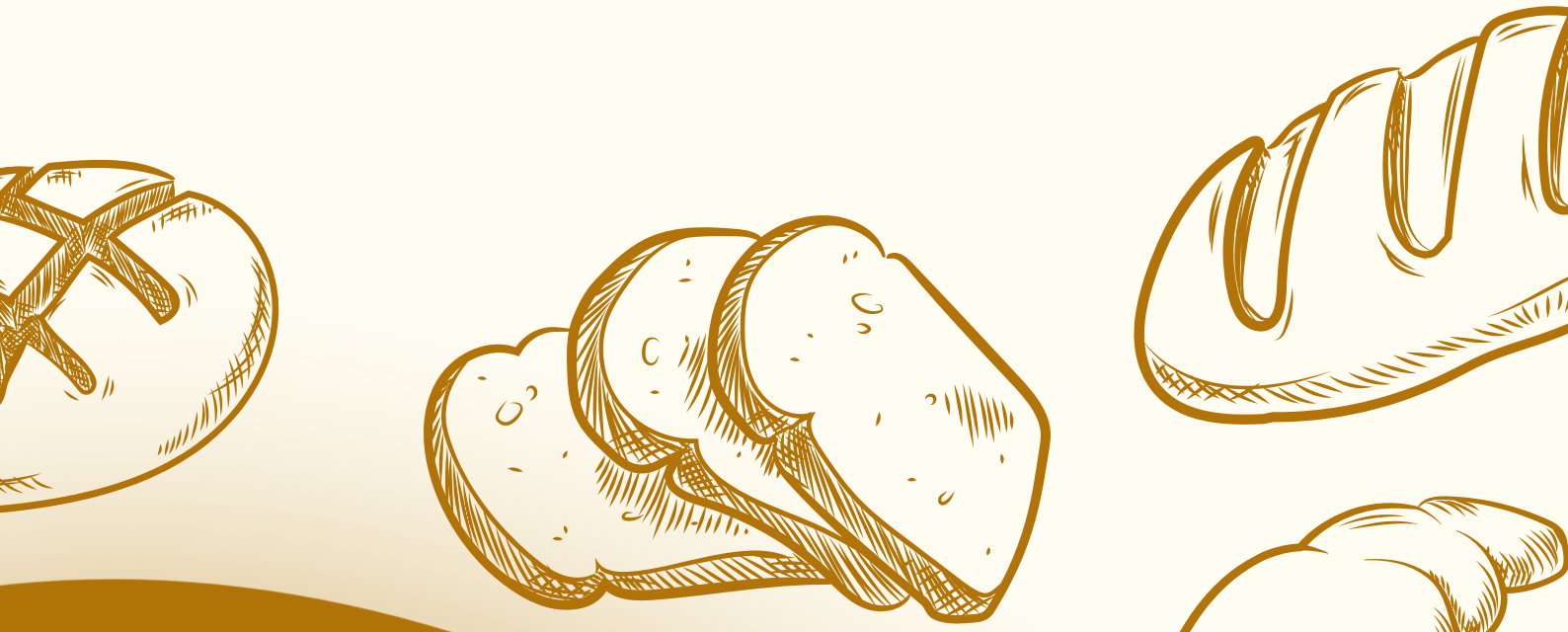
KING ABDULAZIZ UNIVERSITY
FACULTY OF ECONOMICS AND ADMINISTRATION
MANAGEMENT INFORMATION SYSTEM DEPARTMENT
MIS373- 2nd Semester Final Project



BakeCraft

Bake it your way

Portfolio Project of Human-Computer Interaction (MIS 373)
Design Thinking Framework

By:

[redacted]

Section: CA
Instructor: Dr. Zeina Alkubaisy

Prototype link

Table of content

1.Introduction 06
1.1 The purpose of the mobile app 07
1.2 The target audience and their needs 07
1.3 Goals and Objectives 08
2.Empathize 09
2.1 Pilot Study on Research Methods 10
2.1.1 Questionnaire pilot study 10
2.1.2 Interview pilot study 12
2.1.3 Observation pilot Study 14
2.2 The Research Methods 18
2.2.1 The Observation 18
2.2.2 The questionnaire 20
(Table Questionnaire Objectives Mapping: Phrases, Objectives, and Approaches)
2.2.3 The interview 24
(Table Interview Objectives Mapping: Phrases, Objectives, and Approaches)
2.3 Aggregate Empathy Maps 27
2.3.1 Aggregate Empathy Maps 27
2.3.2 Interview Analysis(Cross-Interviews) 28
2.4 User persona 29
2.4.1 Persona 1 (User journey, Empathy Map) 29
2.4.2 Persona 2 (User journey, Empathy Map) 32
2.4.3 Persona 3 (User journey, Empathy Map) 35
2.4.4 Persona 4 (User journey, Empathy Map) 38
2.5 Pain points and opportunities 41
3. Define 42
3.1 User Problems and Challenges 43
3.2 The 5 W's Framework 44
3.3 Problem Statements 45
3.4 Hypothesis Statements If Then 45
3.5 Value Proposition 47
3.5.1 Features and Benefits 47
3.5.2 Application Value 49
3.5.3 Meeting Users' Needs 49
3.5.4 Official Value Proposition 50
3.6 Problem Statement to Guide the Development Process 50
4. Ideate 51
4.1 Competitive Audits 52
4.1.1 Direct Competitors (Overview, Aspects, Research) 52
4.1.2 Indirect competitors (Overview, Aspects, Research) 52
4.1.3 Analyzing the findings 55
4.1.4 Summarize Findings 56

Table of content

4.2 Brainstorming..... 57
4.2.1 How Might We..... 57
4.2.2 Potential solutions that align with the defined problem statement..... 61
4.2.3 Evaluating ideas..... 62
4.2.4 Goal statements..... 63
4.3 Mapping design features to each persona in a table to fulfill their needs, wants, and desires..... 65
5. Prototype..... 66
5.1 Storyboard Sketching and Wire-framing..... 67
5.1.1 Storyboard: Big-picture and close-up..... 68
5.1.2 Low-fidelity wireframes and sketches..... 71
5.2 Prototyping..... 74
5.2.1 Mock-up prototypes..... 74
6. Test and feedback..... 80
6.2 Usability Testing..... 81
6.2.1 usability testing sessions with users..... 81
6.2.2 pain points, usability issues, and areas for improvement..... 83
6.3 Feedback Analysis..... 85
6.3.1 Analyze the feedback and observations..... 80
6.3.2 Make informed design and functionality decisions (changes based on user input)..... 86
Conclusion..... 89
References..... 90
Appendix..... 91

List of table

Table 1: Observation objectives and approaches..... 18
Table 2: Questionnaire objectives and approach..... 20
Table 3: Interview objectives and approaches..... 24
Table 4: User journey (persona 1)..... 30
Table 5: User journey (persona 2)..... 33
Table 6: User journey (persona 3)..... 36
Table 7: User journey (persona 4)..... 39
Table 8: Competitive audits..... 52
Table 9: Direct competitors (Aspects)..... 53
Table 10: Indirect competitors (Research)..... 54
Table 11: Indirect competitors (Aspects)..... 55
Table 12: Potential solutions for problem statements..... 61
Table 13: Design features for each persona..... 65
Table 14: Usability Testing1..... 81
Table 15: Usability Testing2..... 82

List of figures

Figure 1: Aggregate empathy map..... 27
Figure 2: Aggregate empathy map of persona 1..... 31
Figure 3: Aggregate empathy map of persona 2..... 34
Figure 4: Aggregate empathy map of persona 3..... 37
Figure 5: Aggregate empathy map of persona 4..... 40
Figure 6: Problem statement of persona 1 45
Figure 7:if/then statement of persona 1 45
Figure 8: Problem statement of persona 2..... 45
Figure 9:if/then statement of persona 2..... 45
Figure 10: Problem statement of persona 3..... 46
Figure 11:if/then statement of persona 3..... 46
Figure 12: Problem statement of persona 4..... 46
Figure 13:if/then statement of persona 4..... 46
Figure 14:crazy 8 - persona 1..... 57
Figure 15:crazy 8 - persona 2..... 58
Figure 16: crazy 8 - persona 3..... 59
Figure 17: crazy 8 - persona 4..... 60
Figure 18: Goal statement- persona 1 63
Figure 19: Goal statement- persona 2..... 63
Figure 20: Goal statement- persona 3..... 64
Figure 21: Goal statement- persona 4..... 64
Figure 22: Storyboard:Big-picture - persona 1..... 67
Figure 23: Storyboard:Close-up - persona 1..... 67
Figure 24: Storyboard:Big-picture - persona 2..... 68
Figure 25: Storyboard:Close-up - persona 2..... 68
Figure 26: Storyboard:Big-picture - persona 3..... 69
Figure 27: Storyboard:Close-up - persona 3..... 69
Figure 28: Storyboard:Big-picture - persona 4..... 70
Figure 29: Storyboard:Close-up - persona 4..... 70
Figure 30-32: Wireframes..... 71
Figure 33-66: Mock-ups Prototype..... 74
Figure 67: User Feedback on Time-Consuming Product Customization. 86
Figure 68: Personalized Customization Packs Feature..... 86
Figure 69: User Difficulty in Repeated Bakery Searches..... 87
Figure 70: Favorite Bakeries Feature Implementation..... 87
Figure 71: Manual Search for Dietary-Friendly Bakeries..... 88
Figure 72: Dietary and Bakery Search Filtering Feature..... 88
Figure 73-93: Data Analysis..... 119

Introduction

In today's digital marketplace, many consumers are looking for products that match their personal tastes and dietary preferences. This is especially noticeable in the food industry, where people increasingly prefer customized baked goods instead of standard options. However, customers often face difficulties finding bakeries that can meet their specific requirements, such as particular flavors, fillings, sizes, designs, or dietary needs like low sugar or gluten-free products. At the same time, many small bakeries and home-based producers struggle to reach customers who are actively searching for such customized products.

To address this gap, this project proposes a mobile application that acts as a mediator between customers and bakeries. Through the application, users can specify the exact type of baked goods they want by selecting details such as flavor, size, filling, sweetness level, dietary preferences, and design. The request is then sent to bakeries capable of fulfilling the order, allowing them to review and decide whether to accept it.

This project follows the Human–Computer Interaction (HCI) design thinking framework, which focuses on understanding user needs and creating a simple, intuitive, and user-centered experience. The report documents the process of user research, idea generation, prototyping, and usability testing in order to design an application that simplifies the process of ordering customized baked goods while also supporting local bakeries in reaching a wider audience.

1.1 The purpose of the mobile application

The mobile application's goal is to make it simpler for consumers to locate bakeries or shops that take orders for personalized baked items. Users of the app can request customized bakery goods according to their dietary requirements, preferred ingredients, or designs. Additionally, it seeks to streamline the process of informing bakeries of these requests and setting up delivery, giving customers a more practical and effective way to purchase personalized baked goods.

1.2 The target audience and their needs

The intended audience for BakeCraft was identified through a combination of secondary data such as health reports on dietary trends and allergies and a preliminary survey conducted with potential users. The survey results highlighted a significant gap in the market for personalized bakery products.

Our audience is highly diverse, spanning a wide age range from young adults to the elderly. This includes health-conscious individuals, those managing medical conditions (such as diabetes or obesity), and users with specific dietary restrictions like lactose or gluten intolerance. Furthermore, the audience includes elderly users or individuals with physical sensitivities who require specific food textures, such as softer bread or pre-sliced portions for easier consumption.

Based on our preliminary survey and research, their needs include:
Advanced Customization: Intuitive tools to modify ingredients (e.g., removing allergens, reducing sugar, or substituting eggs) to fit personal health plans.

Accessibility and Inclusivity: A simple, high-contrast interface designed to be user-friendly for all ages, especially the elderly, ensuring a frustration-free ordering process.

Texture and Physical Accommodation: Options to request specific preparation styles, such as extra-soft textures or bite-sized pieces for those with chewing difficulties.

Locality and Freshness: A seamless connection with nearby partner bakeries to ensure customized orders are prepared fresh and delivered promptly.

Nutritional Transparency: Clear, real-time information regarding the nutritional value and ingredients of their customized "BakeCraft" creations.

The app meets these documented needs by offering a robust customization engine and an accessible interface that bridges the gap between specific health requirements and local bakery expertise.

1.3 Goals and objectives

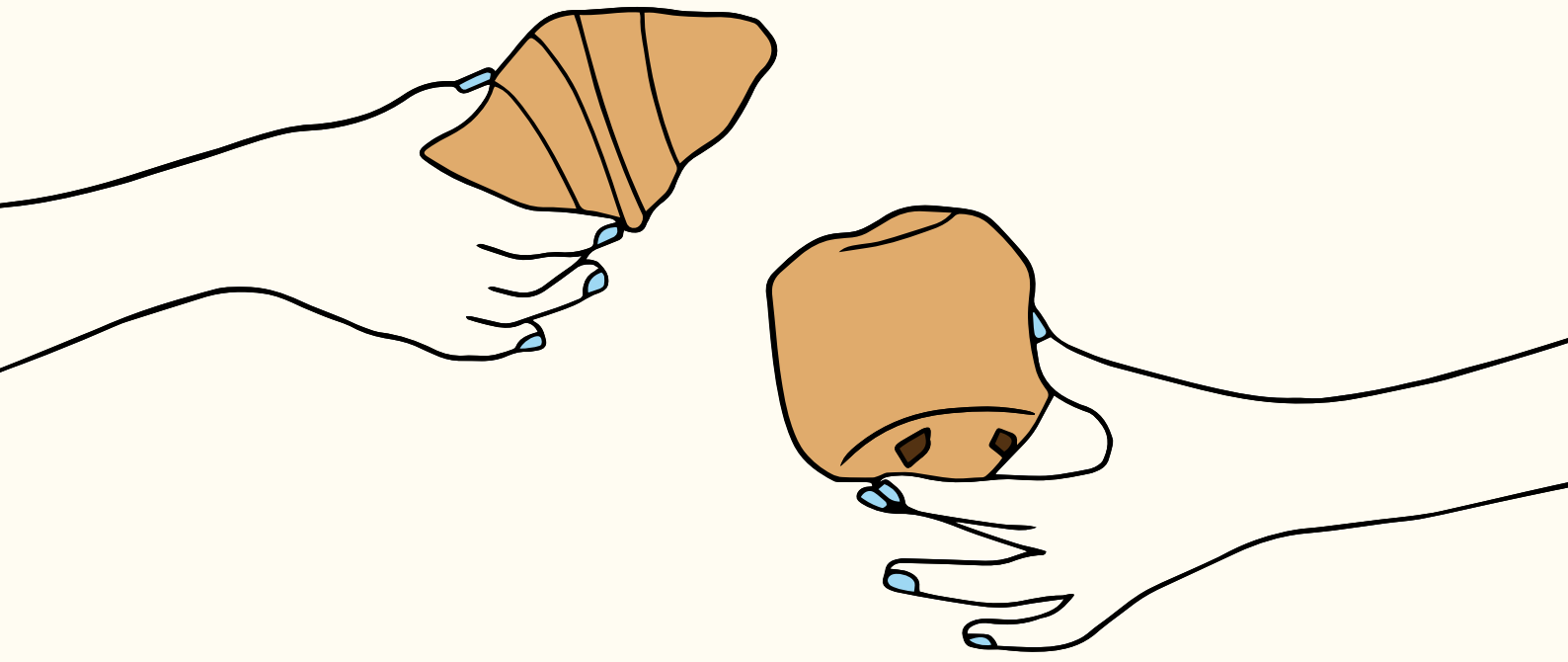
Goal:

To develop a mobile application that make users capable of finding and customising baked goods depending on their preferences and choices, conect the user with suitable shops or bakeris that are able to receive the order and prepare it. And delivering the order at the end which ensure a smooth and efficient ordering experience from customization to final delivery.

Objectives:

- Allow users to customize baked products based on their preferences.
- Connect customers with suitable bakeries that can fulfill their orders.
- Help users to discover the bakeries that offer specialized options.
- Help bakeries in receiving more customers and manage customised orders by providing them with a simple platform.
- Ensure a smooth ordering process from product customization to order completion and delivery.

Stage 1 : Empathy



2.1 Pilot Study on Research Methods

2.1.1 Questionnaire pilot study

2.1.1.1 Objectives

The questionnaire pilot study aims to assess users' requirements, their preferred choices and their difficulties which they encounter when they used the first version of the survey . The study seeks to get the users' feedback and how to analyse them in order to improve the questionnaire to meet their needs.

2.1.1.2 Sample size

We collected a variety of responses from a sample of 10 people.

2.1.1.3 Recruitment

Online platforms which included WhatsApp groups and social media served as the recruitment method for participants.

2.1.1.4 Procedure

The questionnaire was distributed using **Google Forms**. Participants were asked to complete the survey which contains multiple-choice questions at a time that suited them and then provide feedback after completing it.

2.1.1.5 Feedback questions

The feedback questions are asked to users after giving them a sample of the questionaire to help us improve our survey. These are the questions:

Do you have any suggestions or ideas to improve the app?
Were the instructions and overall structure easy to follow?
How long did it take you to complete the questionnaire?
Do you have any suggestions to improve the survey?
Was the title font clear and easy to read?
Was it clear to exit the questinnaire if you chosed not to prticipate?

Data Analysis

We conducted an analysis of feedback from some of the survey pilot study participants, identifying common problems they encountered while using it. The responses were categorized into areas such as question clarity, structure, answer choices, and overall design. We used their comments and suggestions to refine and improve the survey, making it clearer and more flexible to meet their needs.

2.1.1.7 Iterative change

Several questions were changed and modified to become more effective and useful. For example, one of the Respondents recommended us to make the font of title bigger and to add Arabic subtitle. Another change is in the first question, it was modified so that if someone does not want to participate in the survey, they can go directly to submitting it. Also, many of the questions were only one choice, so that more than one option can be chosen in order to make the topic more flexible for the respondent. In addition, the second question, which asks about age, had an option added for over 50 to include the largest number of individuals.

2.1.1.8 Conclusion

Overall, the survey was full of valuable information and one of the most helpful tools for understanding people's thinking, preferences, and how to design an application that meets their needs. In particular, their feedback allowed us to consider aspects we might not have otherwise considered. We will strive to meet these requirements as much as possible.

2.1.2 Interview pilot study

Objectives

The main goal of the pilot interviews was to assess the effectiveness and clarity of the interview questions while obtaining preliminary information about the difficulties consumers have while looking for bakeries that provide customized or dietary-specific goods. Additionally, the survey sought to determine what users needed in terms of safe ingredient alternatives and subscription services that could improve ease.

Sample size

A pilot research was first carried out with a small sample of three individuals between the ages of 18 and 60 in order to evaluate and improve the interview questions. To guarantee that the questions were understandable and inclusive of all viewpoints, the participants came from a range of backgrounds. This first sample was meant to validate the interview framework before it was applied more widely to users who wanted to customize their bakery orders rather than to reflect a particular demographic.

Recruitment

Individuals were contacted and asked to participate in a study on a customized baking application in order to enroll participants. Prior to the interviews, each participant was made aware of the goals of the study and granted consent for their answers to be recorded.

Procedure

After participants agreed to participate, individual semi-structured interviews were undertaken. Each session lasted roughly ten to fifteen minutes. In order to enable a methodical and thorough study of the data gathered, the interviews were audio recorded. This made it possible to identify important patterns and conclusions from the participants' answers.

Feedback questions

The following feedback questions were asked to participants after completing the interviews to evaluate the clarity, structure, and overall effectiveness of the interview questions:

Were all the questions clear and easy to understand?
Did the order of questions feel logical and natural?
Was the number of questions appropriate?
What would you change to improve the interview questions?

Data Analysis

The responses from the participants were categorized into key concepts that reinforced the main goal of the study by looking at the data from the recordings. Their experiences purchasing personalized baked goods, difficulties with dietary requirements like lactose or gluten intolerance, and their readiness to use an application that streamlines the procedure were the main topics.

Iterative change

Based on the feedback received from participants, several improvements were made to the interview questions to enhance clarity, flow and oveall usability.
Some questions were simplified to reduce confusion, as partcipants indicated a need for clearer wording and more real-life examples such as lactose-free or suger-free options, minor adjustments were also made to avoid repetition and make the questions more concise and easier to understand.
As a result, the final version of the interview questions became clearer, more structured, and easier for participants to respond to.

Conclusion

In conclusion, the purpose of the interviews was to learn more about the hidden difficulties that people face while placing orders for personalized baked items. To obtain a range of viewpoints, ten people were interviewed. The final version of the BakeCraft app was able to capture users' requirements, desires, and ideas for how to make it since the pilot interviews improved the clarity and organization of the questions.

2.1.3 Observation pilot Study

Objectives

The aim of this observational pilot study was to understand how users behave when searching for bakeries, particularly when looking for customized baked goods. It also explored users' familiarity with digital platforms and identified key factors influencing their decision-making process, such as ratings, photos, pricing, and delivery options.

Sample Size

The observation was conducted with three participants who regularly purchase customized bakery products, allowing for focused insights into relevant user behaviors.

Recruitment

Participants were recruited using convenience sampling and represented a range of ages and backgrounds to capture varied user perspectives.

Procedure

Participants were asked to simulate the process of ordering a customized baked product (e.g., a cake) using their usual method and preferred applications. During the observation, the researchers focused on the following:

- How users search for bakeries
- Which applications or platforms they use
- How they compare different options
- What information they prioritize (e.g., price, ratings, images, ingredients)
- Any difficulties or points of hesitation during the process

2.1.3 Observation pilot Study

Feedback (First observation)

The initial observation was conducted to understand user behavior when interacting with existing bakery or food delivery applications. It revealed that users tend to focus heavily on visual content, such as product images, as well as delivery-related details like time and cost. The predefined observation categories, including “Focus” and “Hesitation,” were effective in identifying moments of user uncertainty, particularly when users were exploring customization options or reviewing order details. Some participants appeared confused when trying to modify their selections, highlighting a potential need for clearer customization features.

Data Analysis (First observation)

The analysis showed that approximately 80% of users prioritized price and distance when choosing a bakery. Additionally, repeated tapping or hesitation was observed when users expected certain elements (such as images) to be interactive. The observation also suggested that lengthy registration processes may act as a barrier, as users showed reluctance when required to input extensive personal information before completing an order.

Iterative Change (First observation)

Based on the pilot observation, several design directions were identified for the proposed “BakeCraft” application. These include emphasizing clear and visible customization options, enhancing interactive cues for visual elements, and simplifying the onboarding process by considering features such as guest checkout. These insights will guide the next stages of the design process and help ensure that the application aligns with real user behaviors and expectations.

2.1.3 Observation pilot Study

Feedback (Second observation)

The second observation focused on how users interact with bakery or food delivery applications when they have specific dietary preferences. Participants emphasized the importance of clear ingredient and nutritional information, particularly for those with dietary restrictions. Some users felt overwhelmed by the large number of available options and expressed a preference for a simpler and more guided customization process. Others highlighted the value of well-organized categories and recommendations to support faster and more confident decision-making.

Data Analysis(Second observation)

The analysis revealed that users with dietary requirements tend to spend more time reviewing product details to ensure suitability. A recurring pattern showed that excessive choices and complex layouts led to hesitation and decision fatigue. In contrast, clear visuals and structured information were found to enhance user trust and improve the overall decision-making process.

Iterative Change(Second observation)

Based on these insights, simplifying the customization process and organizing options into clear, intuitive categories are essential design considerations. Additionally, incorporating recommendation features and improving the clarity of ingredient information can help reduce confusion and support more efficient user decisions. These directions will guide the development of the proposed application concept.

2.1.3 Observation pilot Study

Feedback (Third observation)

The third observation highlighted that pricing plays a significant role in shaping user behavior, often triggering strong emotional responses. Participants compared multiple options, paying close attention to prices while briefly viewing product images. Nutritional information received minimal attention. Additionally, users appeared overwhelmed by cluttered interfaces, which made it difficult to locate preferred options and sometimes led them to reconsider their choices.

Data Analysis (Third observation)

The findings indicate that price is a primary driver in user decision-making, outweighing other factors such as visual appeal or nutritional content. Complex and crowded interfaces introduced noticeable hesitation and confusion, demonstrating that usability issues can significantly impact user choices. Overall, users prioritized affordability and ease of navigation over detailed product information.

Iterative Change(third observation)

To address these findings, reducing visual clutter and simplifying the interface structure are key priorities. Presenting pricing information clearly and prominently can better support user decision-making. Additionally, nutritional information could be redesigned in a more engaging and accessible way to encourage user interaction. These improvements aim to create a smoother and more user-friendly browsing experience.

Conclusion

Overall, the observation pilot study provided valuable insights into user behavior when interacting with existing bakery and food delivery applications. Across the observations, price sensitivity, ease of navigation, and clarity of information consistently influenced user decisions. Users showed a preference for simple, well-structured interfaces with clear customization options, while complex layouts and excessive choices led to hesitation and frustration. Additionally, users with dietary needs placed greater emphasis on ingredient transparency, highlighting the importance of trust and clarity. These findings will inform the next stages of the design process and guide the development of a more user-centered and intuitive application experience.

2.2 Research Methods

2.2.1 The Observation

Objective	Observed Behavior	Approach
Understand users' actual behavior when purchasing baked goods	Users repeatedly zoom in on product images and check the ingredients before adding items to the cart	Direct observation
Identify key decision-making factors (price, ratings, distance, delivery time)	Users toggle between different bakeries to compare delivery fees and estimated arrival times for the same product type	Direct observation
Detect usability issues in current ordering methods or apps	Users exhibit 'rage clicking' on non-intractive icons or struggle to find the 'edit customization' button after adding an item	Indirect observation
Explore trust concerns when ordering customized products	Users check reviews and hesitate before confirming	Indirect observation
Identify barriers that prevent users from using mobile applications	Users abandon the process when the registration requires too many personal information or when encountering a complex check out flow	Direct observation

Table 1 - Observation objectives and approaches

2.2 Research Methods

2.2.1 The Observation

Objective	Observed Behavior	Approach
Observe how users handle dietary needs and allergy-related decisions	Users showed strong interest in customization, especially in choosing ingredients, fillings, and dietary options	Direct observation
Map the full user journey from search to order completion	Users followed a clear journey from searching to order completion, starting with familiar platforms such as social media, delivery apps, or known bakeries	Indirect observation
Analyze how users interact with product information (images, ingredients, nutrition)	During journey tracking, users compared options based on price, distance, and availability before proceeding	Direct observation
Observe emotional reactions during the ordering process	Emotional reactions such as hesitation and slight frustration were observed when information was unclear or when too many options were presented	Indirect observation
Evaluate how users respond to multiple options and complexity Users feel overwhelmed or simplify choices Observe decisions	Some users felt overwhelmed by multiple choices and tended to simplify their decisions or select default options	Direct observation

Table 1 - Observation objectives and approaches

2.2 Research Methods

2.2.2 The Questionnaire

Phrase	Objective	Approach
How often do you buy baked goods?	To understand the frequency with which users purchase baked goods in order to estimate the level of demand for bakery products and the potential usage of the application.	Multiple Choice
Where do you usually buy baked goods?	To identify the common sources where users purchase baked goods and understand their current purchasing behavior. In order to know which Source to add.	Checkbox
Have you ever ordered customized baked goods (e.g., cake with a name, special design)? If yes, how did you place your order?	To determine whether users have prior experience with ordering customized baked goods and to understand the methods they use.	Multiple Choice (Yes, No) & Checkbox
What is the biggest difficulty you face when ordering customized baked goods?	To identify the main challenges and pain points users encounter when ordering customized baked goods.	Checkbox
What matters most to you when choosing a bakery?	To have a better understanding on the factors that effect their buying decisions	Checkbox

Table 2 - Questionnaire objectives and approaches

2.2 Research Methods

2.2.2 The Questionnaire

Phrase	Objective	Approach
What type of customization do you prefer when ordering baked goods? (You can select more than one)	To identify the types of customization options users prefer when ordering baked products.	Checkbox
Do you want to see a preview of the product before ordering?	To evaluate whether users are interested in viewing a visual preview of customized baked goods before confirming their orders.	Multiple Choice (Yes, No)
How would you like to customize your order?	To know their preferred way to describe customised orders in order to use it in the app.	Checkbox
Which of the following features would you find most useful in a mobile app for ordering customized baked goods?	To identify the most important features users expect in a mobile application for ordering customized baked goods.	Checkbox
Would you like the app to save your previous customized orders for easier reordering in the future?	To determine whether users prefer having a feature that saves their previous orders to simplify future purchases.	Multiple Choice (Yes, No)

Table 2 - Questionnaire objectives and approaches

2.2 Research Methods

2.2.2 The Questionnaire

Phrase	Objective	Approach
How useful would it be if the app suggested bakeries based on your dietary restrictions or preferences?	To determine whether consumers value tailored bakery suggestions based on their dietary requirements and preferences.	Multiple Choice
Would you prefer the app to include a feature that allows you to communicate directly with the bakery for special customization requests?	To evaluate whether consumers desire a feature that allows them to communicate directly with bakers for personalized orders.	Multiple Choice (Yes, No, Maybe)
Would it be helpful if the app allowed you to view pictures of customized baked goods previously made by the bakery?	To determine if consumers choose a bakery based on visual representations of prior customized orders.	Multiple Choice
How important is it for you to see bakery ratings before ordering?	To assess the degree to which consumers rely on ratings when choosing a bakery.	Multiple Choice
What is your preferred method for receiving your. order?	To determine the delivery method that users prefer (e.g., pickup versus delivery).	Multiple Choice

Table 2 - Questionnaire objectives and approaches

2.2 Research Methods

2.2.2 The Questionnaire

Phrase	Objective	Approach
Would you like to track your order in real-time?	To determine how crucial real-time order tracking is to enhancing customers' ordering experiences.	Multiple Choice (Yes, No)
Would you prefer notifications about your order status?	To figure out whether users are interested in being informed about the status and updates of their orders.	Multiple Choice (Yes, No)
Do you have any suggestions or ideas to improve the app? (Optional)	To collect further user suggestions and qualitative input in order to help the application's future enhancements.	Short Answer

Table 2 - Questionnaire objectives and approaches

2.2 Research Methods

2.2.3 The Interview

Phrase	Objective	Approch
Q1: Do you usually buy baked goods from bakeries?	To understand whether users typically purchase baked goods from bakeries	Close- ended
Q2: How do you usually find bakeries?	To explore the methods users use to discover bakeries that make us have better understanding of their behaviour	Close-ended
Q3: How important are dietary options such as low sugar, gluten-free, or special ingredients to you?	To understand the importance of dietary preferences and special ingredient options when users choose baked goods	Open-ended
Q4: Have you ever ordered customized baked goods before? If yes, can you describe your experience?	To understand the main pain points and the good things regarding their experience that we could learn from.	Open-ended
Q5: Do you face any difficulties when trying to find bakeries that offer customized baked goods?	To identify the challenges users encounter when searching for bakeries that provide customized baked products.	Close-ended

Table 3 - Interview objectives and approaches

2.2 Research Methods

2.2.3 The Interview

Phrase	Objective	Approch
Q6: What kind of customization options would you like to have when ordering baked goods?	To explore users' expectations and preferences regarding customization options when ordering baked goods.	Open-ended
Q7: When customizing your bread (like removing sugar or adding fillings), do you prefer to choose from a ready-made list or write your own notes to the baker?	To determine how customers prefer to submit customisation requests (writing their own notes or choosing from predefined options) when placing baked goods orders.	Close-ended
Q8: If an ingredient you want to remove (like eggs) is essential to the bread's structure, would you like the app to suggest a "safe alternative" (like flaxseed)?	To determine whether users would appreciate the app's recommendation of safe substitutes for necessary ingredients while eliminating them from baked items.	Close-ended
Q9: Since customized bread takes time to bake, how long are you willing to wait for a "made-from-scratch" order compared to a regular one?	To assess how long customers are willing to wait for personalized, homemade baked items in comparison to standard orders.	Open-ended

Table 3 - Interview objectives and approaches

2.2 Research Methods

2.2.3 The Interview

Phrase	Objective	Approach
Q10: For customized healthy bread that you might need daily, would you be interested in a "Weekly Subscription" where your specific loaf is delivered fresh at a set time?	To find out whether customers would be interested in a weekly subscription service that would send personalized, healthful bread on a regular basis.	Close-ended
Q11: If there was a mobile app that connects you with bakeries that prepare baked goods based on your preferences, would you use it? Why or why not?	To estimate how likely users are to utilize a smartphone app that links them to bakeries that provide customized baked items.	Open-ended
Q12: What is the one must-have feature you need in the app to satisfy your bakery needs?	To identify the most crucial feature that people look for in an application to fulfill their demands related to bakeries.	Open-ended

Table 3 - Interview objectives and approaches

2.3 Aggregate Empathy Maps

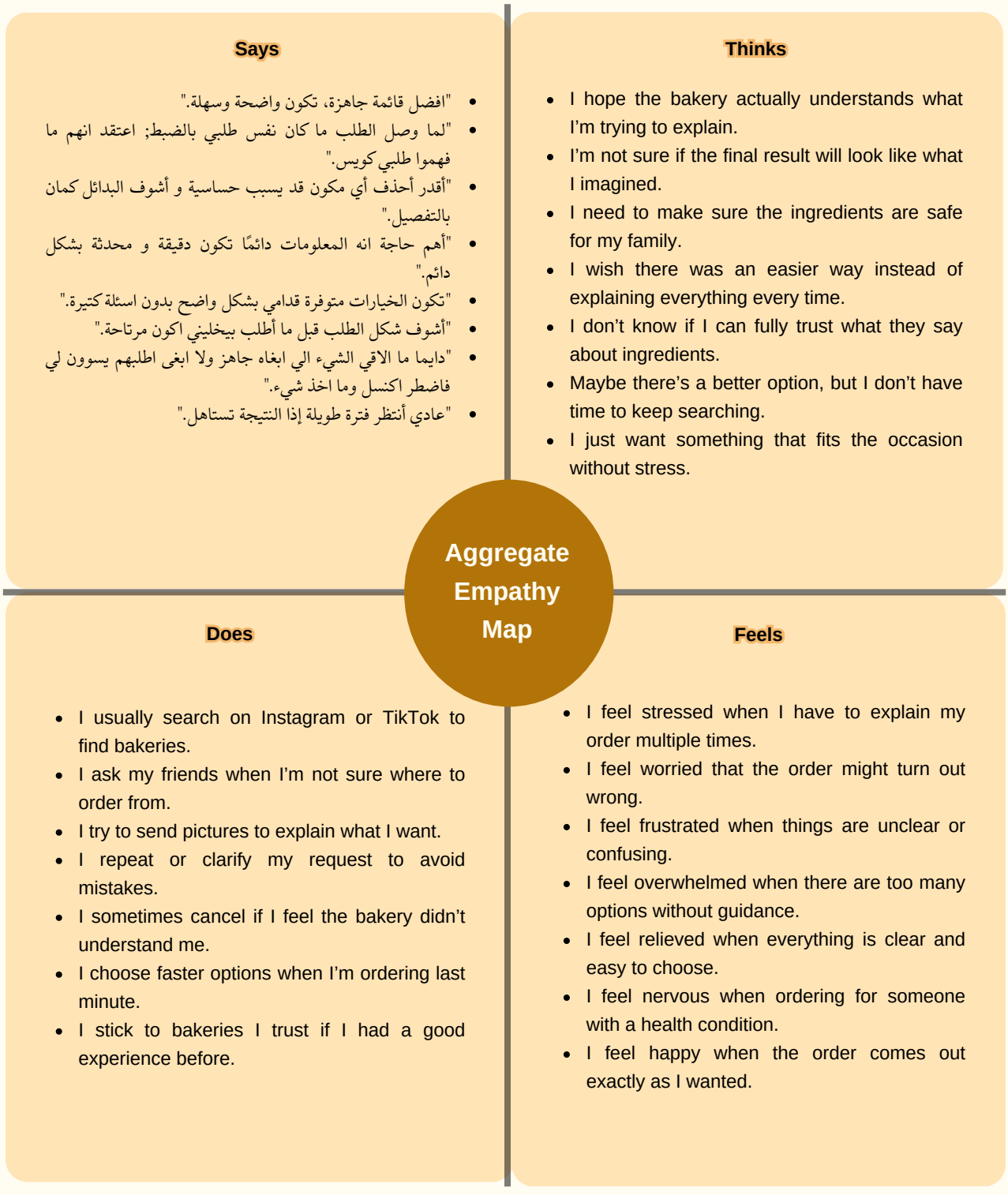


Figure 1 - Aggregate empathy map

2.3.2 Interview Analysis (cross-interviews)

In analyzing the interview responses, several common patterns were identified regarding users' experiences with baked goods and customization. Most participants showed a strong interest in customizing baked goods, whether for health-related reasons such as low sugar or allergies, or simply for personal preferences like choosing ingredients, fillings, and taste. This indicates that customization is a key need for a wide range of users, not limited to those with strict dietary restrictions.

It was also observed that many users face difficulties when trying to find bakeries that offer reliable customization services. Common challenges include limited availability of customized options, difficulty in clearly communicating specific requests, and lack of trust in whether the bakery will meet their expectations. In addition, issues related to delivery time, location, and high prices were frequently mentioned, which negatively affect the overall experience.

Participants also showed different preferences in how they want to customize their orders. Some preferred selecting from ready-made options for simplicity and speed, while others preferred writing personal notes to have more control over their requests. This highlights the importance of providing a flexible system that supports both methods. Moreover, there was clear interest in features such as suggesting alternative ingredients, especially for users with allergies or specific dietary needs.

Finally, the results reflect a clear need for a simple and user-friendly solution that connects users with bakeries offering customized baked goods. Users expect features such as clear ingredient information, reliable delivery, reasonable pricing, and filtering options based on distance or preferences. Overall, the data emphasizes the importance of creating a platform that simplifies customization, improves communication, and provides a trustworthy and convenient experience.

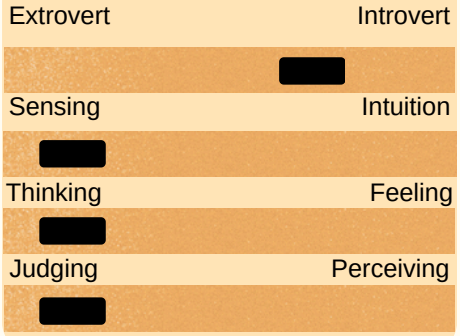
2.4 User Persona (1)

The Health-Conscious User

"I want to clearly see what I'm eating and customize it to fit my diet."

Name: Mohammed Al-Harbi
Age: 35
Occupation: Engineer
Education: Student of mechanical engineering at KAU.
Personality: ISTJ
Status: married

Personality



BIO

Mohammed is is an athletic and health-conscious user who prefers baked goods that match his diet, such as low sugar, high protein, or whole wheat. He often spends time checking ingredients and nutritional details before making a decision.

Tech Level

Medium.

Uses delivery apps and social media.

Comfortable but not expert

Pain Points

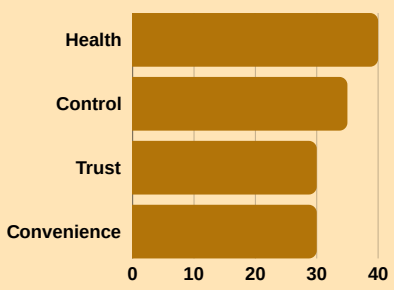
Difficulty finding bakeries that meet health requirements.

Hard to clearly describe customization to bakeries.

Lack of clear nutritional information in apps.

Too many options cause confusion.

Motivations



Goals

Find healthy baked goods (low sugar, high protein, whole wheat).

Access clear ingredient and nutritional information

Customize ingredients based on dietary needs

Receive reliable and high-quality products

2.4.1 User journey - persona 1

The Health-Conscious User

ACTION	Discover Bakeries	Search & Filter Products	Customize Baked Items	Place Order & Payment	Track & Receive Order
TASK LIST	1. Ask friends or family for recommendations. 2. Search on social media (e.g., Instagram, Snapchat). 3. Use Google Maps to find nearby bakeries. 4. Check ratings and comments.	1. Browse bakery profiles and menus. 2. Look for information about healthy or dietary options. 3. Compare prices and product quality. 4. Check photos and previous customer reviews	1. Contact the bakery via WhatsApp or phone. 2. Explain dietary needs. 3. Discuss flavor, size, design, and ingredients. 4. Exchange images or reference designs. 5. Wait for confirmation of feasibility and price.	1. Confirm order details. 2. Choose delivery or pickup. 3. Select payment method. 3. Receive order confirmation.	1. Wait for updates from the bakery. 2. Receive the order via delivery or pickup. 3. Check if the product matches dietary and design requirements. 4. Provide feedback informally through messages or social media.
FEELING ADJECTIVE	Curious and hopeful about finding healthy options.	Focused but slightly overwhelmed due to scattered and unclear information.	Careful and reassured when clear information is available; concerned about ingredient accuracy.	slightly anxious about order acceptance.	Concern that the final product may not match the requested customization.
PAIN POINTS	Difficulty finding bakeries that offer healthy or customized products.	1. Lack of clear nutritional information. 2. Hard to compare options efficiently	1.Communication gaps and misunderstanding. 2.Time consuming. 3. back-and-forth messaging.	Uncertainty about whether the bakery can meet health requirements.	1. Lack of real-time tracking. 2. Risk that the final product does not match the request. 3. No structured feedback or rating system.
IMPROVEMENT OPPORTUNITIES	1. Centralized platform listing verified bakeries. 2. Smart search with location-based results.	1. Advanced dietary filters. -2.Transparent ingredient and nutritional labels. 3. Easy comparison of bakeries.	1.Include an intuitive customization interface with predefined dietary options and visual previews. 2.Direct in-app communication with bakeries.	Enable direct chat with bakeries and Automatic order summaries and confirmations.	1.Real-time order tracking. 2.Quality assurance badges. 3.Integrated rating and feedback system.

Table 4 - User journey (persona 1)

2.4.1 Empathy Map

Persona (1) The Health-Conscious User

Figure 2 - Aggregate empathy map for persona 1

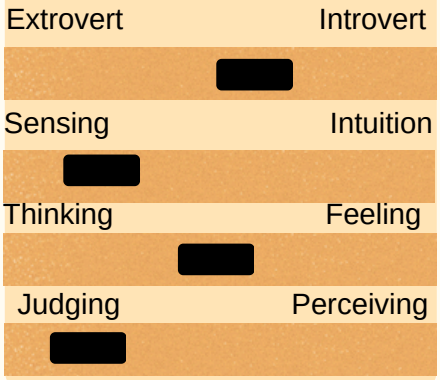
2.4.2 User Persona (2)

The Allergic user

"I just want to order without having to ask too many questions about my order's ingredients."

Name: Noor Ibrahim
Age: 24
Occupation: Senior student
Education: Studying interior design at University of Jeddah
Status: Single

Personality



BIO

Noor enjoys trying different bakeries but has difficulty ordering because she has lactose intolerance and gluten allergy, and she has to double-check everything on the menu. She incessantly checks ingredient labels before eating anything new.

Tech level

Medium - High

Skilled with both daily and professional / creative apps.

Quick to learn new interfaces

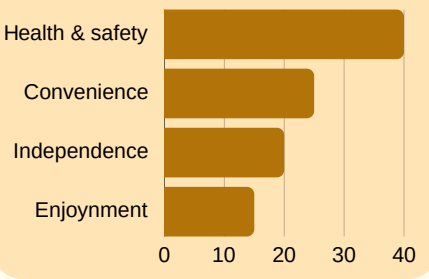
Pain Points

There's no option to filter bakery items by allergy type before ordering

Ordering feels uneasy and time-consuming when compared to others

Substitutes (e.g., oat milk, gluten-free flour) may be difficult to find or unavailable

Motivations



Goals

Avoid allergens that could cause serious health conditions
Save a personal allergy profile so she doesn't have to enter it every time
Provide safe alternatives to critical ingredients

2.4.2 User journey - persona 2

The Allergic user

ACTION	Discover bakeries	Browsing & Filtering	Order Customization /substitution request	Order Confirmation and payment	Track and recieve order
TASK LIST	1. Opens the app 2. Goes over a long list of bakeries that don't have any allergy indicators.	1. Manually searches each bakery's menu for items marked "lactose free," "gluten-free," etc. 2. Attentively reads descriptions	1. Finds something she likes, but the list of ingredients is not comprehensive. 2. Tries to ask for a replacement, but there aren't many options available. 3. spends a long time comparing the product's ingredients to what she knows is safe for her.	1. Reviews her order information many times before verifying 2. Payment proceeds without confirming whether her request for a replacement was properly submitted.	1. Waits for her order while questioning whether the bakery recalled her customization.
FEELING ADJECTIVE	Overwhelmed and uncertain; there are too many possibilities, but none of them feel safe or reliable enough for her to firmly choose.	She is frustrated because she is taking far too long to find a safe option.	She is anxious and tense since she doesn't think her request was fully comprehended.	Hesitant and nervous	She is concerned the entire time until she physically examines the meal to make sure it appears right.
PAIN POINTS	Overwhelmed by the variety of bakery options and the lack of obvious allergy-friendly signs	Some bakeries claim to be allergy-friendly but don't provide enough information, and filters aren't specific enough.	There are limited options for substitution, and there is no assurance that the replacement is entirely safe.	There is no way to send a special allergy request or message to the bakery before confirming	Direct contact with the bakery is limited in case of special allergy concerns.
IMPROVEMENT OPPORTUNITIES	Show an "allergy-friendly" tag on bakeries that allow complete customisation.	Include particular allergy filters (lactose-free, gluten-free) and clearly defined bakery-level indicators.	Provide preset allergy-safe alternatives with a safety confirmation message after each substitution.	Include a clear allergy summary card before the final checkout and payment.	Send a notification when the bakery certifies that the order was prepared according to the allergy preferences.

Table 5 - User journey (persona 2)

2.4.2 Empathy Map

Persona (2) The Allergic User

Figure 3 - Aggregate empathy map for persona 2

2.4 User Persona (3)

The Convenience Seeker User

"I need a quick and simple way to order without wasting time searching or messaging multiple bakeries."

Name: Noura Ahmed

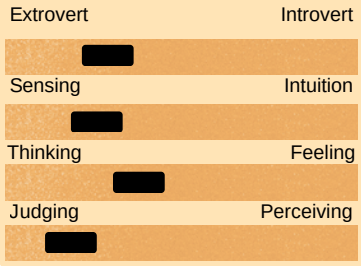
Age: 35

Occupation: Project Manger

Education: Student of Management at KSU.

Status: Married and a mother of two

Personality



BIO

Sara is a busy working mother managing both her career and family responsibilities. She frequently orders baked goods for her children, guests, or small occasions. Due to her tight schedule, she relies heavily on quick and convenient methods like Instagram, WhatsApp, and delivery apps. She prefers fast, simple, and reliable ordering rather than complex customization

Tech Level

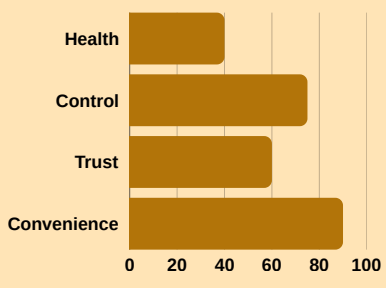
High

Frequent use of delivery platform and mobile apps

Pain Points

- Needs to switch between multiple platforms (Instagram, WhatsApp)
- Slow response time from bakeries
- No clear or fast way to compare options (price, delivery, availability)
- Ordering process is time-consuming and repetitive
- No easy way to reorder previous item

Motivations



Goals

- Order baked goods quickly with minimal effort
- Save time during the search and ordering process
- Get reliable and fast delivery
- Easily reorder from previous experiences

2.4.3 User journey - persona 3
The Convenience Seeker user

ACTION	Discover Bakeries	Browsing & Filtering	Contact & Order	Order Confirmation & payment	Wait & recieve order
TASK LIST	1. Opens social media 2. Scrolls through bakery posts or ads 3. Asks friends or family for recommendations	1. Opens multiple bakery accounts 2. Checks photos, highlights, and menus 3. Looks for prices and delivery info 4. Tries to compare between bakeries	1. Sends messages via WhatsApp or Instagram 2. Asks about price, availability, and delivery 3. Waits for replies 4. Repeats questions with multiple bakeries	1. Confirms order details through chat 2. Chooses delivery or pickup 3. Sends payment 4. Waits for confirmation	1. Waits for updates from the bakery 2. Receives the order 3. Checks if everything is correct
FEELING ADJECTIVE	Busy and eager to quickly find a suitable bakery	Slightly overwhelmed and impatient while comparing options	Frustrated and rushed due to slow responses	Uncertain but hopeful that everything is correct	Concerned about timing but relieved when the order arrives on time
PAIN POINTS	Too many scattered sources (social media, chats), and takes time to find a suitable option	Hard to compare options efficiently, Missing or unclear information (price, delivery time)	Slow response time, Repetitive communication, Time-consuming back-and-forth messaging	No structured order summary, Risk of misunderstanding details, No clear confirmation system	No real-time tracking,Unclear delivery timing, Limited communication after ordering
IMPROVEMENT OPPORTUNITIES	Centralized platform for bakery discovery Personalized suggestions based on previous orders	Easy comparison between options Organized display of prices and delivery details	Faster communication channels, Standardized order request format, Reduced need for repeated questions	Clear order summary before payment, Structured confirmation process, Transparent pricing and details	Real-time tracking system, Clear delivery updates, Better communication during delivery

Table 6 - User journey (persona 3)

2.4.3.2 Empathy Map

Persona (3) The Convenience Seeker user

Figure 4 - Aggregate empathy map for persona 3

2.4 User Persona (4)
Customization & Occasion User

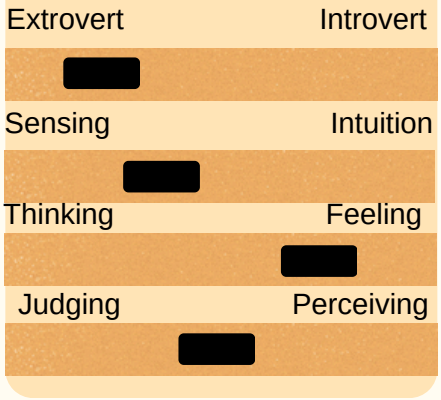
"I want to customize my order and trust it's perfect for any occasion."

Name: Layla Mohammed.
Age: 30 years old.
Occupation: Administrative specialist.
Education: Bachelor's degree in business administration.
Status: Single.

BIO

Layla is a health-conscious individual who values quality and freshness above all. she frequently hosts family gatherings and social events where standard, "ready-made" bakery items don't always meet her specific standards or dietary needs.

Personality



Pain Points

Limited Options Most bakeries only sell ready-made products rather than preparing items based on personal requests.

Fear that bakeries will not be able to fulfill specific wishes exactly as requested.

High delivery prices and long arrival times that affect the freshness and quality of the baked goods.

It takes a long time to find the "right store" that offers the specific customization options she needs.

Tech Level

High level.

Uses delivery apps and social media.

An expert who is comfortable with techs

Goals

To easily order baked goods with specific modifications without the hassle of long searches.

Impress guests at occasions with unique, high-quality products that aren't found on standard menus.

To find a "comprehensive" app that categorizes bakeries based on ratings, distance, and their ability to fulfill custom orders.

Motivations

2.4.4 User journey - persona 4

Customization & Occasion User

ACTION	Searches for bakeries	Browsing & Filtering	Order Customization /substitution request	Order Confirmation and payment	Track and recieve order
TASK LIST	1. Realizes they need baked goods for an event 2. Searches on Instagram, Google, or asks friends 3. Browses different bakery pages 3. Reads reviews and feedback	1. Reviews menus, photos, and prices 2. Looks for customization or allergy-related details 3. Compares options across different bakeries	1. Contacts bakeries via Instagram or WhatsApp 2. Asks about ingredients and customization options 3. Waits for responses 4. Shares specific requirements	1. Confirms order details through chat 2. Provides customization requests 3. Reviews details 3. Completes the order	1. Waits for updates 2. Receives the order 3. Checks product and packaging 3. Confirms it matches expectations
FEELING ADJECTIVE	Interested and exploring	Attentive and selective	Engaged and careful	Careful and attentive	Anticipating and observant
PAIN POINTS	1. Information is scattered across multiple platforms 2. Takes time to go through different sources 3. Not always clear which bakeries meet specific needs	1. Ingredient details are sometimes limited 2. Information varies between bakeries 3. Comparing options requires effort	1. Response times can vary 2. Needs to repeat similar questions 3. Some answers may not be detailed enough	1. Details may rely on manual communication 2. No structured summary of the order 3. Limited confirmation regarding specific requirements	1. Limited visibility on order status 2. Packaging may not always include detailed labels 3. Needs to manually verify details upon receiving
IMPROVEMENT OPPORTUNITIES	More organized access to bakery information Clearer overview of available options Easier way to narrow down choices	More consistent presentation of information Clearer details about ingredients and customization Simpler comparison between options	More efficient communication process Clear and consistent responses Reduced need for repeated inquiries	Clearer overview of order details More structured way to review requests Better confirmation of requirements before payment	More visibility during the delivery stage Clearer labeling of products Additional reassurance at handoff

Table 7 - User journey (persona 4)

2.4.4 Empathy Map

Persona (4) Customization & Occasion User

Figure 5 - Aggregate empathy map for persona 4

2.5 Pain points and opportunities

Pain points

Users across different personas experience a fragmented and inefficient bakery ordering process that requires navigating multiple platforms such as social media, messaging apps, and maps, leading to wasted time and increased effort. A major challenge is the lack of clear, consistent, and detailed information, whether related to ingredients, nutritional values, allergy considerations, pricing, or delivery details, which creates confusion and makes it difficult to compare options or make confident decisions. Communication with bakeries is often slow and unstructured, relying on back-and-forth messaging that leads to misunderstandings, repeated inquiries, and uncertainty about whether specific requirements—such as dietary restrictions, allergy concerns, or customization requests—are fully understood and applied. Additionally, users face limited support for personalization and filtering, making it harder to find suitable options that meet their specific needs. The absence of a clear order summary, confirmation system, and real-time tracking further reduces trust in the process, as users remain unsure about order accuracy, preparation, and delivery timing. Overall, the experience is perceived as time-consuming, mentally demanding, and unreliable, especially for users who value convenience, safety, customization, and quality.

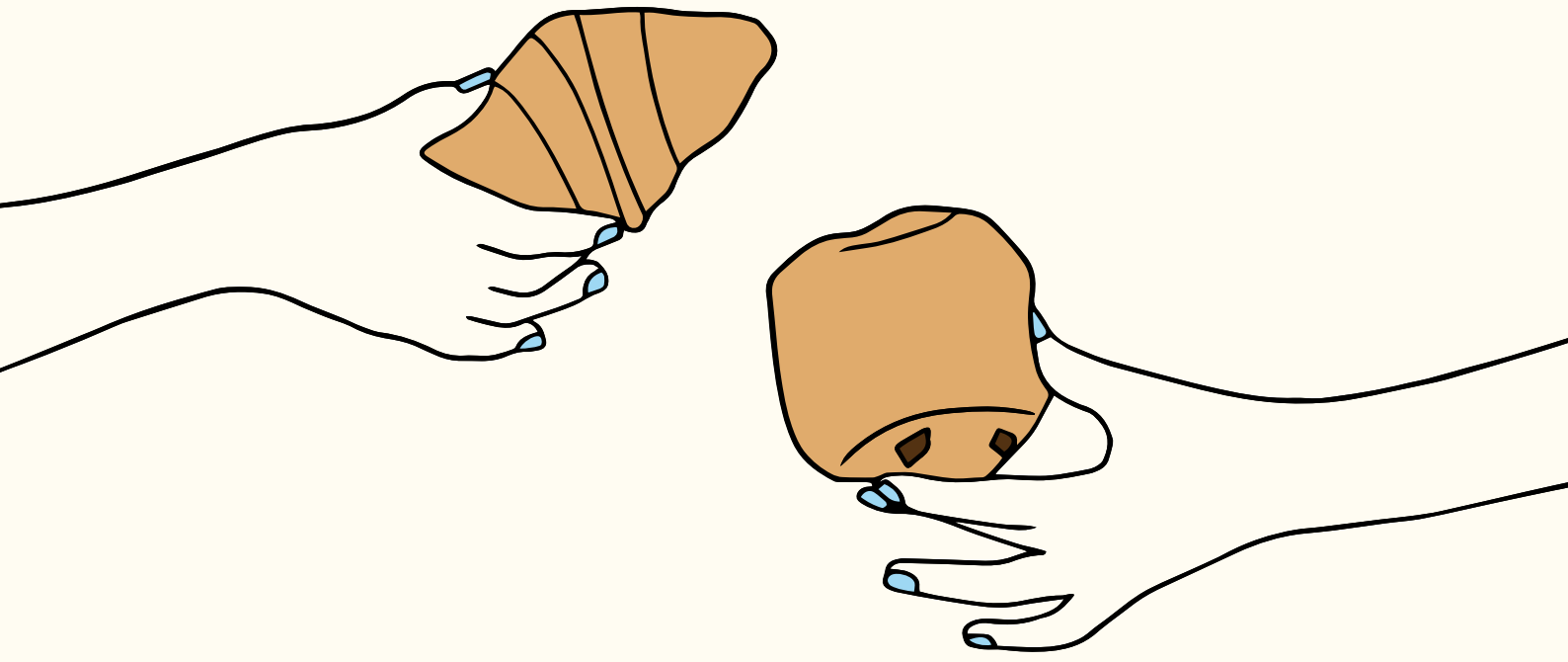
opportunities

The identified challenges highlight an opportunity to create a more adaptive and personalized bakery ordering experience that responds to different user priorities. The platform can support informed decision-making by presenting structured and easy-to-read information about ingredients, dietary categories, and preparation options. Introducing intelligent filtering and preference-based personalization can help users quickly navigate choices without feeling overwhelmed.

In addition, enabling seamless customization in a clear and guided manner allows users to confidently adjust orders to match their health requirements or occasion-specific needs. From a usability perspective, simplifying the overall flow of the application and reducing unnecessary steps can significantly improve efficiency for users who value speed and convenience. Features such as saved orders, preference memory, and consistent interface patterns can further enhance ease of use.

Moreover, improving communication between users and bakeries, along with setting clear expectations for delivery time and product quality, can strengthen trust and reliability. Altogether, these improvements position the application as a practical, flexible, and dependable solution that accommodates a wide range of user behaviors and expectations.

Stage 2 : Define



3.1 User problems and challenges

- Health-conscious consumers often struggle to find baked goods that meet their dietary needs, such as low-sugar or high-protein options. Many platforms lack clear and detailed nutritional information, making it difficult for these consumers to make suitable choices. Even when customization options are available, they are often vague and limited, forcing users to spend extra time explaining their requirements, which is frustrating and inefficient.

- Users seeking convenience often face difficulties when placing orders that are time-consuming or require navigating multiple social media platforms. Another challenge is the difficulty of comparing numerous options due to the slow response times of some bakeries. Consequently, the overall experience becomes repetitive and inefficient, failing to meet their need for a quick and simple ordering process.

- Due to insufficient ingredient information, no allergy-related filtering, and few alternatives available, allergic users must deal with needless difficulties while placing orders. What should be a straightforward process becomes difficult and time-consuming in the absence of a trustworthy method to verify that their demands were fulfilled.

- Platforms that exclusively provide conventional pre-made products frequently underserve users who are looking for highly customized and fresh baked foods. Finding the ideal bakery becomes time-consuming and unreliable without a way to select bakeries based on their customisation capabilities. This is particularly difficult when placing orders for significant events like social gatherings or family get-togethers when accuracy and quality are crucial.

3.2 The 5 W's Framework

Who: The problem affects different users who buy baked goods who include health-conscious customers and people who have dietary restrictions which include allergies and low-sugar needs and customers who want quick access to products. All users between different age groups and varying levels of technical skills need a straightforward system that enables them to order personalized baked goods.

What: The main points Users struggle with when ordering these customized baked goods are problems that we will seek to solve, include: limited customization options, unclear ingredient and nutritional information, and the need to communicate with bakeries via various social media channels to deliver requirements. The ordering process is often tiring and confusing, and lacks clear confirmation or tracking.

Where: Users typically use the product in everyday environments such as their homes, workplaces, or while using their phones. Some of them are often in uncomfortable or urgent situations, such as planning an event or placing a quick order, so we will strive to provide a seamless and hassle-free experience.

When: Users use the app to order customized baked products, whether they are planning ahead of time for a special occasion such as a family gathering or function, or they are placing a quick order on a regular basis. The need is especially apparent when users are short on time and cannot afford to go through a long and unreliable ordering procedure.

Why: The existing ordering experience for personalized baked goods is disjointed and frustrating. Users must browse across several platforms and social media pages to find a bakery that meets their requirements, with no guarantee of ingredient transparency, customization correctness. The software exists to streamline and centralize this process into a dependable and inclusive platform.

How: The app links users with bakeries that provide customization, allowing them to choose by dietary restrictions, check ingredient details, and safely place orders via a simple and user-friendly interface.

3.3 Problem Statements

3.4 Hypothesis Statements If Then

Persona 1

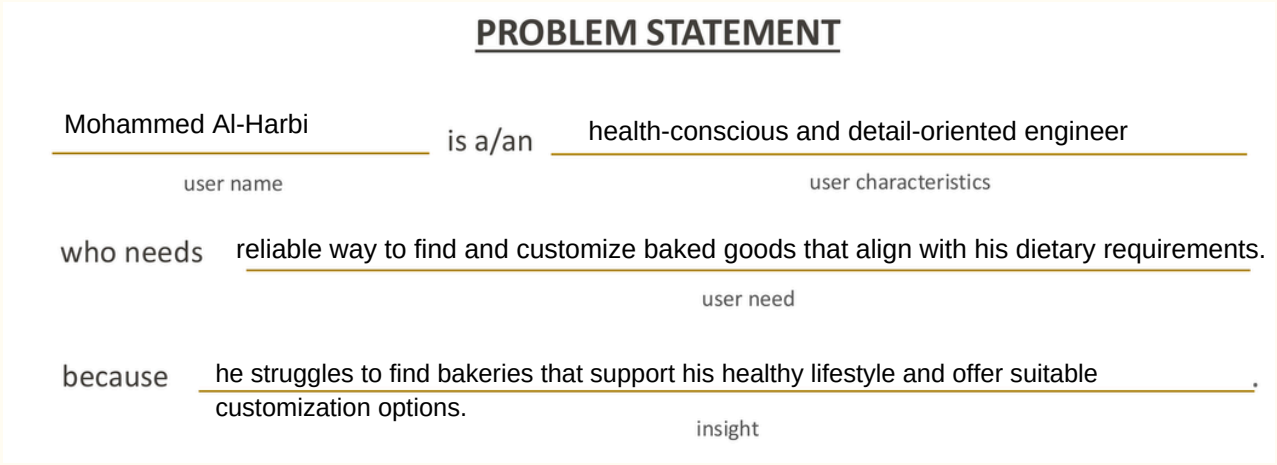


Figure 6 - problem statement of persona 1

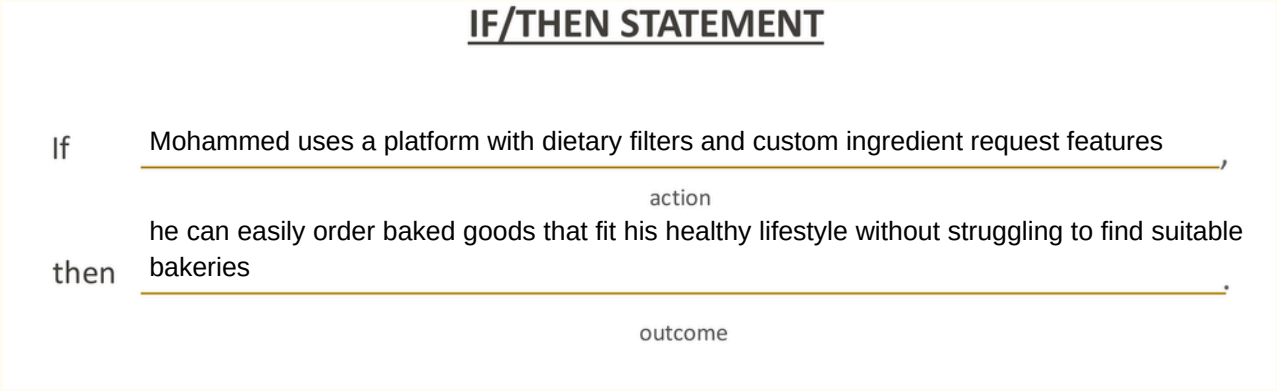


Figure 7 - if/then statement of persona 1

Persona 2

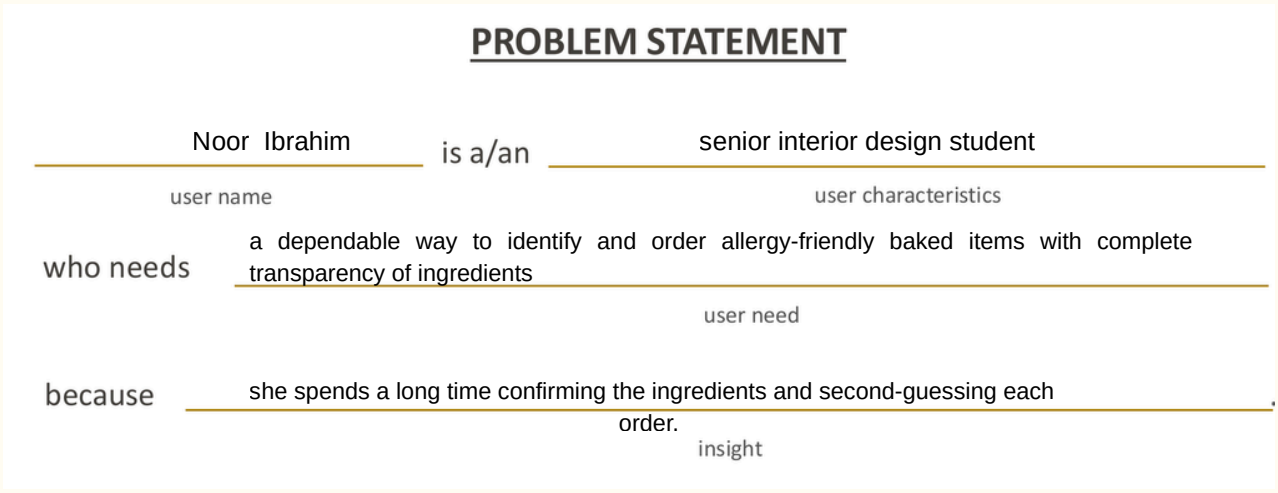


Figure 8 - problem statement of persona 2

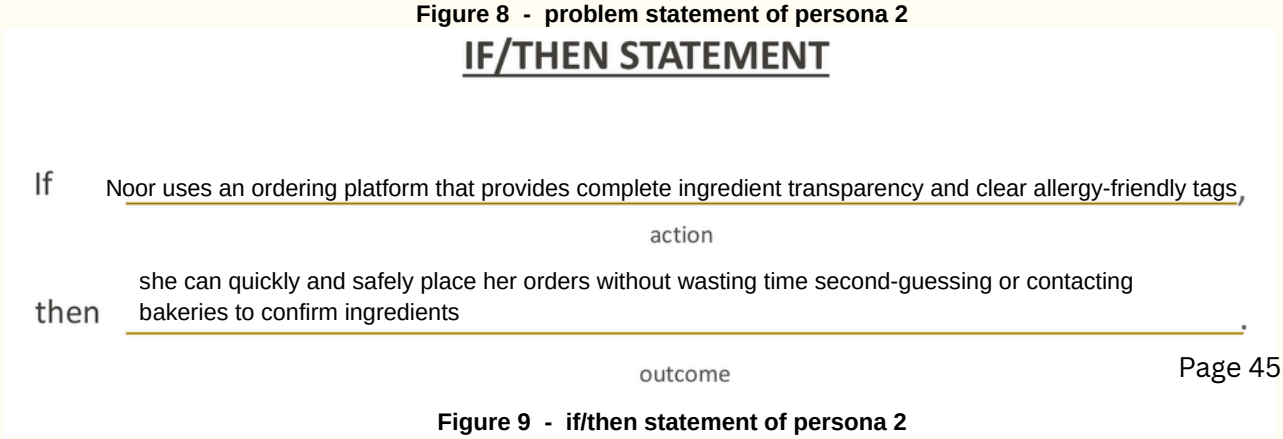


Figure 9 - if/then statement of persona 2

Persona 3

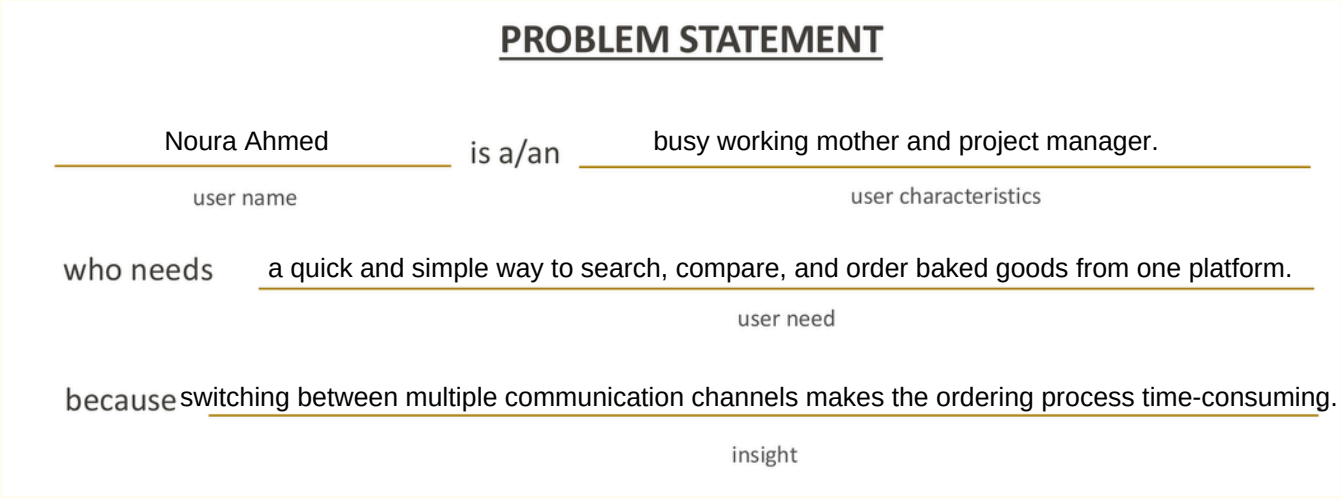


Figure 10 - problem statement of persona 3

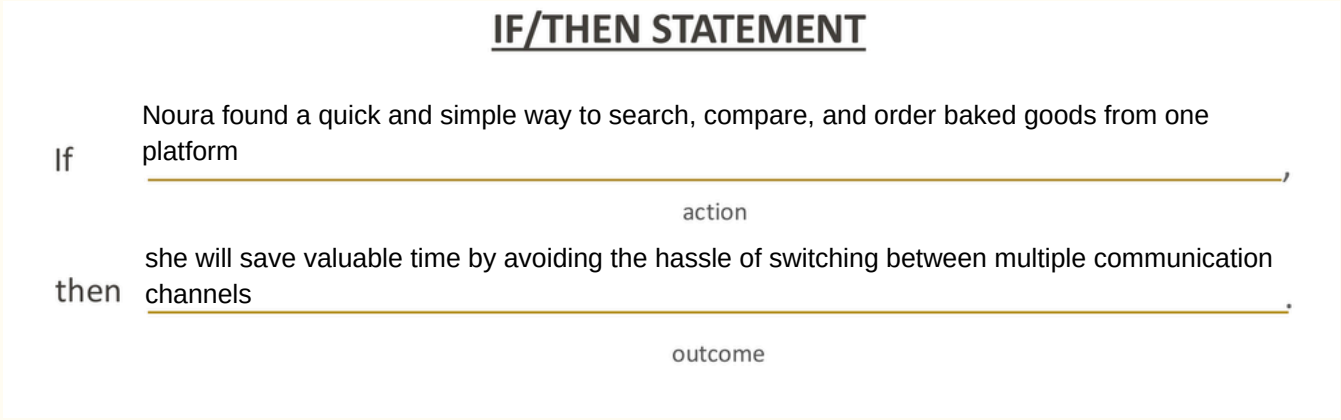


Figure 11 - if/then statement of persona 3

Persona 4

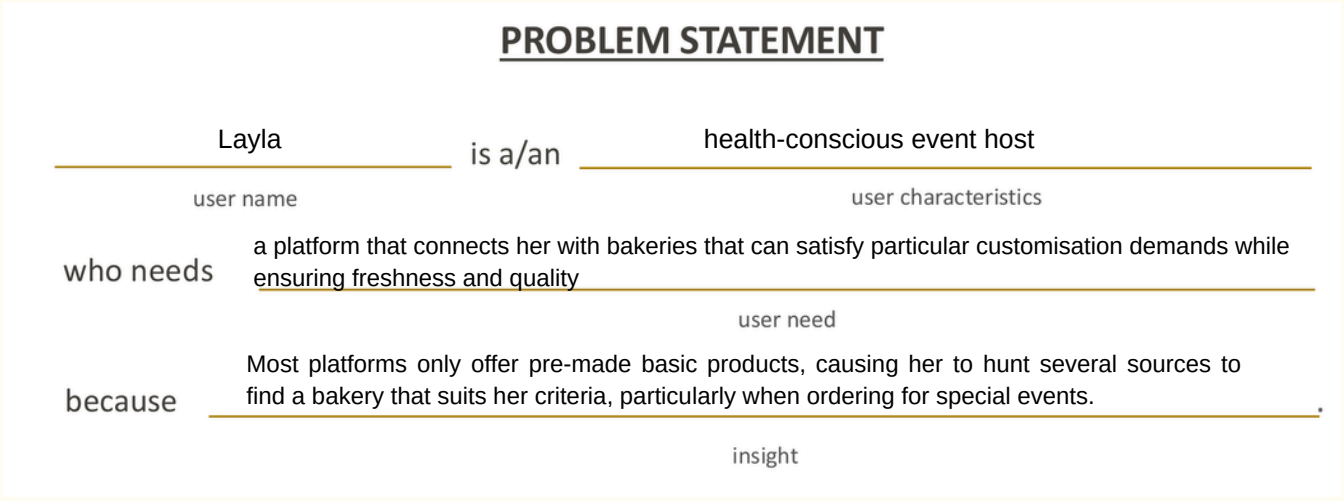


Figure 12 - problem statement of persona 4

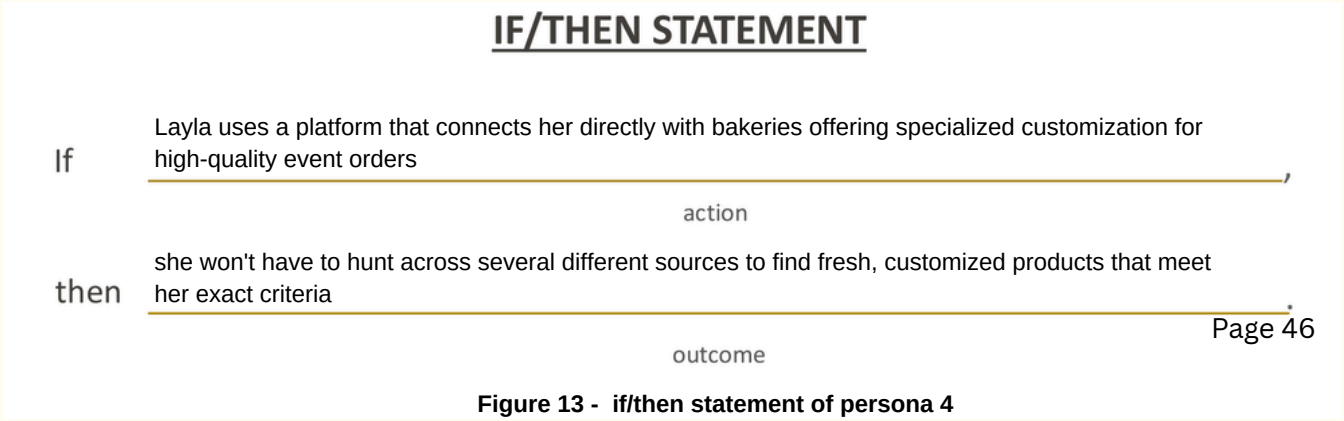


Figure 13 - if/then statement of persona 4

3.5 Value Proposition

3.5.1 Features and Benefits

- **Features**

Customizable ingredients *(e.g., sugar level, flour type, fillings)*

Allergy-based filtering *(e.g., dairy-free, gluten-free)*

Bakery discovery based on preferences

Easy bakery discovery

Ratings and reviews for bakeries

Estimated preparation and delivery time

Weekly subscription option

All-in-one bakery platform

Reorder previous items

Price comparison between bakeries

Order tracking and live status

Clear product information

Allergy profile saving

Delivery time scheduling

Personalized bakery recommendations

Availability & stock updates

Favorite bakeries list

Custom order notes

Flexible customization controls

Discounts and offers

Multiple payment methods

User-friendly interface

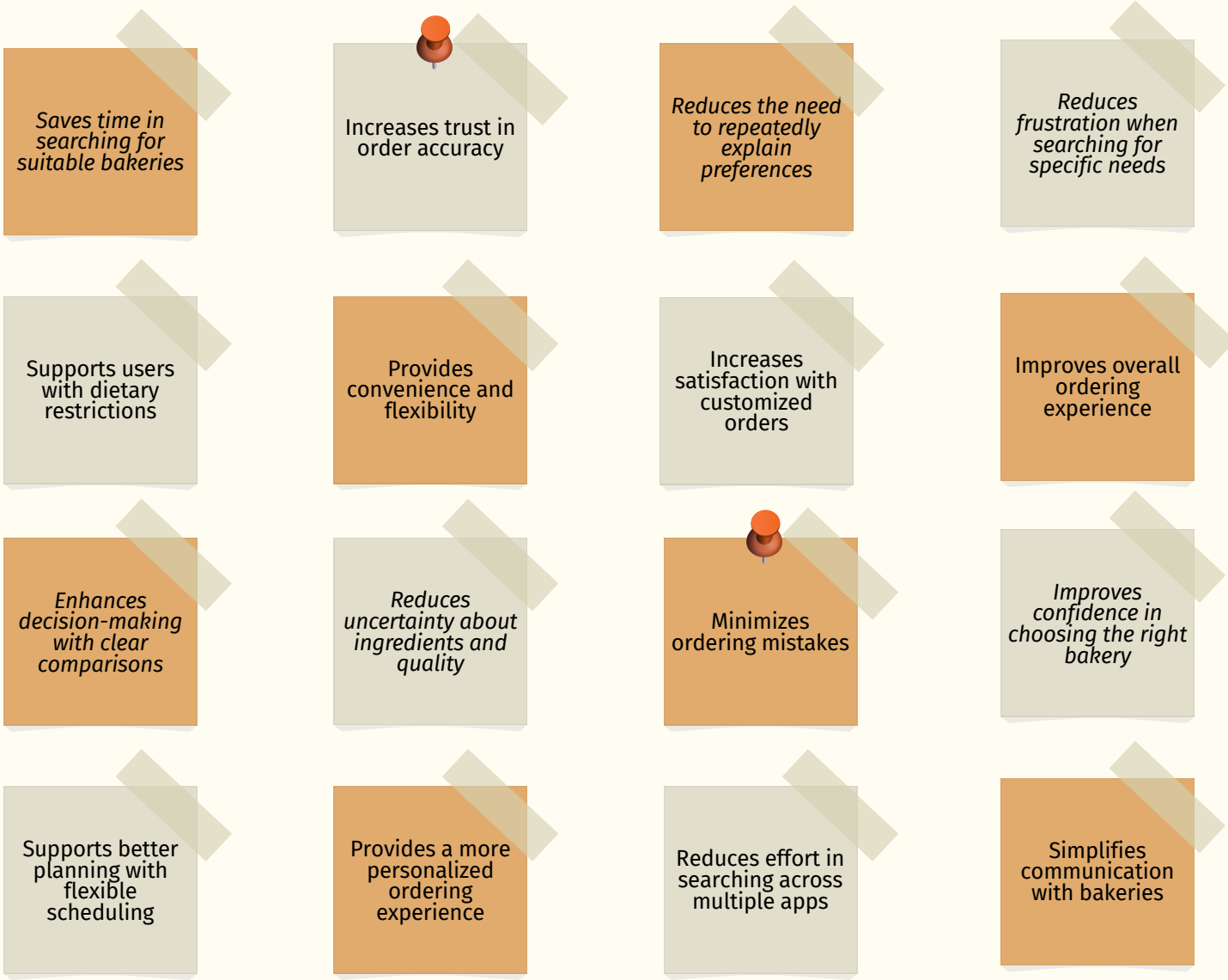
Multi-language support

Trusted bakery profiles

3.5 Value Proposition

3.5.1 Features and Benefits

- **Benefits**



3.5 Value proposition

3.5.2 Application Value

The application simplifies the process of finding and ordering customized baked goods by connecting users with bakeries that match their preferences. Instead of searching across multiple delivery apps to find suitable bakeries, the platform brings together a wide variety of bakeries in one place, making it easier to explore available options. Users can quickly find products that match their needs, and if they cannot find exactly what they are looking for, they can customize their orders based on their preferences. This transforms a fragmented and time-consuming process into a centralized, efficient, and user-friendly experience.

3.5.3 Meeting Users' Needs

Mohammad Al-harbi

- Customizable ingredients *(e.g., sugar level, flour type, fillings)*
- Clear product information
- *Reduces uncertainty about ingredients and quality*
- *Improves confidence in choosing the right bakery*

Noor Ibrahim

- Allergy-based filtering *(e.g., dairy-free, gluten-free)*
- *Allergy profile saving*
- Supports users with dietary restrictions
- Minimizes ordering mistakes

Noura Ahmed

- *Easy bakery discovery*
- Reorder previous items
- *Saves time in searching for suitable bakeries*
- Improves overall ordering experience

Layla Mohammed

- Custom order notes
- *Personalized bakery recommendations*
- Increases satisfaction with customized orders
- Increases trust in order accuracy

3.5 Value proposition

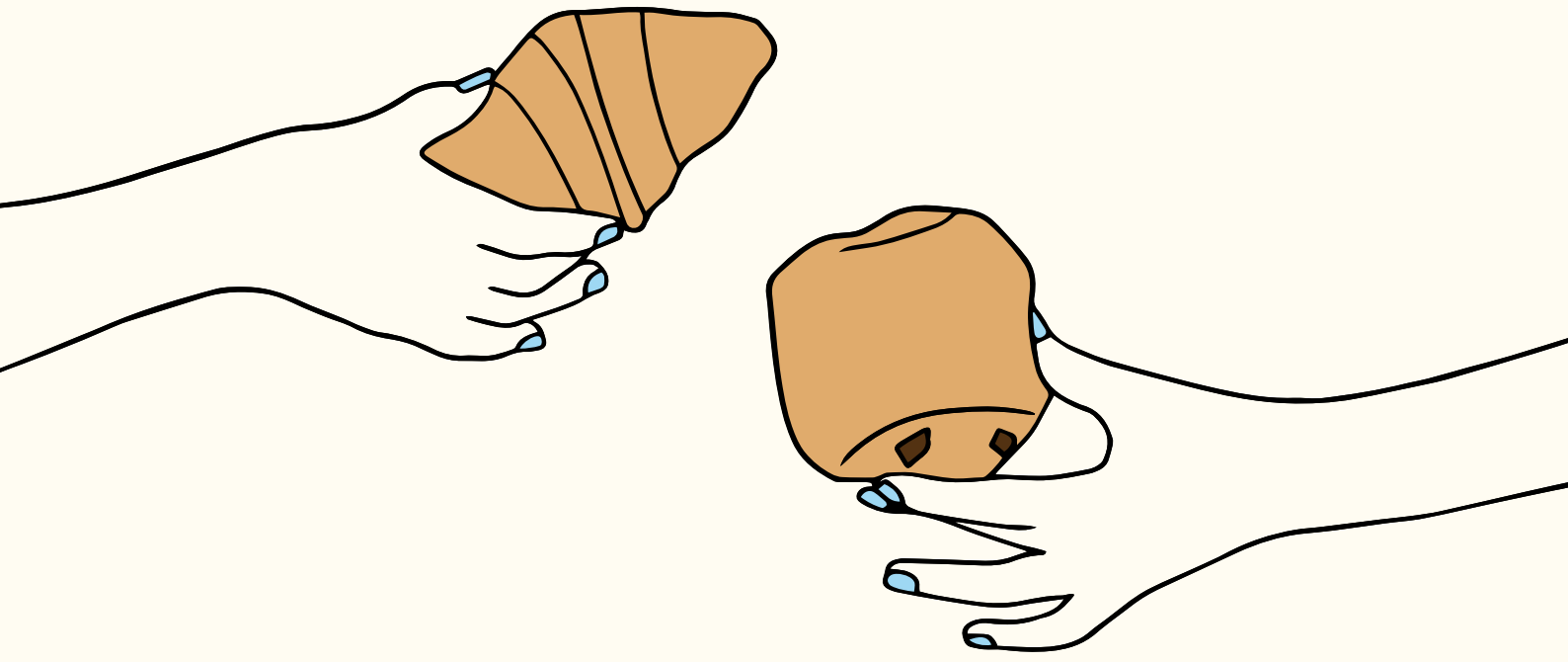
3.5.4 Official Value Proposition

- **Centralized choice:** Brings multiple bakeries and options into one place, reducing the need to search across different apps.
- **Personalized orders:** Allows users to customize ingredients, fillings, and dietary preferences easily.
- **Effortless experience:** Minimizes repeated explanations and simplifies the ordering process.
- **Trusted decisions:** Provides clear information, ratings, and transparency to increase confidence.
- **Flexible convenience:** Supports different user needs through features like subscriptions and reorder options

3.6 Problem statement to guide the development process

Users need a centralized and reliable platform that allows them to easily discover, compare, and customize baked goods based on their dietary preferences, time constraints, and specific requirements. Currently, users struggle with scattered bakery options across multiple platforms, limited customization features, lack of clear ingredient information, and time-consuming communication with bakeries. This results in frustration, uncertainty, and inefficiency in the ordering process. Therefore, there is a need for an integrated solution that simplifies discovery, ensures transparency, supports customization, and provides a seamless and time-efficient user experience.

Stage 3 : Ideate



4.1 Competitive Audits

4.1.1 Direct Competitors (Overview)

4.1.2 Indirect competitors (Overview)

Direct competitors	Indirect competitors
Mrsool is a delivery application that allows users to order from various stores and restaurants by directly communicating with delivery drivers. It offers flexible ordering but lacks a structured system for advanced product customization, especially for baked goods.	**Salla** is an e-commerce platform that enables businesses to create online stores and sell products, including baked goods. Although it supports online ordering, it does not focus on real-time customization or connecting users with multiple sellers in a single, unified platform
HungerStation is a popular food delivery platform that connects users with a wide range of restaurants and bakeries. While it provides a smooth ordering and delivery experience, it offers limited customization options for products compared to specialized solutions.	**Instagram (Home-based bakery accounts)** Instagram is widely used by home-based businesses and small bakeries to showcase and sell their products. Customers can request customized baked goods through direct messages, but the process is informal, unstructured, and lacks features like filtering, comparison, and order tracking.

Table 8 - Competitive audits

4.1.1 Direct Competitors (Aspects)

Aspect	مرسول MRSOOL	HUNGER STATION
First impression	Provides a sense of flexibility and freedom in ordering, but may feel slightly unclear for first-time users.	Clear, familiar, and easy to navigate from the first use.
Interaction	Relies heavily on chat-based communication with drivers, making it flexible but less structured.	Smooth and well-structured ordering process with clear steps.
Visual design	Simple and functional, but lacks a modern and visually appealing interface.	Modern, clean, and user-friendly interface.
Content	well-organized, but it depends on user requests rather than clear product listings.	Well-organized menus and categories, but limited in terms of deep customization

Table 9 - Direct Competitors (Aspects)

4.1.1 Direct Competitors (Research)

4.1.2 Indirect competitors (Research)

Names	What works well	what can be improved	Audience needs
MRSOOL	Highly flexible ordering system that allows users to request almost anything. Direct communication with drivers increases convenience and personalization.	Lack of structure makes the process confusing. No clear product listings or customization interface. Ordering depends heavily on manual input.	Users need a more organized system with clear options, pricing, and easier customization without relying entirely on chat.
HUNGER STATION	Smooth and reliable ordering experience with clear categories, menus, and fast delivery. Easy navigation for users.	Limited customization options for products, especially for baked goods. Less focus on personalized orders.	Users want more control over product customization (e.g., ingredients, design, special requests) while keeping the same smooth experience.
	Highly visual platform that helps users discover unique and customized baked products. Direct interaction with sellers.	Unstructured ordering process. No built-in system for filtering, comparing, or tracking orders. Communication can be slow or unclear.	Users need a more organized platform where they can browse options, compare sellers, and place customized orders clearly.
سلة	Provides a structured and professional online store experience. Easy browsing, ordering, and payment process.	Limited support for real-time customization. Each store is separate, so users cannot easily compare multiple sellers in one place.	Users want a unified platform that combines multiple bakeries with flexible customization options and easy comparison.

Table 10 - Indirect Competitors (Research)

4.1.1 Indirect Competitors (Aspects)

Aspect		
First impression	Highly engaging and visually appealing due to its image-focused nature.	Professional and business-oriented, but varies depending on each store.
Interaction	Informal and based on direct messages, making it unsuitable for a structured ordering process.	Follows a standard e-commerce flow for browsing and purchasing products.
Visual design	Strong visual appeal with high-quality images and aesthetic layouts.	Varies by store; some are visually appealing while others are basic.
Content	Rich in visuals and details, but lacks organization for browsing or ordering.	Structured and clear, but not designed for real-time or advanced customization.

Table 11 - Indirect Competitors (Aspects)

4.1.3 Analyzing the findings

The competitive analysis reveals clear differences between direct and indirect competitors in terms of structure, customization, and user experience.

Direct competitors such as Mrsool and HungerStation provide functional delivery services with relatively smooth ordering processes. However, both platforms show limitations in supporting advanced customization. Mrsool offers flexibility through direct communication, but this results in an unstructured and sometimes confusing experience. On the other hand, HungerStation provides a more organized and user-friendly interface, yet it lacks deep customization options, especially for baked goods.

Indirect competitors highlight additional gaps in the market. Instagram enables highly visual product discovery and direct interaction with sellers, but the ordering process is informal, inconsistent, and lacks essential features such as filtering, comparison, and tracking. Salla offers a structured e-commerce experience, but it does not support real-time customization or allow users to easily compare multiple sellers within one platform.

Overall, the analysis shows that current solutions either prioritize flexibility without structure or structure without customization. None of the competitors successfully combine a well-organized system with advanced, real-time customization features tailored for user-specific needs.

4.1.4 Summarize Findings

The findings indicate a significant gap in the market for a unified platform that combines structured browsing, clear product information, and flexible customization.

Users currently face challenges such as:

- Lack of clear and organized customization options
- Dependence on manual communication for special requests
- Difficulty comparing different bakeries in one place
- Limited visibility and tracking during the ordering process

These insights highlight the need for a solution that:

- Provides a structured and user-friendly interface
- Supports advanced and real-time customization
- Enables easy comparison between multiple bakeries
- Improves transparency through clear order details and tracking

This gap represents an opportunity to design a platform that delivers both convenience and personalization in a single, seamless experience.

4.2 Brainstorming

4.2.1 How Might We (HMW) (Persona 1)

- How might we show clear ingredients and nutrition in a simple, easy-to-read way?
- How might we let users customize their order quickly without writing messages?
- How might we help users find suitable healthy options faster?
- How might we filter out irrelevant options automatically based on diet?
- How might we reduce confusion by simplifying choices?
- How might we make communication with bakeries clear and direct?
- How might we build trust by showing reliable and transparent information?

Crazy 8 for persona 1

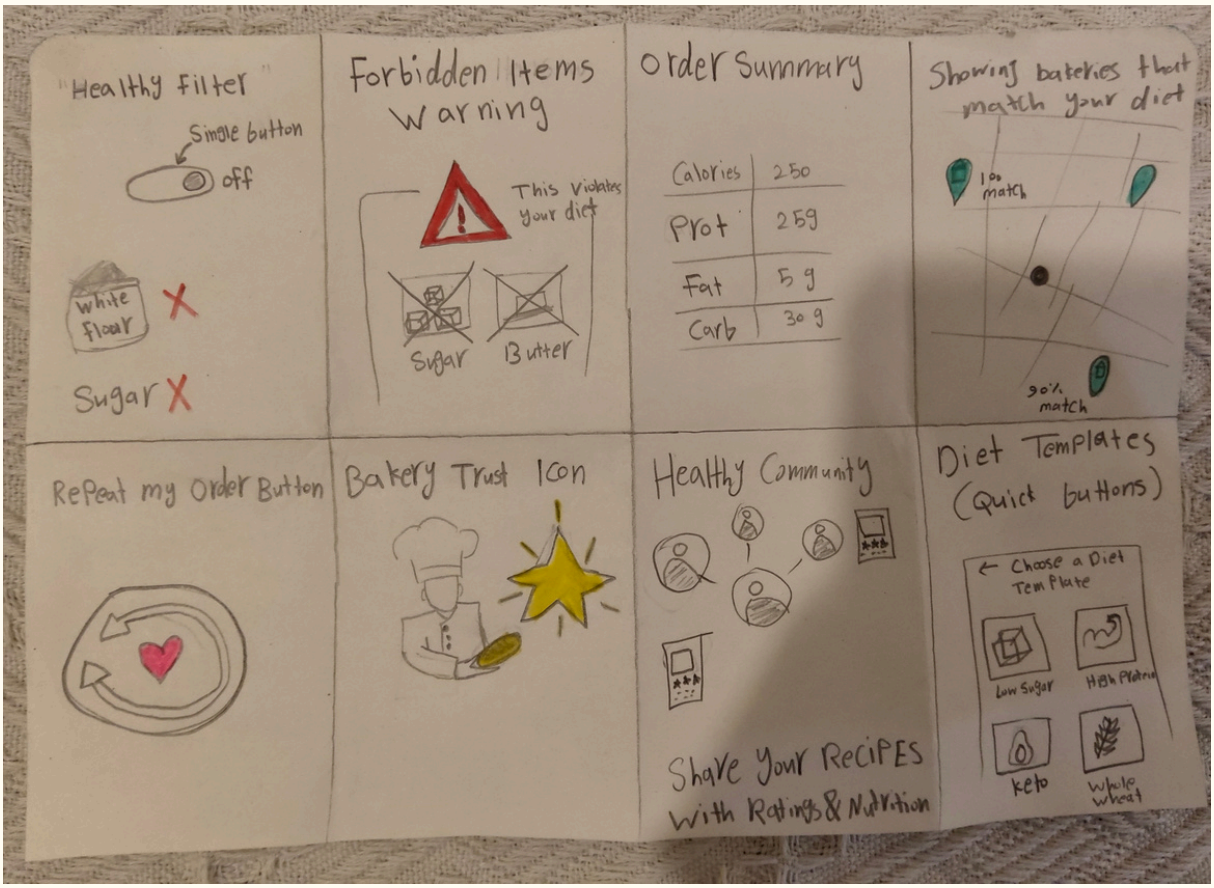


Figure 14 - Crazy 8- persona 1

4.2 Brainstorming

4.2.1 How Might We (HMW) (Persona 2)

- How might we make ingredient information clear for all bakery items without requiring the user to look for it?
- How might we help users locate bakeries that respond to their particular allergies easily?
- How might we allow users to quickly replace risky ingredients?
- How might we ensure allergic users feel secure and in control during the whole ordering process?
- How might we provide customers assurance customization request was recognized and suitably fulfilled?
- How might we minimize anxiety caused by trying a new bakery for the first time?
- How might we Make ordering from a bakery as easy for allergic customers as it is for everyone else?

Crazy 8 for persona 2

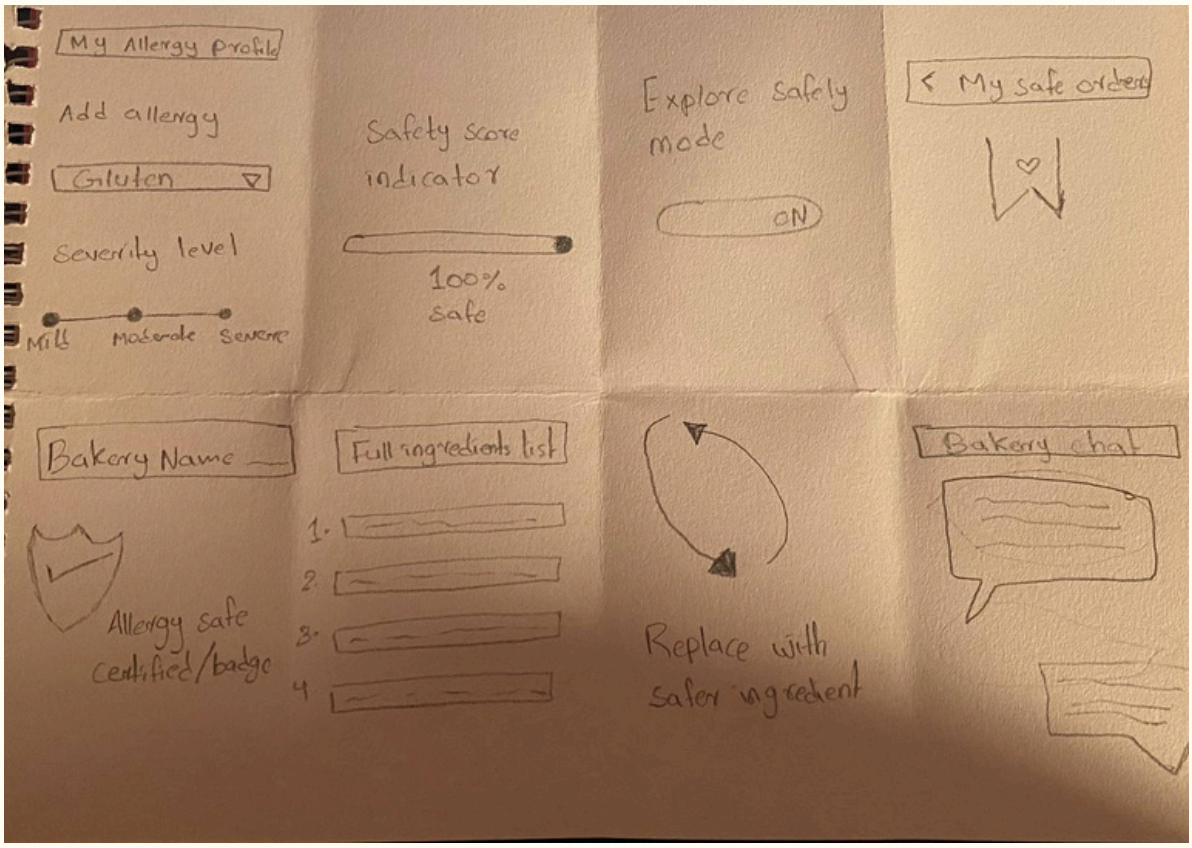


Figure 15 - Crazy 8- persona 2

4.2 Brainstorming

4.2.1 How Might We (HMW) (Persona 3)

- How might we help customers reorder their favorite baked goods more quickly?
- How might we reduce the time spent searching for suitable bakery items?
- How might we make customized bakery ordering easier during busy days?
- How might we simplify the checkout process for frequent users?
- How might we help Nora discover convenient bakery options faster?
- How might we personalize the app experience based on her family's preferences?
- How might we make bakery ordering possible with minimal interaction?
- How might we help busy parents save time when ordering baked goods online?

Crazy 8 for persona 3

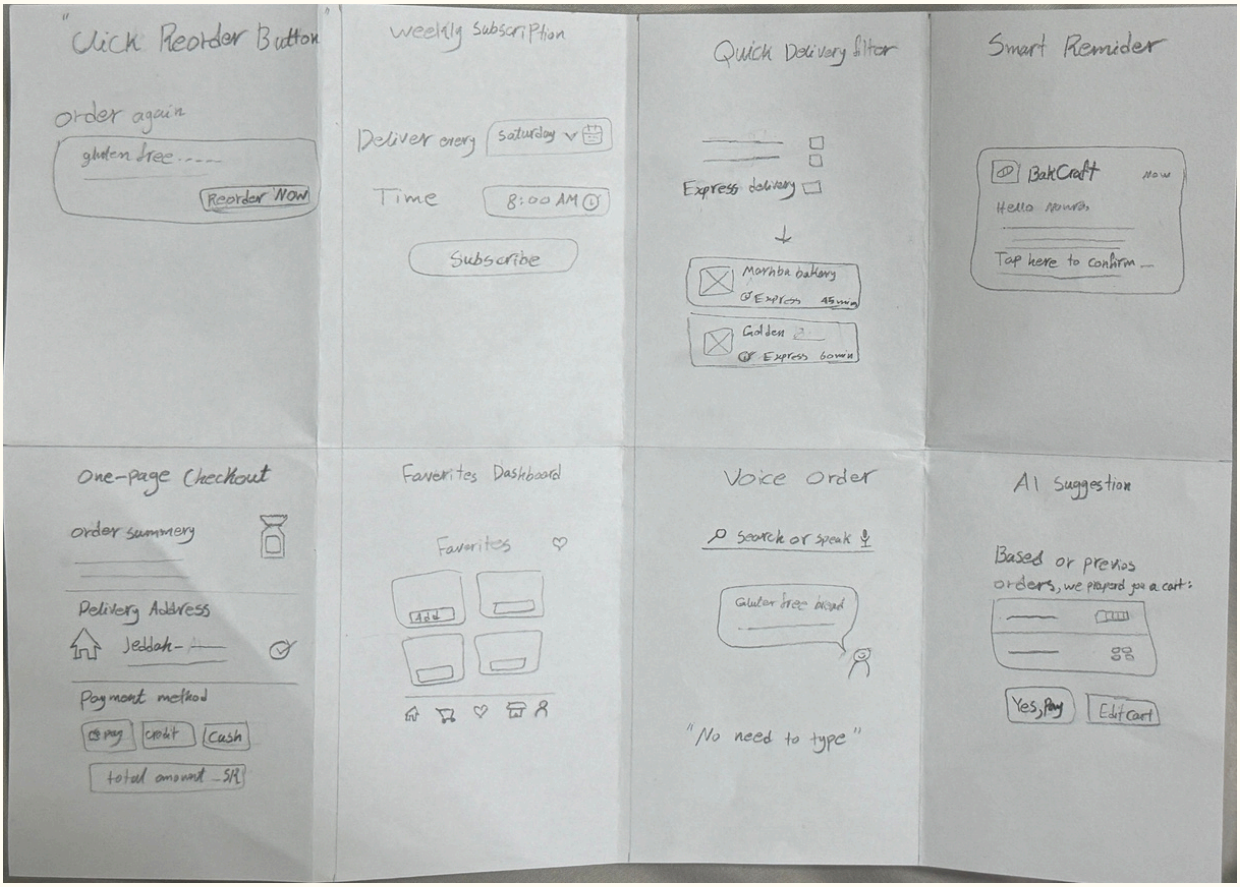


Figure 16 - Crazy 8- persona 3

4.2 Brainstorming

4.2.1 How Might We (HMW) (Persona 4)

- How might we help users easily compare multiple bakeries to reduce decision time?
- How might we simplify the customization process by automatically applying user preferences?
- How might we enable users to quickly reorder their previous customized orders?
- How might we inform users when a bakery has limited customization options?
- How might we recommend suitable bakeries or products based on user needs such as occasion, number of people, and preferences?
- How might we increase user trust by showing real-life reviews and customized product results?
- How might we provide a clear and structured summary of customized orders before confirmation?
- How might we allow users to schedule orders to ensure timely delivery for events?

Crazy 8 for persona 4

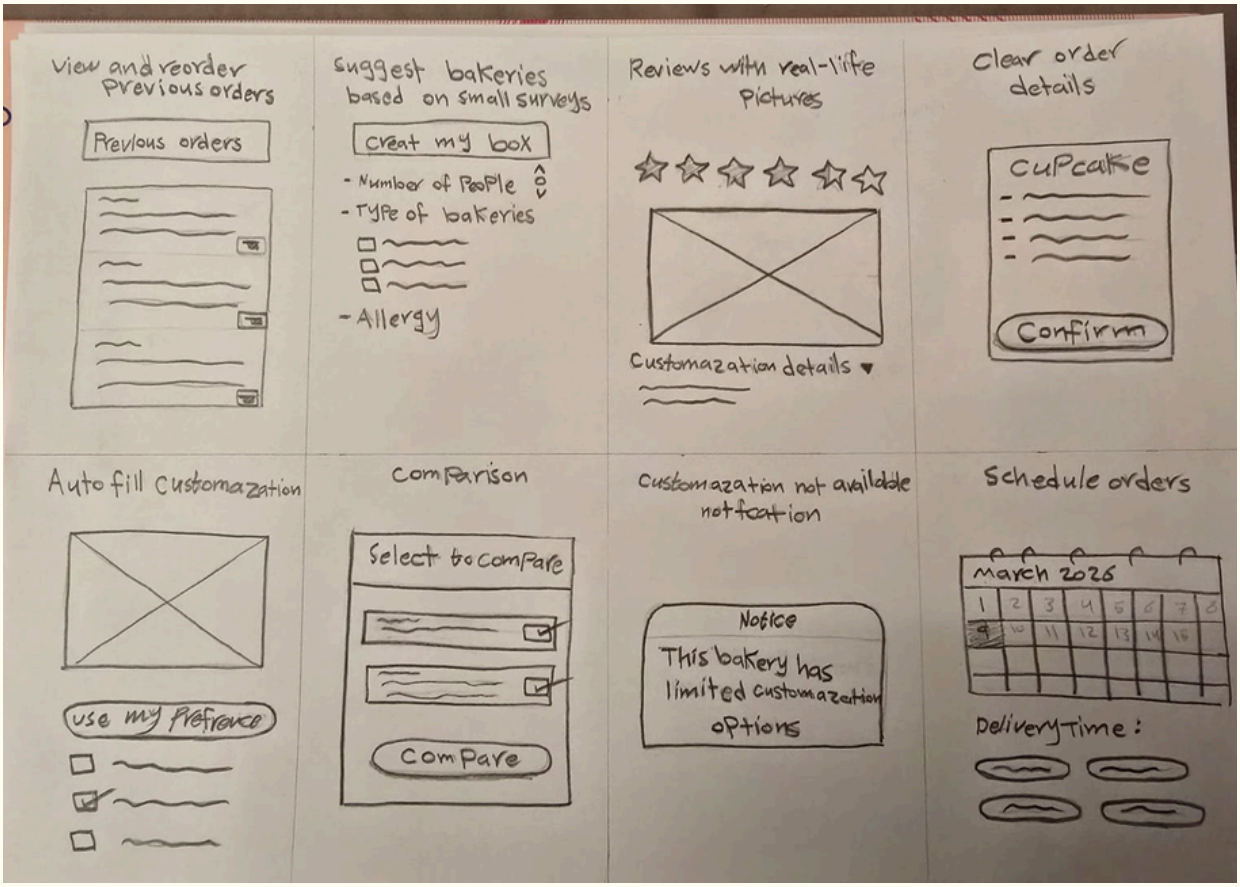


Figure 17 - Crazy 8- persona 4

4.2.2 Potential solutions that align with the defined problem statement

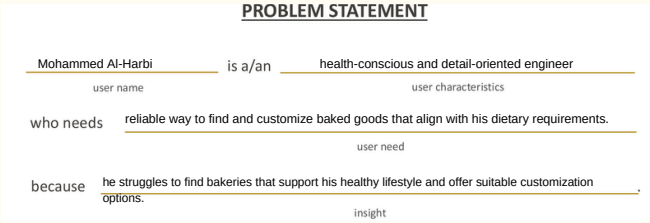
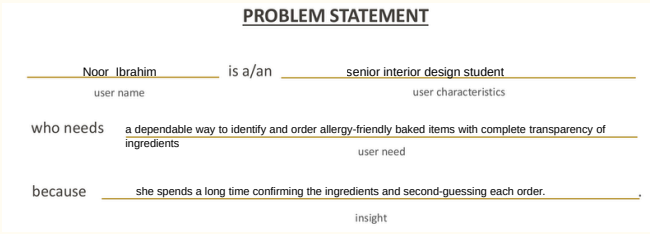
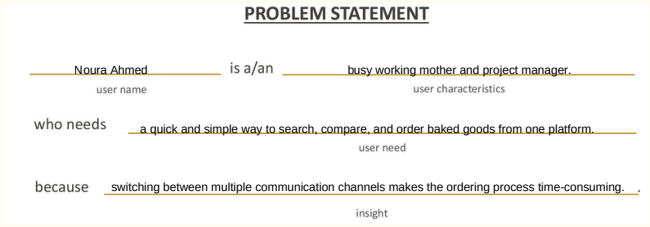
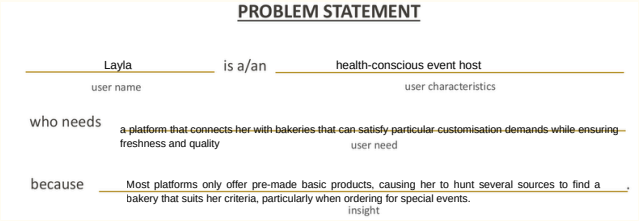
problem statement	Potential solutions
 persona 1	• Filter products based on dietary preferences (e.g., low sugar, gluten-free, keto) • Display clear nutritional information and ingredient details • Allow users to customize ingredients before ordering • Highlight “healthy-certified” bakeries within the app • Provide personalized recommendations based on user preferences
 persona 2	• Show detailed ingredient lists with allergen warnings • Allow users to set allergy preferences in their profile • Automatically filter and label safe products (“safe for you”) • Include verified bakery badges to build trust • Add reviews focused on ingredient accuracy
 persona 3	• Integrate multiple bakeries into one platform • Enable quick comparison (price, delivery time, ratings) • Provide one-tap reorder for previous purchases • Offer fast checkout with saved preferences • Suggest ready-made bundles or quick order options
 persona 4	• Enable bulk and event-based ordering • Provide advanced customization (quantity, flavors, dietary needs, packaging) • Display bakery portfolios (previous work) • Allow scheduling for precise delivery timing • Include direct communication or a structured order request form

Table 12 - Potential solutions for problem statements

4.2.3 Evaluating ideas

Persona 1 – Health-Conscious User

Selected Solution:
Clear nutritional information with dietary preference filters.
Feasibility: High - Easy to implement using product tags and filtering systems.
Desirability: High - Users strongly value clear health and dietary information.
Validity: High - Directly supports users who want healthier bakery choices.

Persona 2 – Allergy-Conscious User

Selected Solution:
Allergy preference profiles with automatic safe-product filtering.
Feasibility: Medium - Requires ingredient categorization and filtering features.
Desirability: High - Reduces anxiety and improves user confidence.
Validity: High - Directly addresses allergy-related ordering concerns.

Persona 3 – Convenience Seeker User

Selected Solution:
One-tap reorder with fast checkout.
Feasibility: High - Can be implemented using saved orders and payment information.
Desirability: High - Saves time and minimizes user effort.
Validity: High - Matches the persona’s need for quick and convenient ordering.

Persona 4 – Event & Customization User

Selected Solution:
Advanced customization with scheduled delivery options.
Feasibility: Medium - Requires customization management and scheduling systems.
Desirability: High - Users planning events need flexibility and personalization.
Validity: High - Directly supports large and customized bakery orders for occasions.

4.2.4 Goal statement

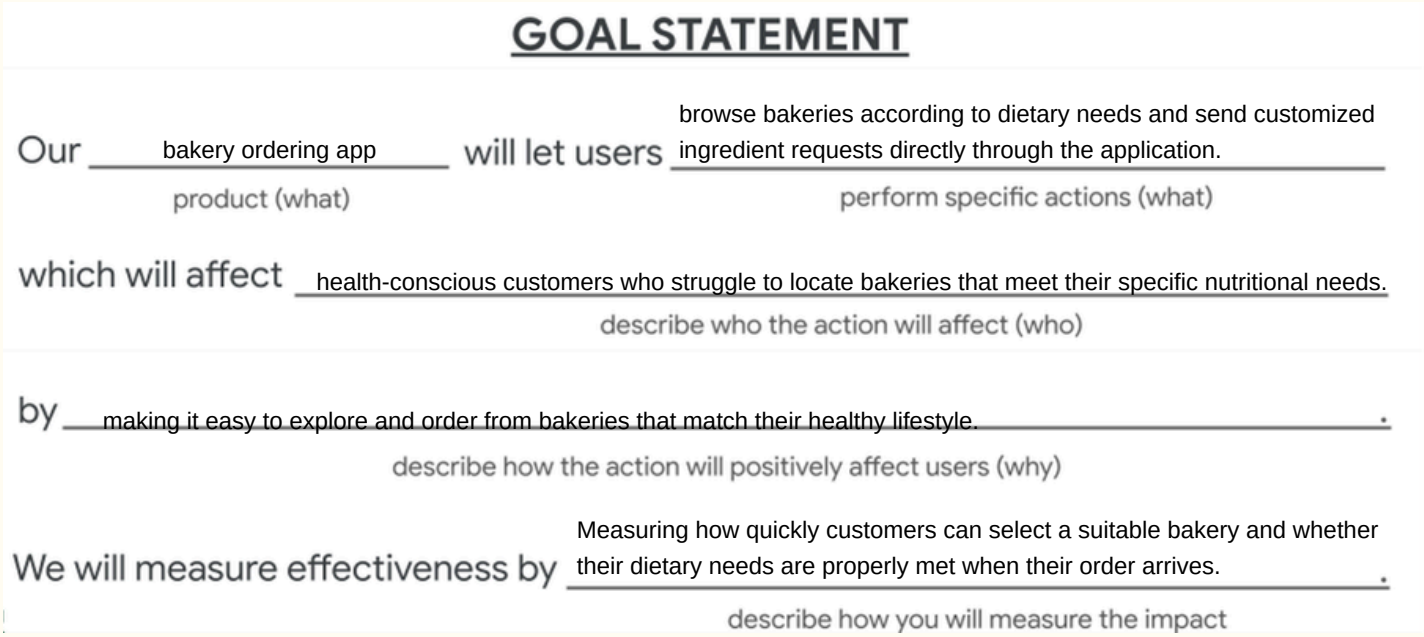


Figure 18 - Goal statement- persona 1

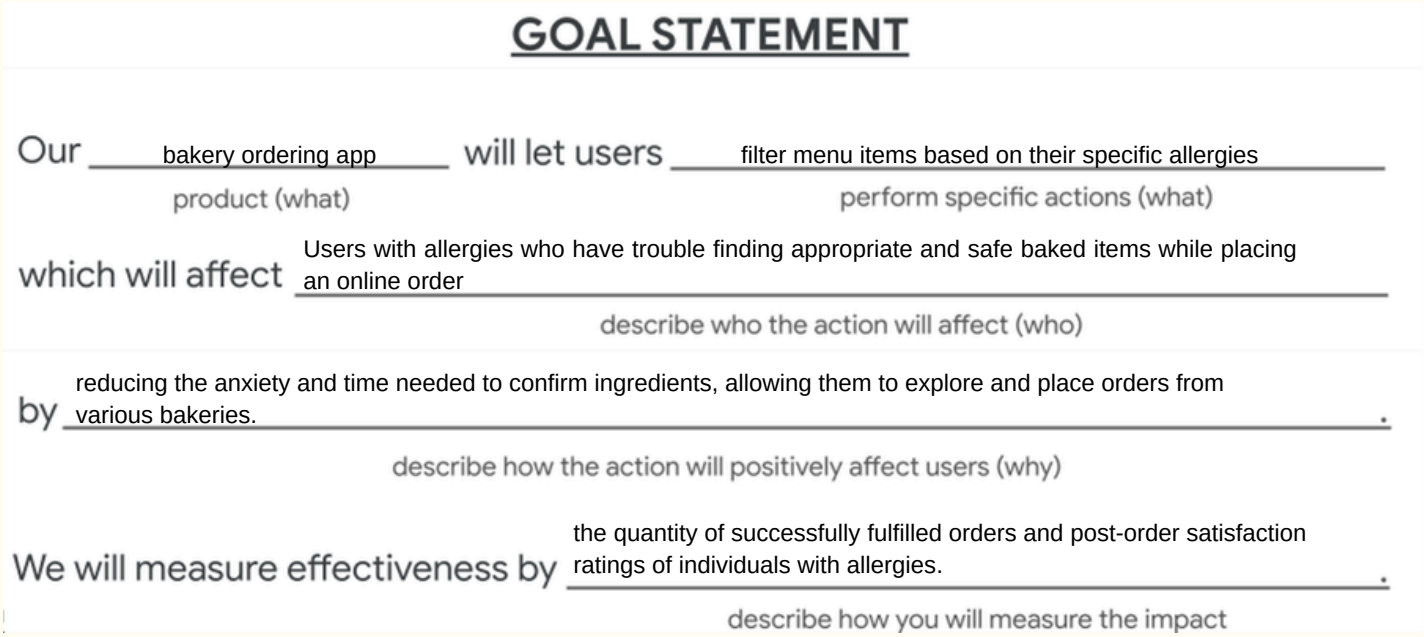


Figure 19 - Goal statement- persona 2

4.2.4 Goal statement

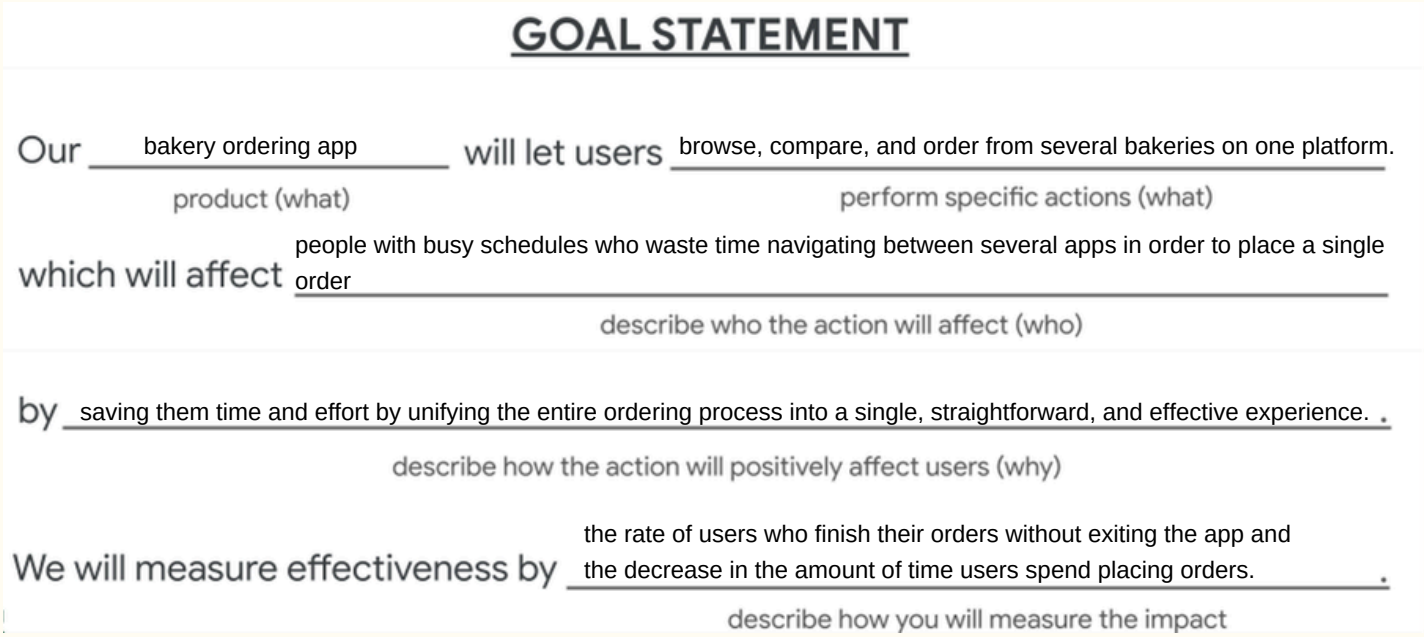


Figure 20 - Goal statement- persona 3

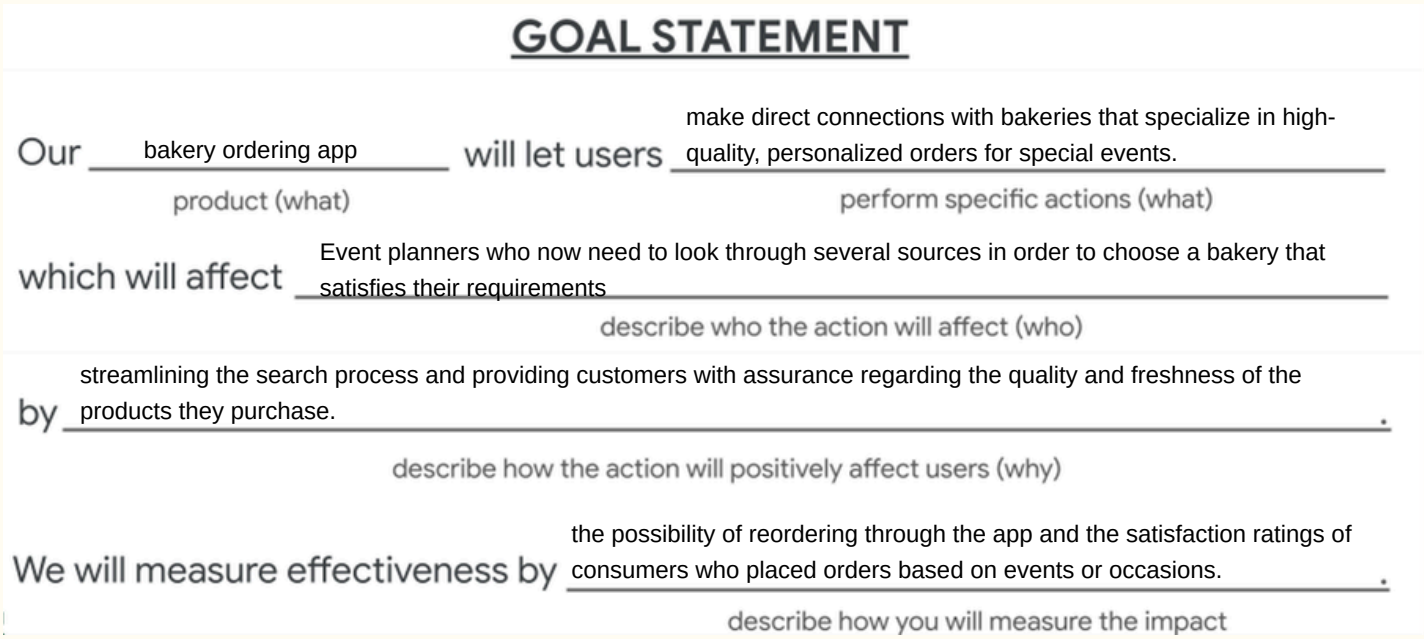


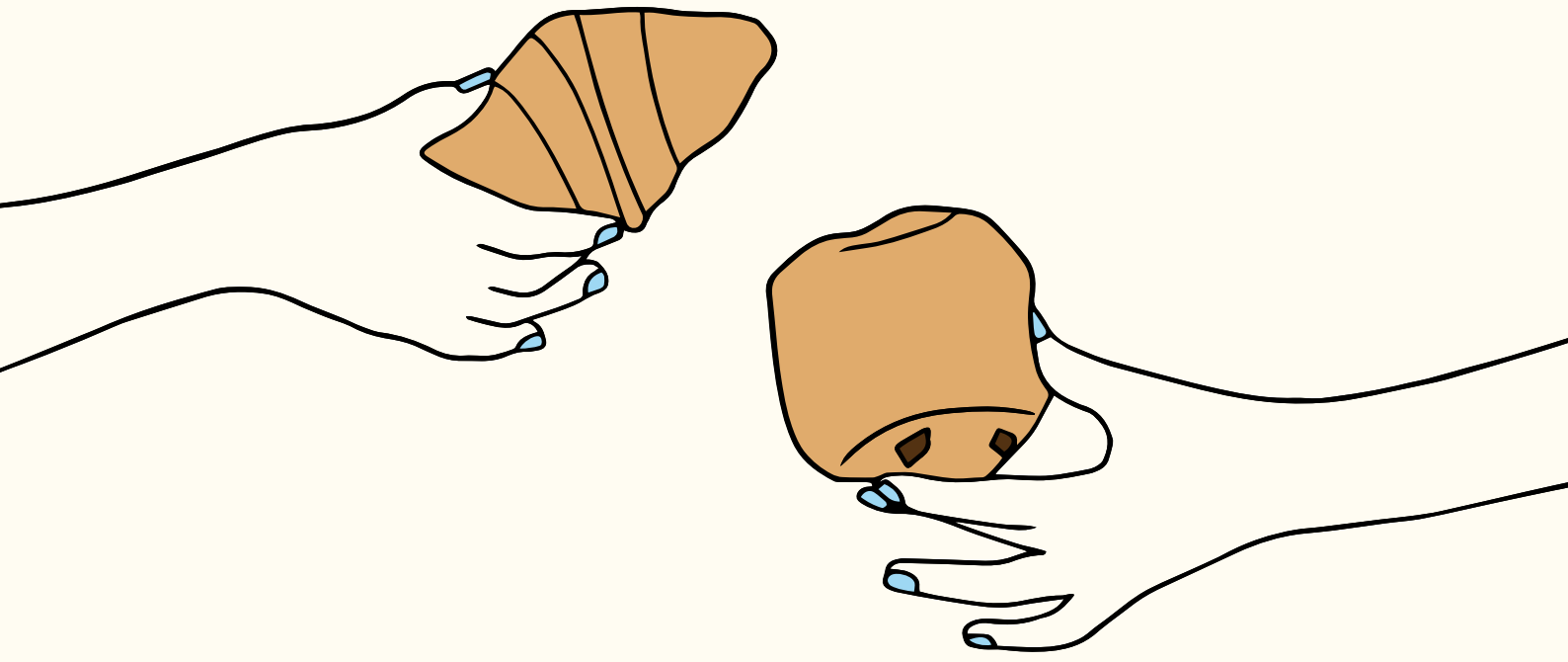
Figure 21 - Goal statement- persona 4

4.3 Mapping design features to each persona in a table to fulfill their needs, wants, and desires

Persona	Needs / Wants / Desires	Design Features
Persona 1 – Health-Conscious User	Wants healthier bakery options and clear nutritional information	Dietary filters, nutritional labels, ingredient details, healthy recommendations
Persona 2 – Allergy-Conscious User	Needs safe bakery products and reassurance about allergens	Allergy filters, allergen warnings, safe-product labels, verified bakery badges
Persona 3 – Convenience Seeker user	Wants faster ordering with minimal effort	One-tap reorder, fast checkout, saved preferences, smart reminders
Persona 4 – Event & Customization User	Needs flexible customization and reliable event ordering	Advanced customization, scheduled delivery, bakery portfolios, bulk orderin

Table 13 - design features for each persona

5. Prototype



5.1 Storyboard Sketching and Wire-framing

5.1.1 Storyboard: Big-picture and close-up (Persona 1)

Scenario: A health-conscious user ordering customized bakery from trusted places

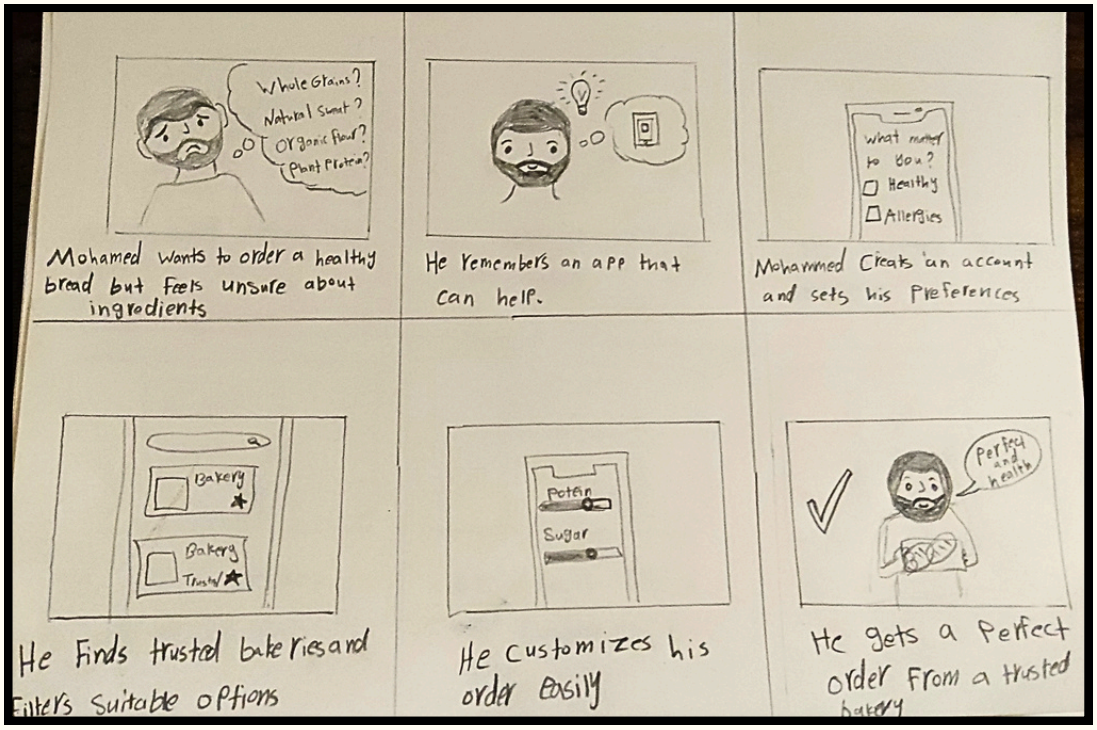


Figure 22 - Storyboard:Big-picture - persona 1

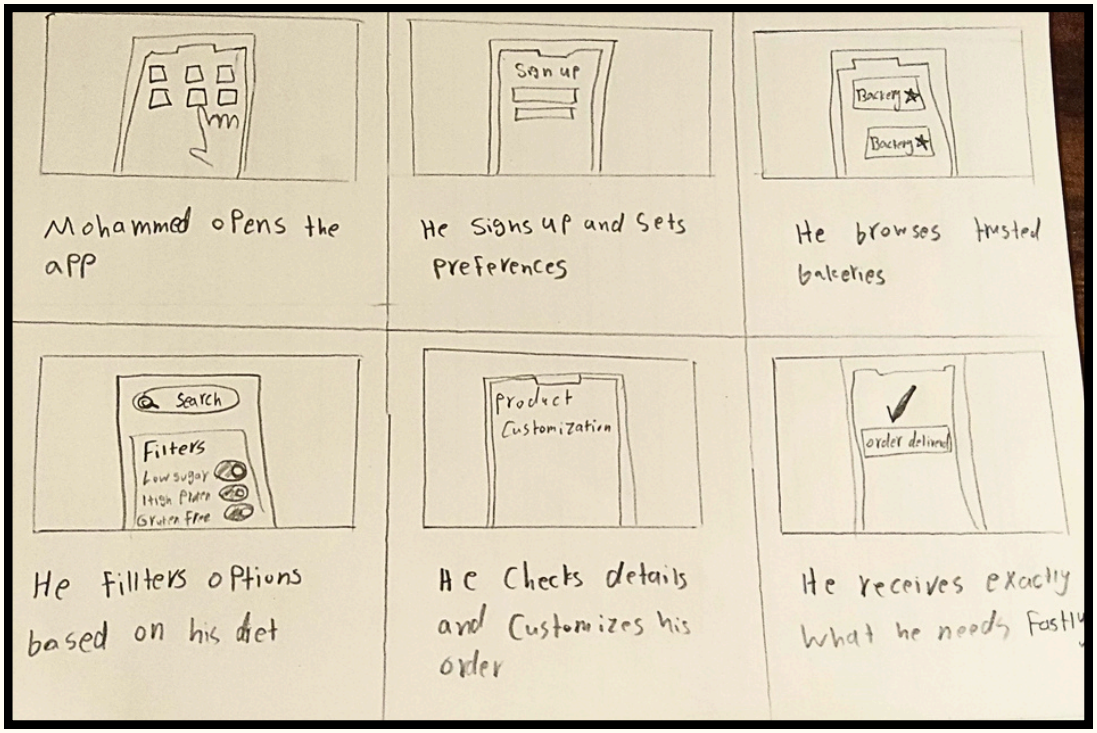


Figure 23 - Storyboard: close-up - persona 1

5.1.1 Storyboard: Big-picture and close-up (Persona 2)

Scenario: An allergic user ordering bakery with safe ingredients based on their preferences.

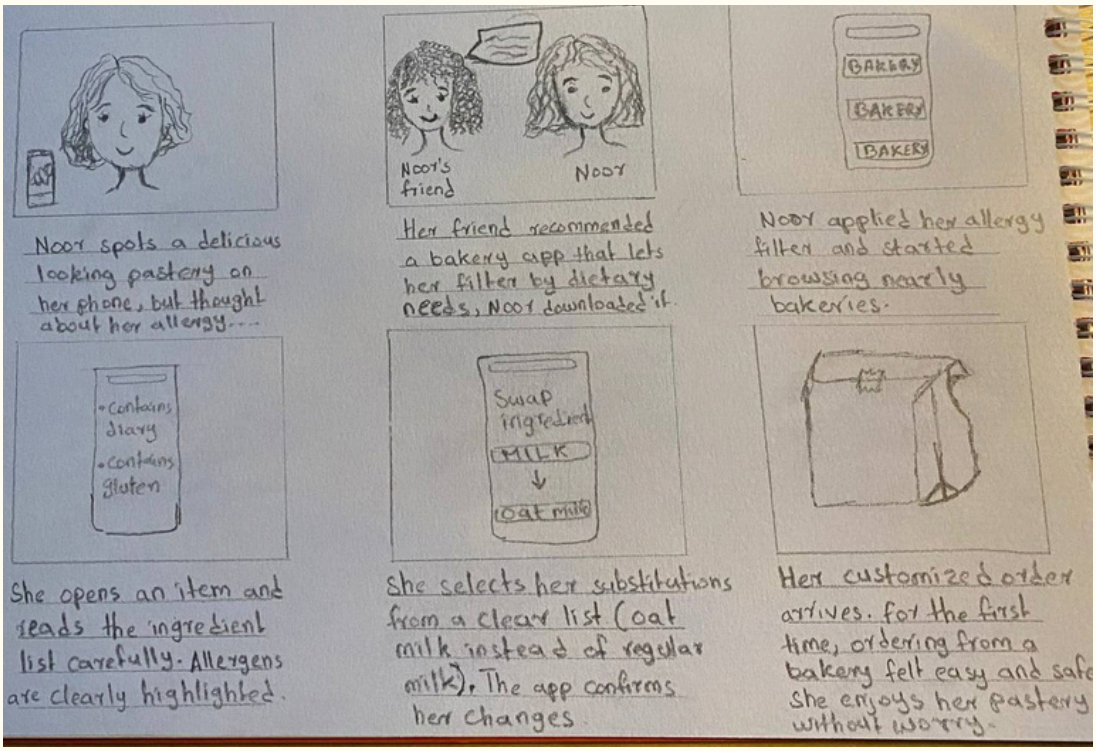


Figure 24 - Storyboard:Big-picture - persona 2

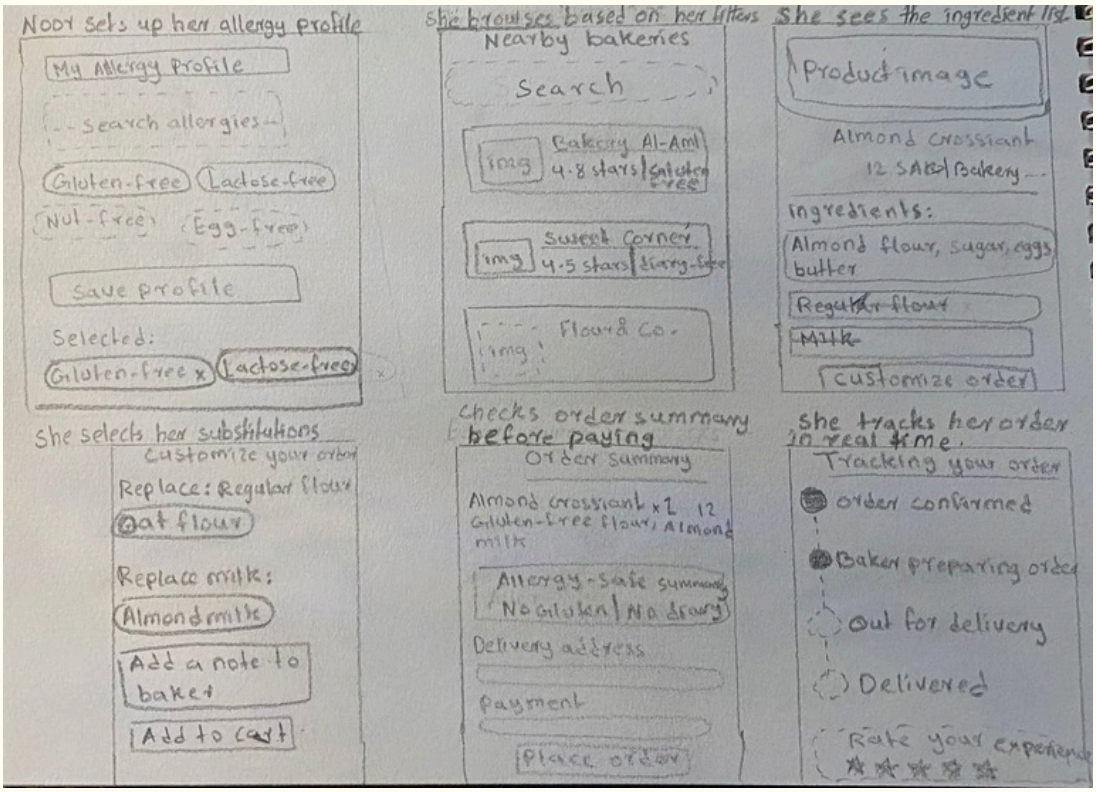


Figure 25 - Storyboard:Close up - persona 2

5.1.1 Storyboard: Big-picture and close-up (Persona 3)

Scenario: A busy mother ordering baked goods quickly and easily with minimal effort through a convenient and time-saving platform.

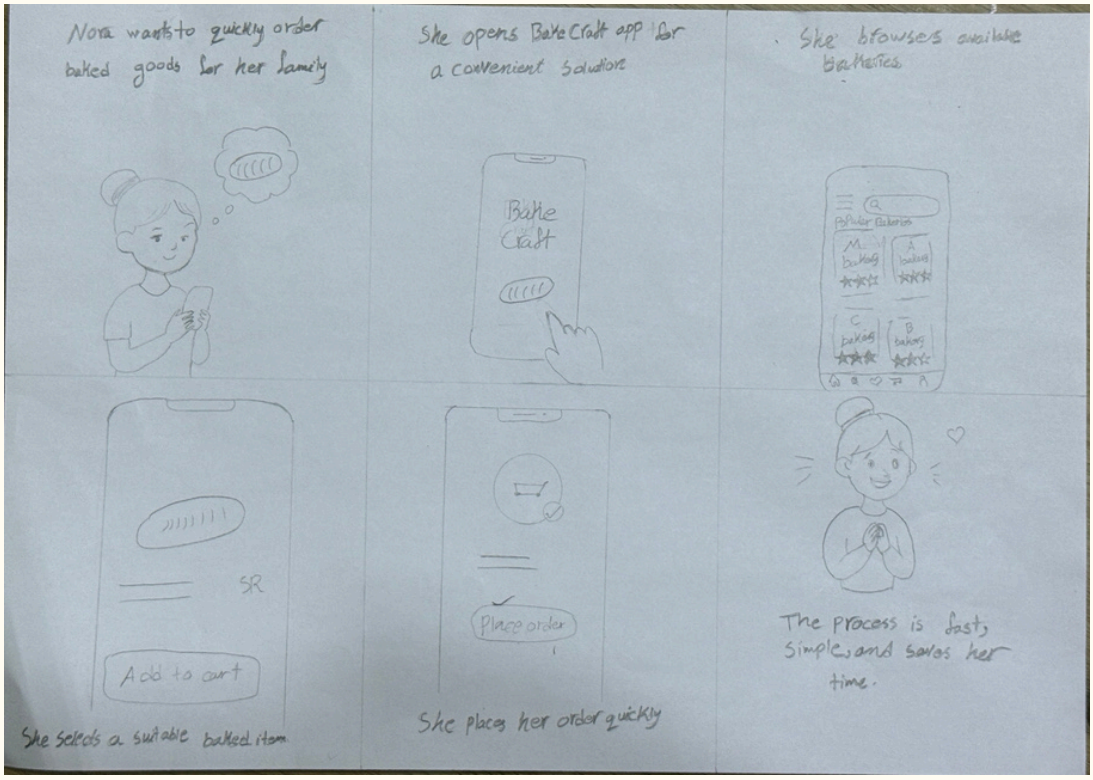


Figure 26 - Storyboard:Big-picture - persona 3

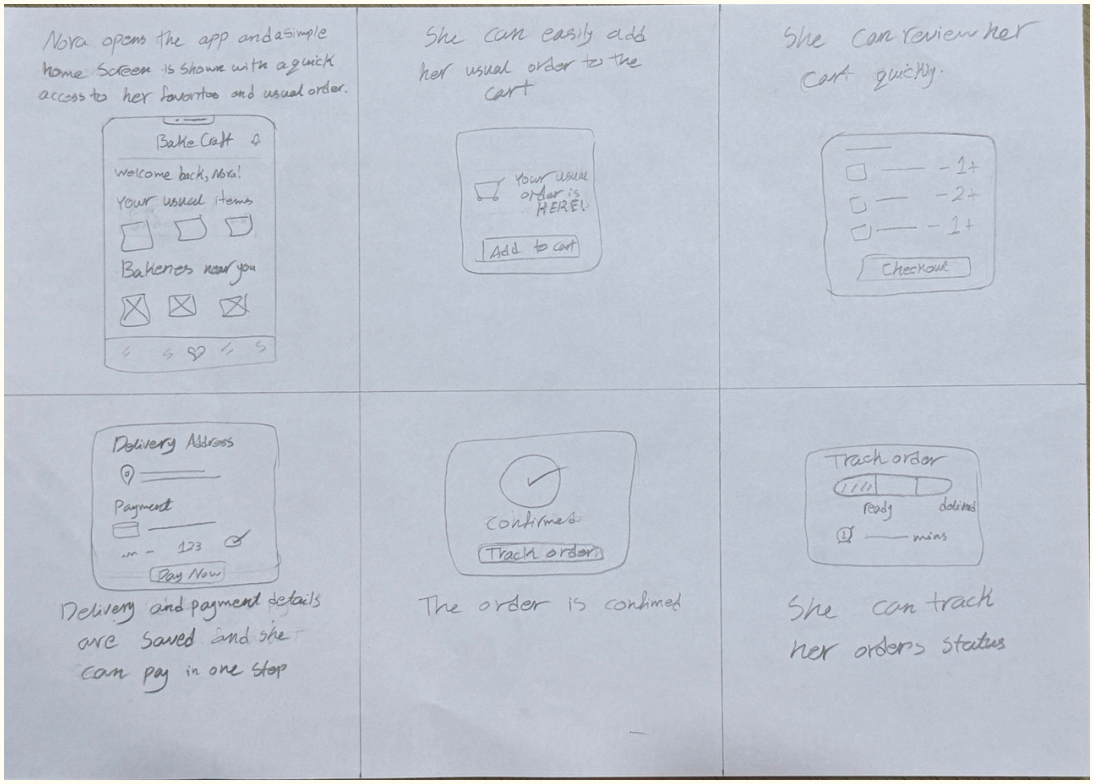


Figure 27 - Storyboard:Close up - persona 3

5.1.1 Storyboard: Big-picture and close-up (Persona 4)

Scenario: Layla uses the app to quickly find and customize baked goods that match her friends' dietary needs, making the ordering process simple and convenient for her gathering.

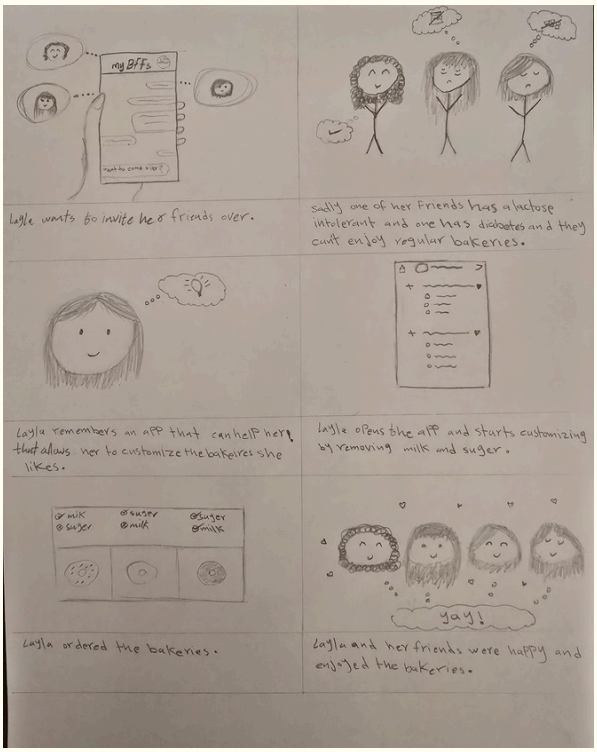


Figure 28 - Storyboard:Big-picture - persona 4

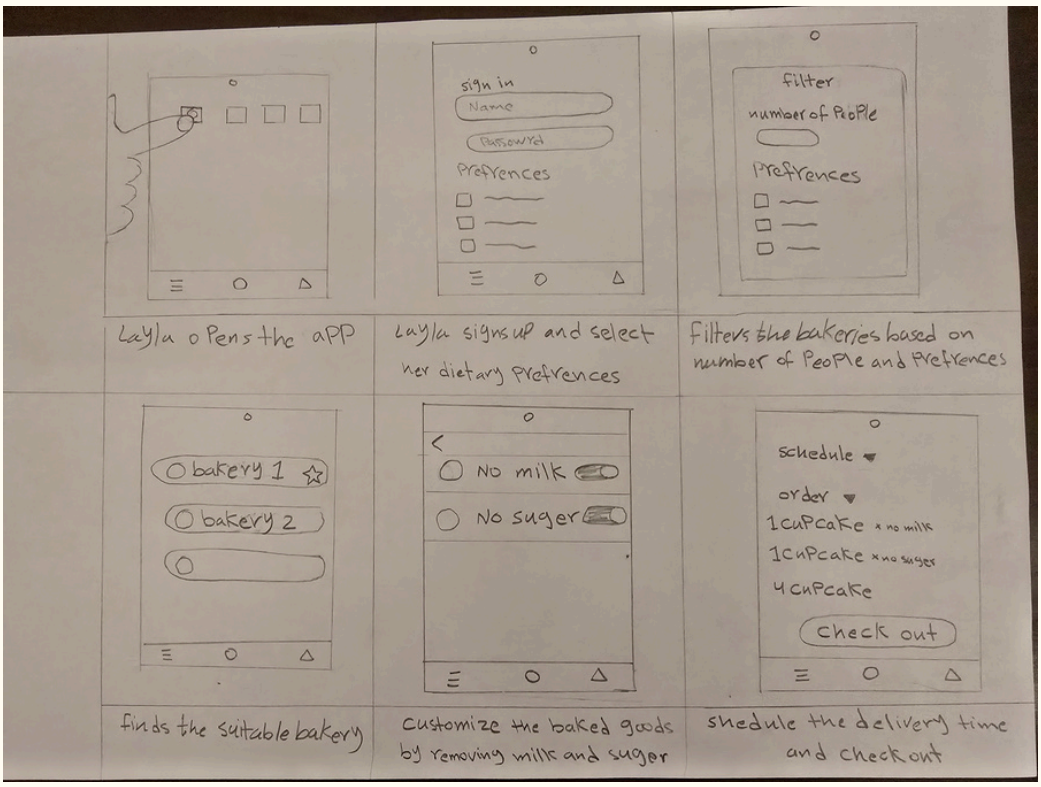


Figure 29 - Storyboard:Close up - persona 4

5.1.2 Lo - fi prototype
Wireframes

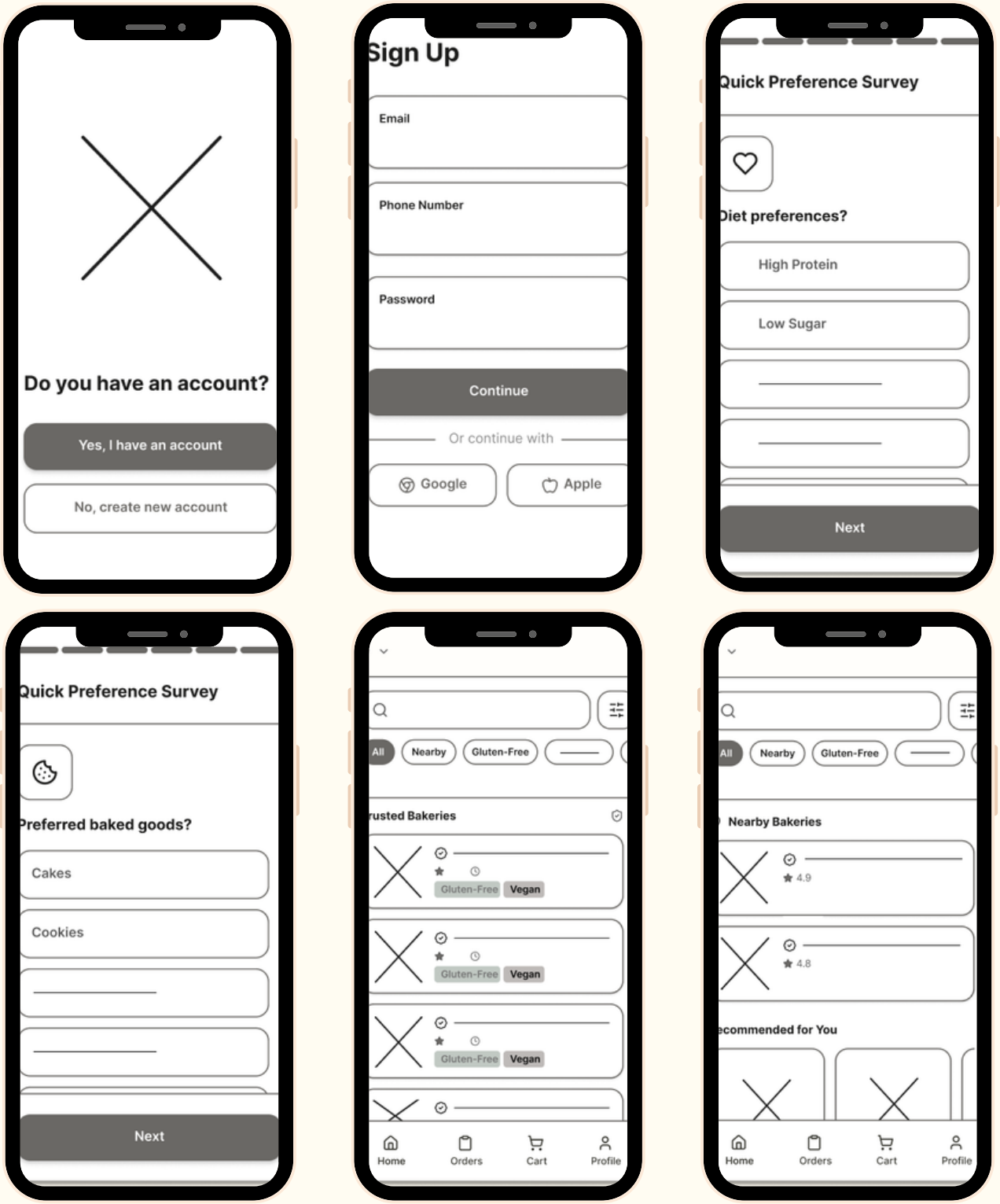


Figure 30 - Wirframes

5.1.2 Lo - fi prototype
Wireframes

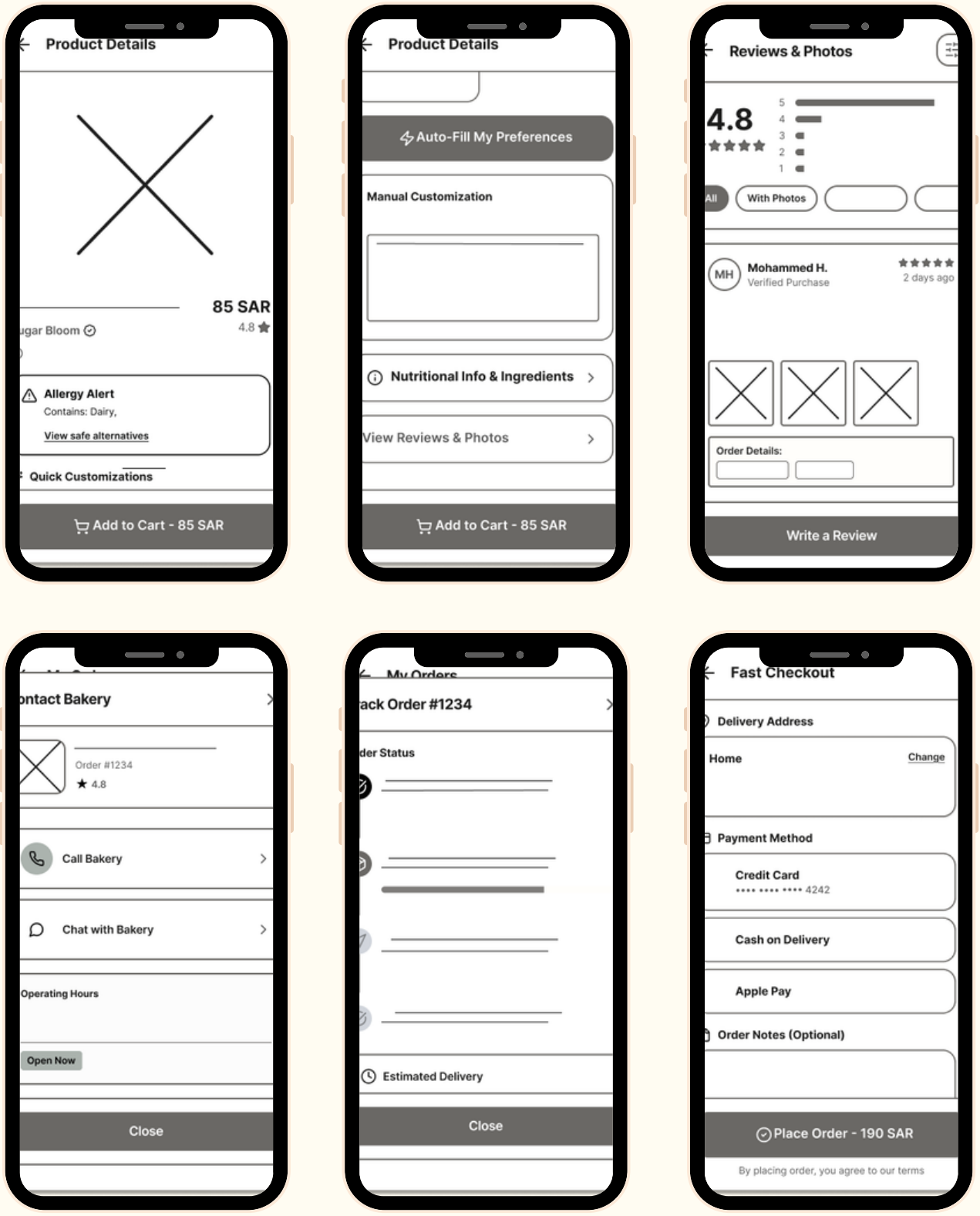


Figure 31 - Wirframes

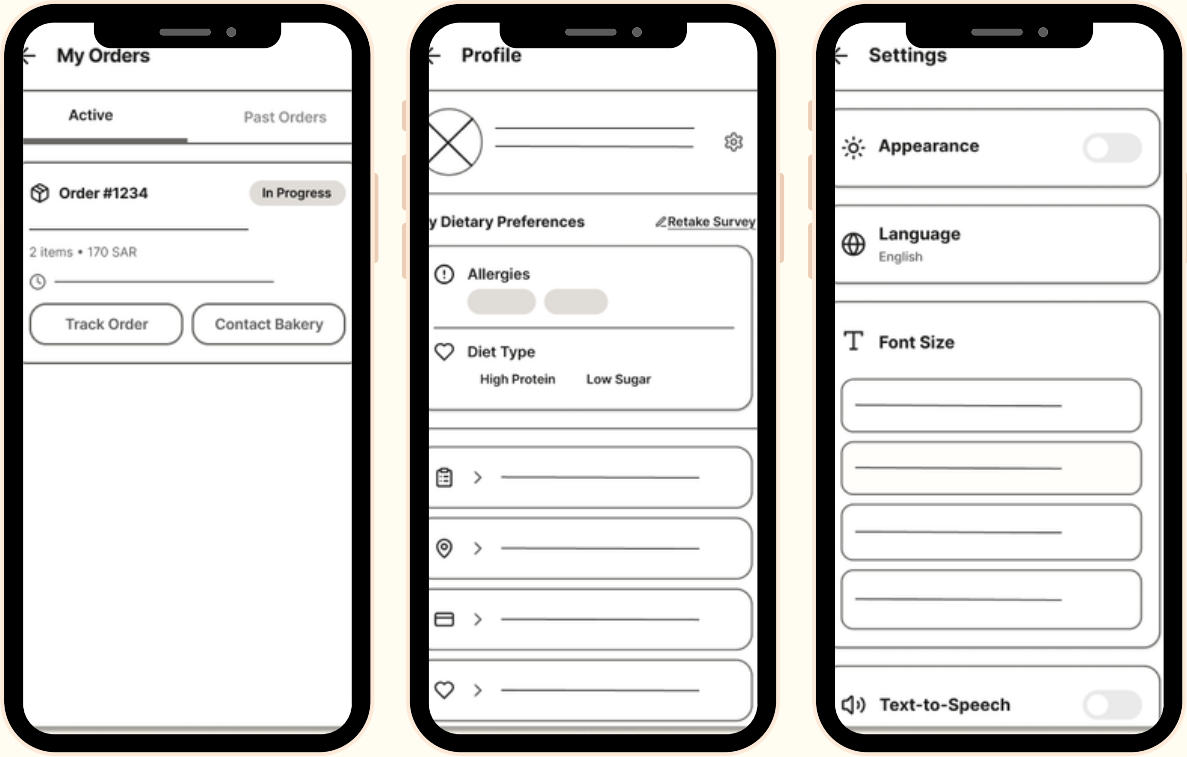


Figure 32 - Wirframes

5.2 Prototyping

5.2.1 Mock - up prototype

Sign in/ Sign up and the Survey

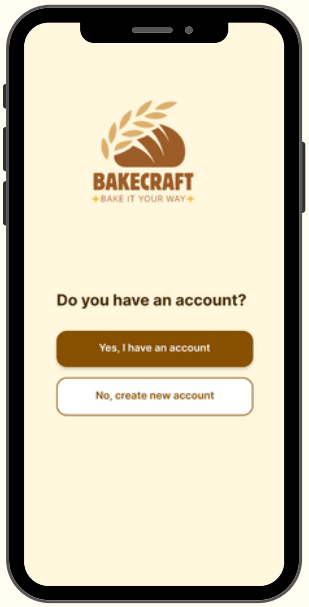


Figure 33 - Welcome / Entry Screen: The BakeCraft logo and asks the user whether they have an existing account or want to create a new one.

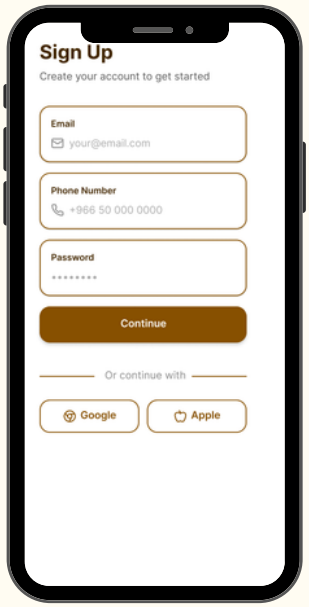


Figure 34 - Sign Up Screen: Allows new users to create an account by entering their Email, Phone Number, and Password, or via social logins (Google/Apple).

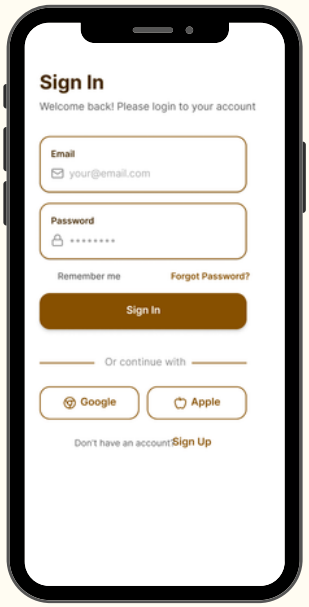


Figure 35 - Sign In Screen: A classic login screen for existing users with email, password, a "Remember Me" toggle, and social login alternatives.

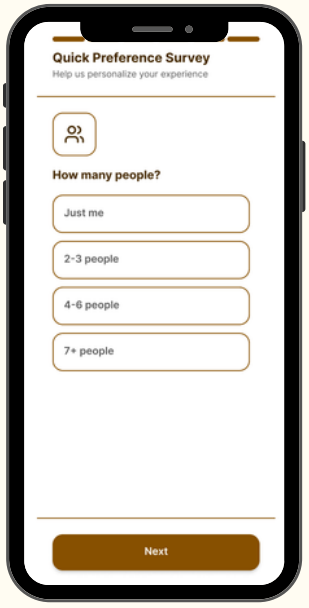


Figure 36 - Survey - Group Size: The first step of the preference survey, asking "How many people? In order to provide family offers

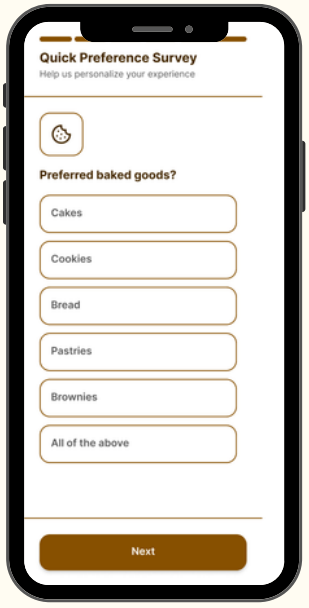


Figure 37 - Survey - Baked Goods:
to customise profile based on Thier preferences

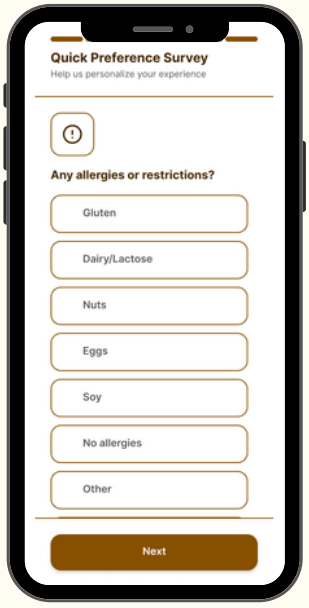


Figure 38 - Survey - Allergies: Asks for any specific allergies or In order for the application to display a warning when the product contains an allergen

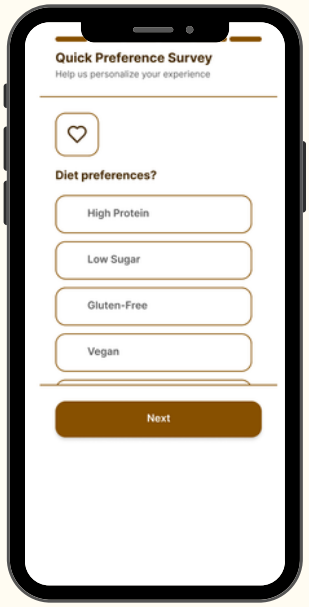


Figure 39- Survey - Diet preferences:
Users select thier dietary preferences to personalize recommendations

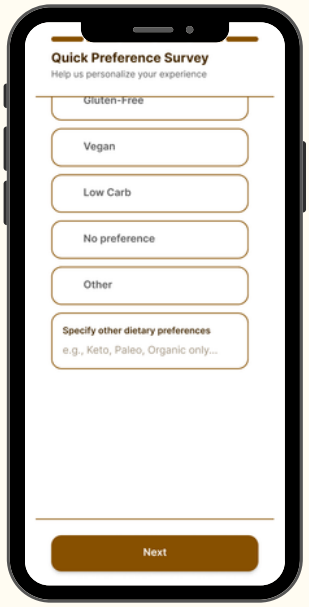


Figure 40- Survey - Diet preferences:
Allows users to choose dietary preferences

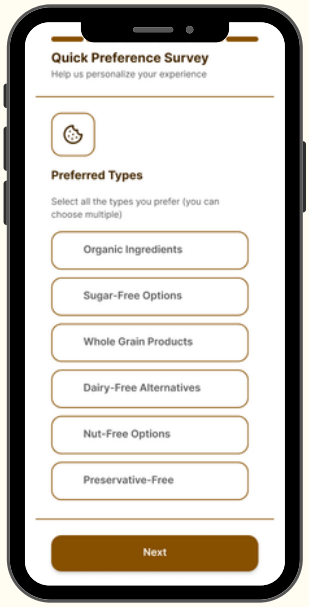


Figure 41- Survey - Preferred Types:
Users can specify additional dietary needs, providing more flexible experience

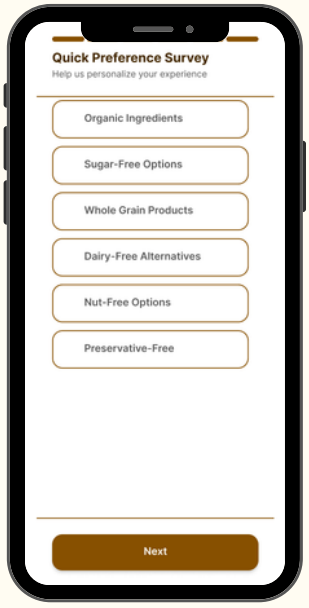


Figure 42- Survey - Preferred Types:
Identifies preferred product

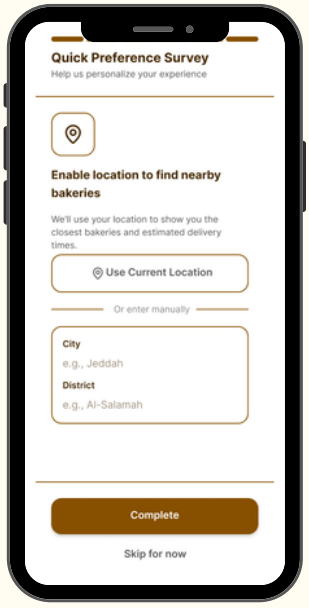


Figure 43- Survey - Enable location to find nearby bakeries:
Allows users to enable location access or enter their location manually to find nearby bakeries

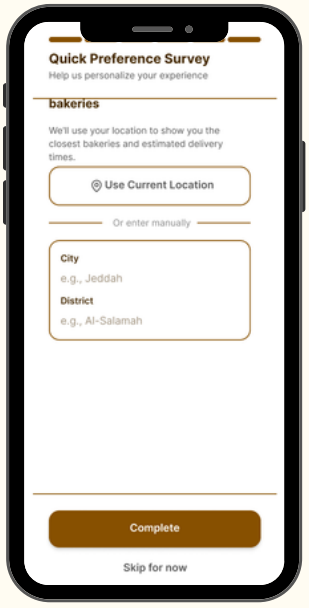


Figure 44- Survey - Enable location to find nearby bakeries:
Enable manually enter their city and district

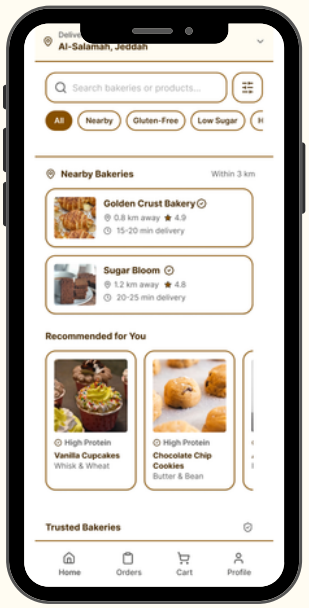


Figure 45- Home Screen: Displays a search bar, dietary filter tags (Low Sugar, Gluten-Free), nearby bakeries, and personalized product recommendations.

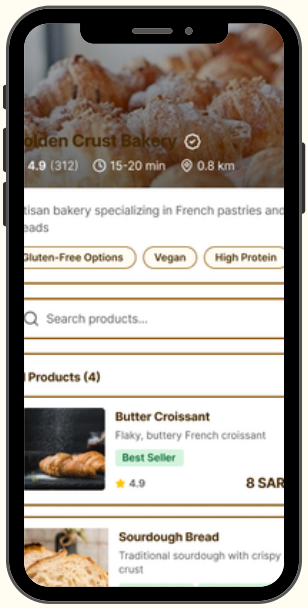


Figure 46- Bakery Profile: Displays a specific bakery's menu, ratings, delivery times, and product categories.

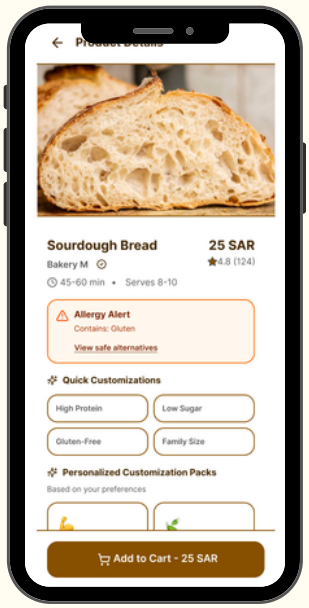


Figure 47- Product Details: Shows an item (e.g., Sourdough Bread) with price, description, quick customization tags, and critical allergy alerts.

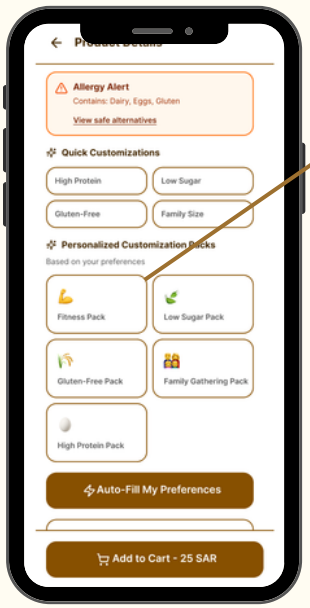


Figure 48- Personalized Packs: Offers customization packages based on user goals that was taken from the survey.

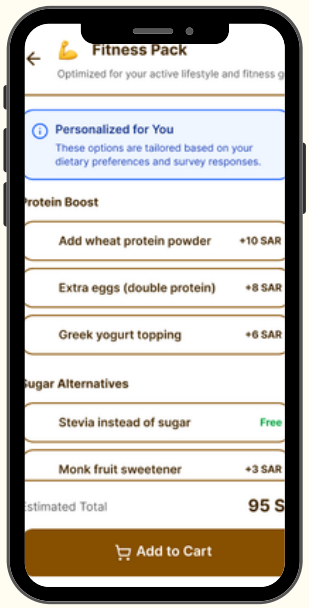


Figure 49- Fitness Pack Details: offer predefined customisation options

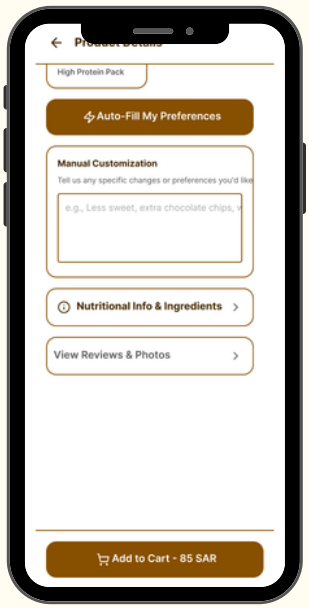


Figure 50- Manual Customization: Provides a manual text input box for users to type custom instructions, alongside an "Auto-Fill My Preferences" button based on the user past history preferences.

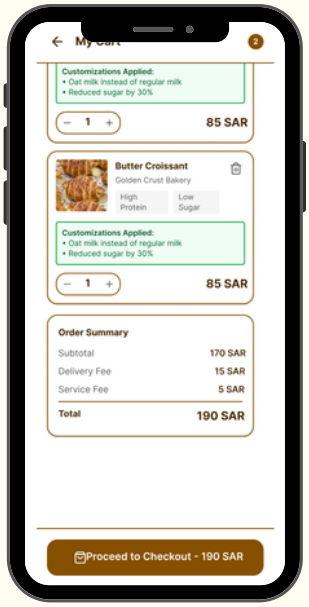


Figure 51 - My Cart: Lists the selected items, applied dietary customizations, quantities, and the order summary total.

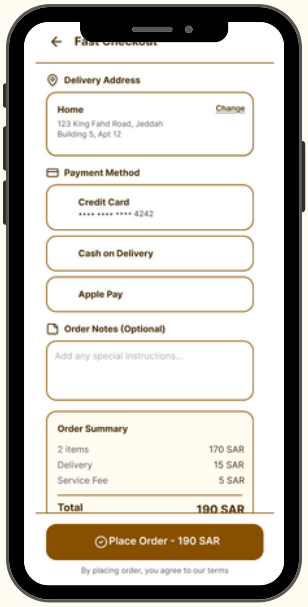


Figure 52 - Fast Checkout: A single-page checkout to confirm the delivery address, choose a payment method (Credit Card, Cash on Delivery, Apple Pay), and place the order.

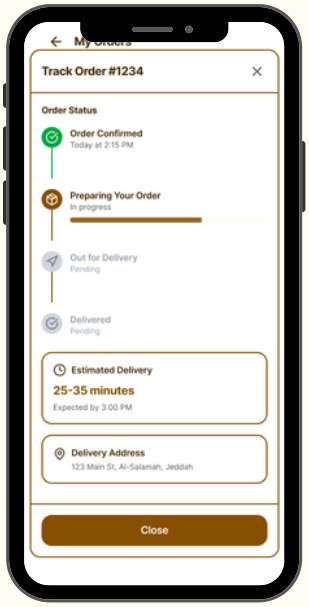


Figure 53 - Track Order: A live progress timeline showing real-time updates (Order Confirmed, Preparing, Out for Delivery, Delivered) and estimated delivery times.

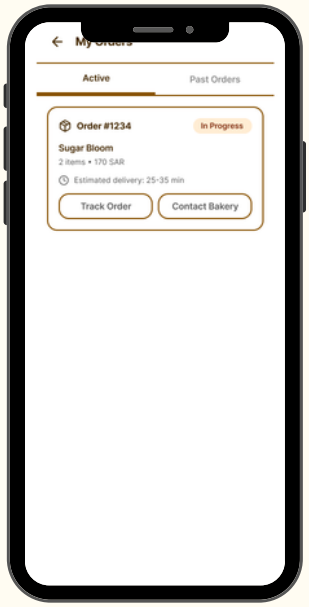


Figure 54 - Active Orders: Displays a list of ongoing orders with quick shortcuts to track the order or contact the bakery.

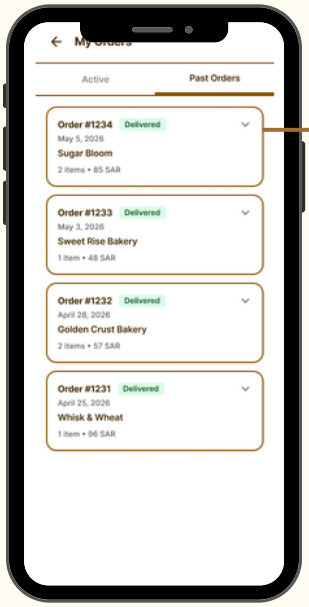


Figure 55 - Past Orders: A history tab showing completed orders with an option to view itemized receipts and a "Repeat Order" button without the need to waste time customising orders again and to make it easy for the user .

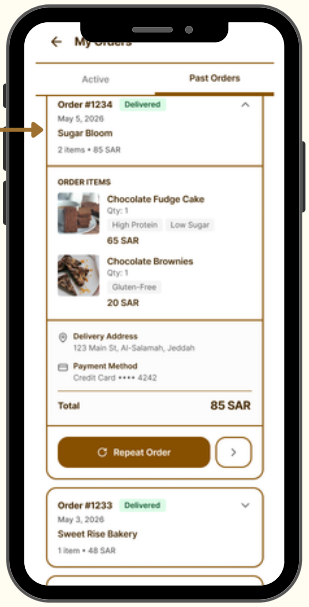


Figure 56 - past order: Provides a detailed information regarding past orders.

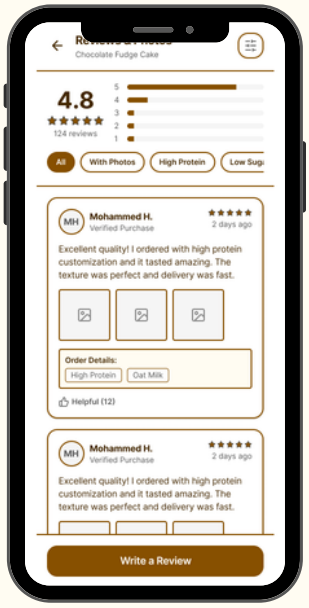


Figure 57 - Allows users to view verified reviews, ratings, and customization experiences from previous customers to support informed purchasing decisions

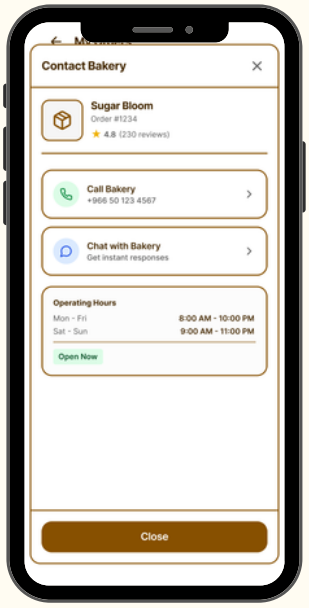


Figure 58 - Provides direct communication options with bakeries through calls or chat, improving customer support and accessibility.

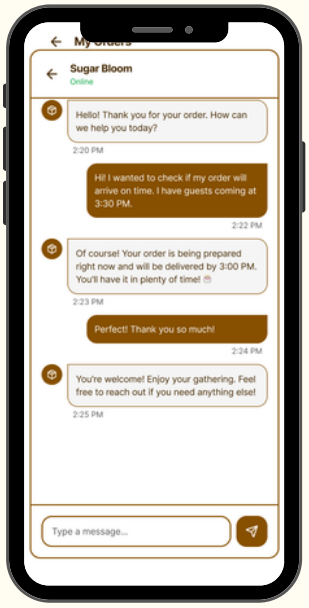


Figure 59 - Allows users to communicate with bakeries in real time to track orders, ask questions, and receive updates.

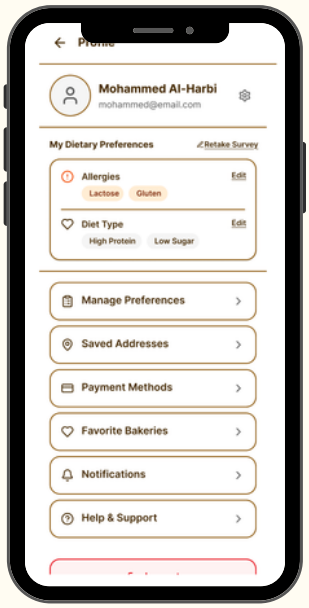


Figure 60 - Centralizes personal settings and preferences for a personalized experience.

Figure 61 - Allows users to adjust language, font size, appearance, and text-to-speech options to improve accessibility and usability.

Figure 62 - Enables users to choose their preferred language for a more comfortable and inclusive experience.

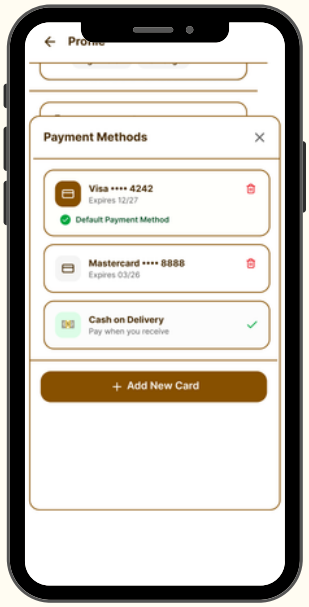


Figure 63 - Payment Methods: Allows users to manage saved credit cards or set a default payment option.

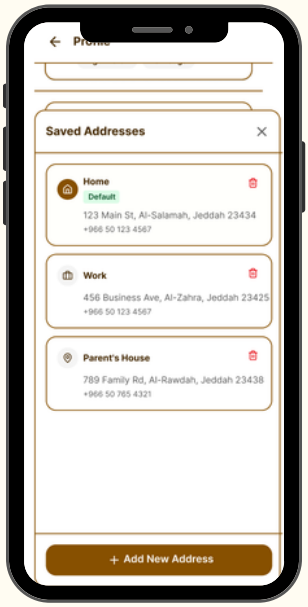


Figure 64 - Saved Addresses: A list of saved delivery locations categorized by labels like Home, Work, or Parent's House.

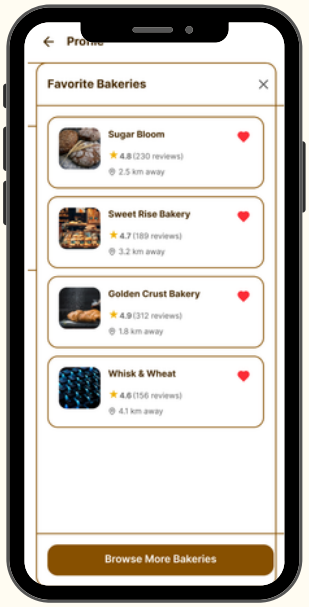


Figure 65 - Favorite Bakeries: A curated list of the user's bookmarked or favorite bakeries for fast access..

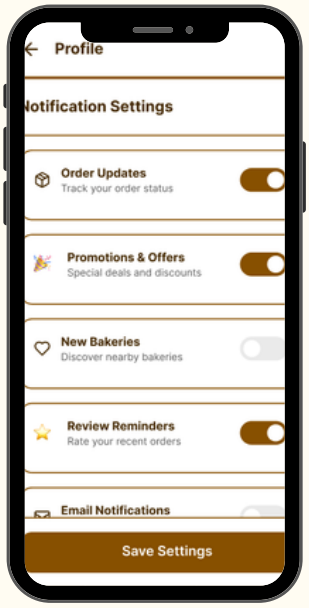
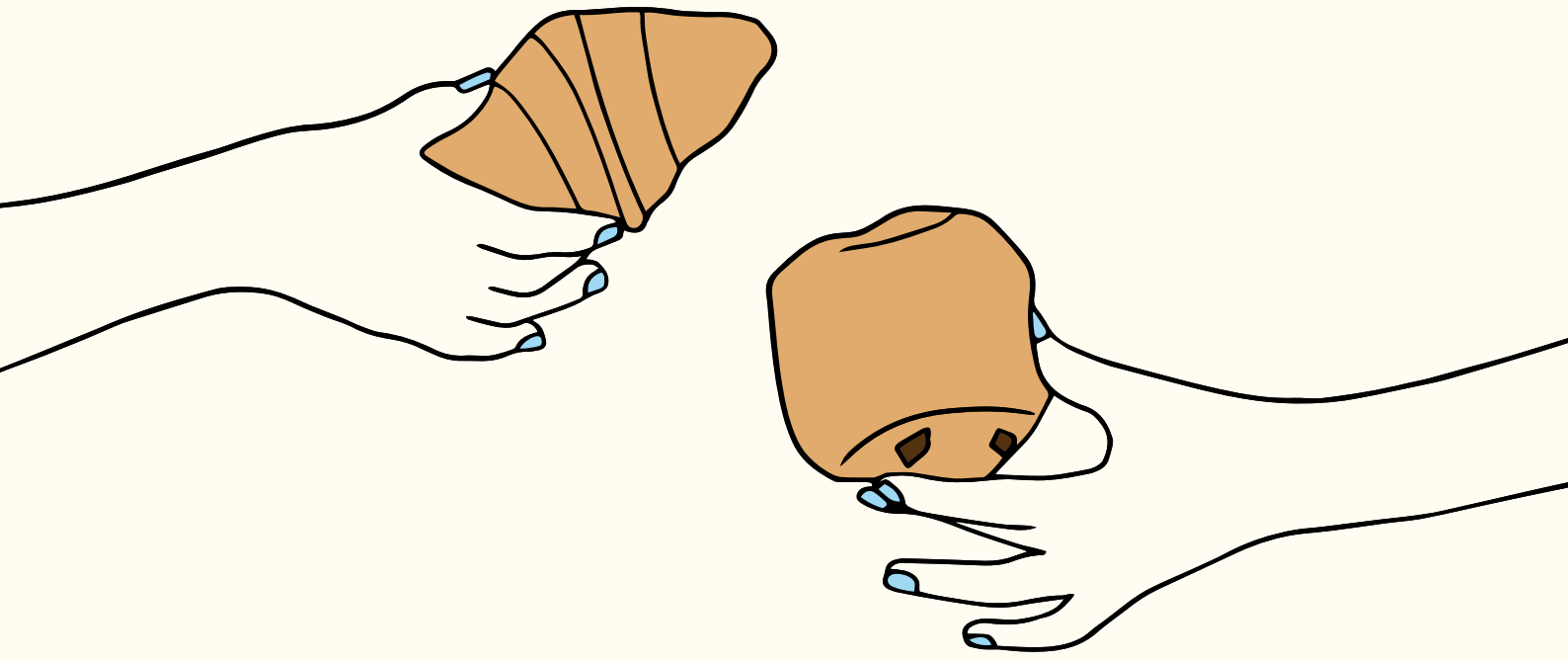


Figure 66- Notification Settings: Fine-grained toggles to control alerts for Order Updates, Promotions & Offers, New Bakeries, and Review Reminders.

6. Test and feedback



6.2 Usability Testing
6.2.1 usability testing sessions with users

Usability Testing 1

Target user group : Allergic users and health-Conscious Users

Number of participants: 4 (Two Allergic user and two Health-Conscious User **)**

Demographics: 2 Male, 2 Female

Task	Completion Status	Time Taken	Difficulty Level	Errors/ Confusion	Notes
Open the app and log in or sign up	Completed	30 - 45 seconds	Easy	None	
do the Survey	Completed	1 minute	Easy	None	
Choose location	Incomplete	20 - 40 seconds	Difficult	the user is unable to have a flexiblety when choosing the location.	The app offer just using the current location and the user couldn't choose the location manually
Apply filtes to search for bakeries	Incomplete	15 - 30 seconds	Easy	There are few Search filter options.	
Opening bakeries	Completed	5 - 10 seconds	Very Easy	None	
Opening a product and applying customisation options	Completed	40 - 70 seconds	Moderate	None	Takes time a little. they mentioned that it's perfect but the reason is because it's a new feature that takes them time to adapt
Adding and viewing cart	Completed	15 - 25 seconds	Very Easy	None	
Checkout and placing the order	Completed	45 - 60 seconds	Very Easy	None	
Track order	Completed	10 - 15 seconds	Easy	None	
Manage past orders	Completed	15 - 30 seconds	Easy	None	
Manage Profile	Completed	1 minute	Easy	little options provided	Unable to see my invoices and no invite friends feature

Table 14 - Usability Testing 1

6.2 Usability Testing

Usability Testing 2

Target user group : convenience seeker user and Event&customization user

Number of participants: 4(Two Convenience-Seeker Users and Two Event & Customization Users)

Demographics: 2 Male, 2 Female

Task	Completion Status	Time Taken	Difficulty Level	Errors/ Confusion	Notes
Open the app and log in or sign up	Completed	30 - 45 seconds	Very Easy	None	
Do the Survey	Completed	1 minute	Easy	None	Survey felt slightly long but clear
Choose location	Completed	20 - 40 seconds	Easy	None	Location access was clear and easy
Apply filtes to search for bakeries	Completed	15 - 30 seconds	Moderate	Minor hesitation while selecting filters	Took time to get used to filters
Opening bakeries	Completed	5 - 10 seconds	Very Easy	None	Bakery pages were visually clear
Opening a product and applying customisation options	Completed	40 - 70 seconds	Moderate	None	Customization options were useful but required attention
Adding and viewing cart	Completed	15 - 25 seconds	Very Easy	None	
Checkout and placing the order	Completed	45 - 60 seconds	Easy	None	
Track order	Completed	10 - 15 seconds	Easy	None	
Manage past orders	Completed	15 - 30 seconds	Easy	None	

Table 15 - Usability Testing 2

6.2 Usability Testing

6.2.2 pain points, usability issues, and areas for improvement

Pain point: Users are forced to use their current GPS location when interacting with the map.

Usability issues: Inability to manually enter or adjust the delivery address on the map, making it impossible to order for other locations or if the GPS is inaccurate.

Areas for improvement: Add a manual address search bar and allow users to pin and adjust their location directly on the map.

Pain point: Lack of financial transparency and order history tracking.

Usability issues: Users are unable to view, access, or download their past and current purchase invoices within the app.

Areas for improvement: Create a dedicated "Invoices" or "Purchase History" section where users can view and download detailed receipts

---

Pain point: Users cannot easily find available discounts or manage their vouchers.

Usability issues: There is no dedicated section or tab for promotional offers, deals, and discount vouchers.

Areas for improvement: Design a prominent "Offers & Vouchers" section on the home screen or profile tab to showcase ongoing bakery deals and saved coupons.

Pain point: Missing social sharing and referral incentives.

Usability issues: There is no "Invite Friends" or referral feature, limiting organic user growth and word-of-mouth marketing.

Areas for improvement: Integrate a "Refer a Friend" feature that gives both the existing and new user a discount code upon signing up.

6.2 Usability Testing

6.2.2 pain points, usability issues, and areas for improvement

Pain point: Users felt that the survey was longer than expected before accessing the main features.

Usability issues: Some users lost interest while answering multiple survey questions during onboarding.

Areas for improvement: Shorten the survey, add a progress indicator, or allow users to skip and complete it later.

Pain point: Users wanted quicker access to bakery categories related to occasions and celebrations.

Usability issues: Browsing through multiple bakery options took additional time before finding suitable products.

Areas for improvement: Add occasion-based categories such as birthdays, weddings, and graduation events.

Pain point: Users spent time comparing customization choices before placing an order.

Usability issues: The large number of customization options required extra attention and slowed decision-making.

Areas for improvement: Group customization options more clearly and add recommended combinations or previews.

Pain point: Users needed time to adapt to the customization feature during their first use.

Usability issues: Some users paused while exploring available options and add-ons.

Areas for improvement: Add brief guidance tips or a simpler customization flow for first-time users.

6.3 Feedback analysis
6.3.1 Analyze the feedback and observations

Target user group 1: Allergic users and health-Conscious Users

1 - Observations:

- The location/map element was the most challenging for users; it was the only unfinished job, rating "Difficult," because the program required GPS use without permitting manual address entry.
- Users complained that there weren't enough filter options in the search filters task, which is crucial for users who need to filter by ingredients or allergens.
- Although Manage Profile was completed, users observed that functionality such an invite friends button and invoice history were lacking.
- The majority of the other tasks (order tracking, checkout, and cart) went out without any issues.

2 - Insights:

- The app's limited filter choices make it less helpful for users whose main goal is to locate bakeries that adhere to tight dietary regulations.
- The lack of referral and invoicing functionality indicates that the profile part is limited in comparison to what the users anticipate from a full app experience.

Target user group 2: convenience seeker user and Event&customization user

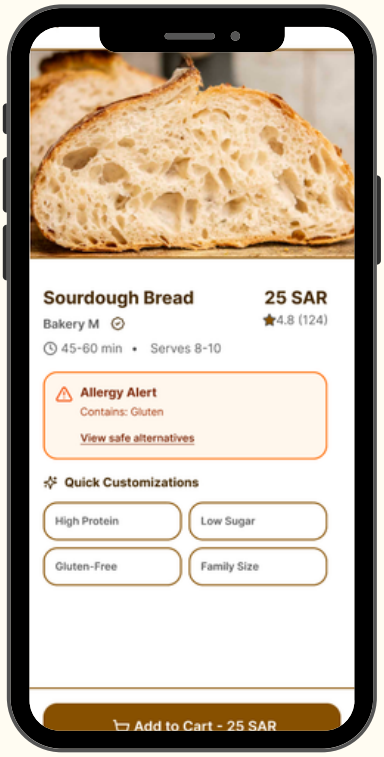
1 - Observations:

- Each task was finished completely, and there were no errors or confusions found in any of the app's primary features.
- Some users reportedly lost interest in the survey during onboarding because it was slightly long.
- When users applied filters to look for bakeries, there was a slight hesitancy.
- Customization options required more attention and took 40–70 seconds to finish; some users paused to go through the available add-ons.

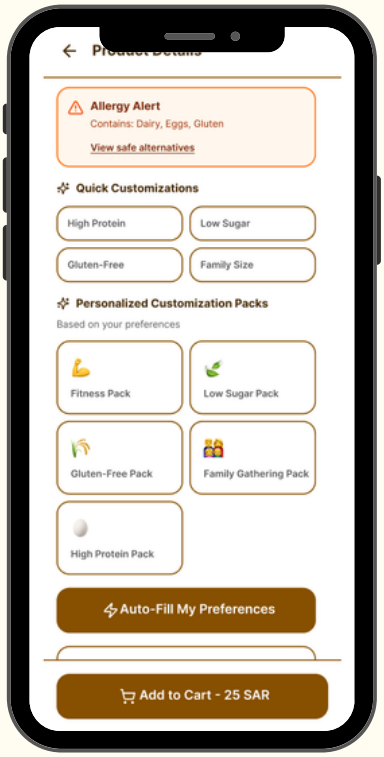
2 - Insights:

- For convenience-seeking customers who anticipate rapid access to the app's primary functions, the duration of the onboarding survey causes needless friction.
- The minimal degree of difficulty associated with applying filters indicates that people browsing with a particular occasion or event in mind may not find the current filter system to be sufficiently straightforward.

6.3.2 Make informed design and functionality decisions

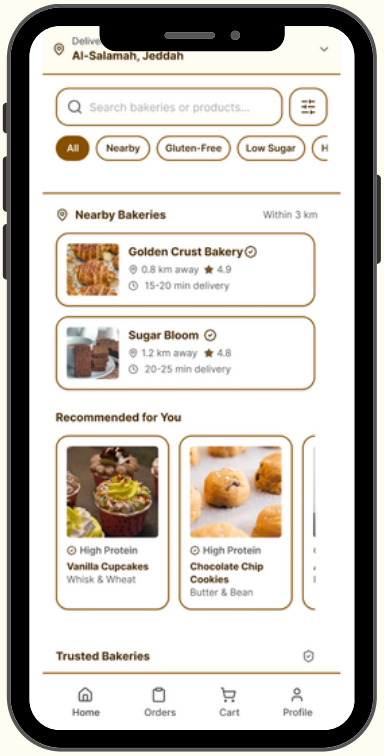


- **Figure 67** - Users reported that customizing products manually took too much time, especially when ordering multiple items.

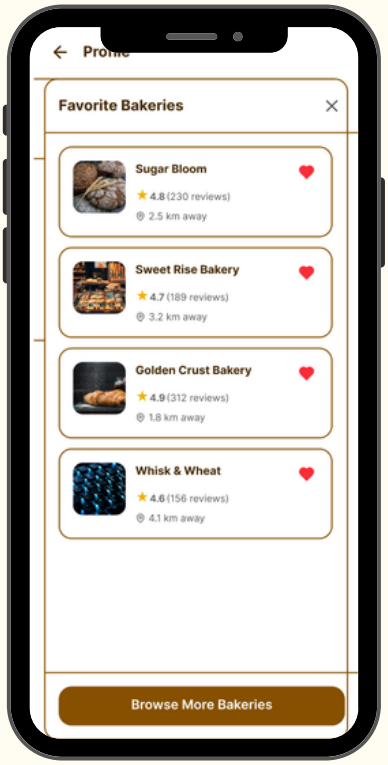


- **Figure 68** - Personalized customization packs were added to provide pre-selected options based on user preferences, making ordering faster and easier.

6.3.2 Make informed design and functionality decisions

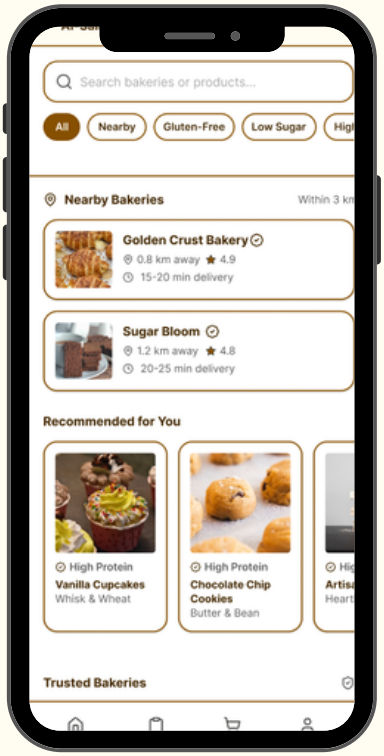


- **Figure 69** - Users had difficulty repeatedly searching for their preferred bakeries whenever they opened the app.

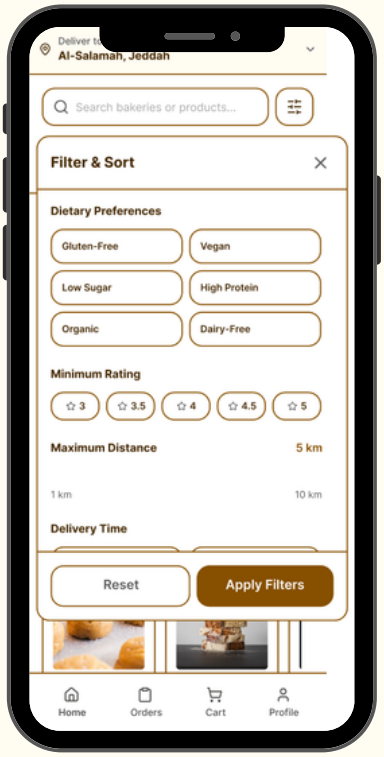


- **Figure 70** - A favorite bakeries feature was added, allowing users to save bakeries and access them quickly through their profile.

6.3.2 Make informed design and functionality decisions



- **Figure 71** - Users struggled to find bakeries matching dietary preferences and had to search manually.



- **Figure 72** - Filter and sorting options were introduced, enabling users to search bakeries by dietary needs, rating, distance, price, and delivery time.

Conclusion

BakeCraft was designed to address common challenges users experience when searching for bakeries that support dietary restrictions, customization needs, and convenient ordering processes. Many users struggle with finding suitable options, communicating specific requests, comparing bakeries, and ensuring products meet their health requirements or personal preferences. Therefore, the application aims to provide a more personalized, accessible, and efficient experience through customization features, allergy support, filtering options, recommendations, and direct communication with bakeries.

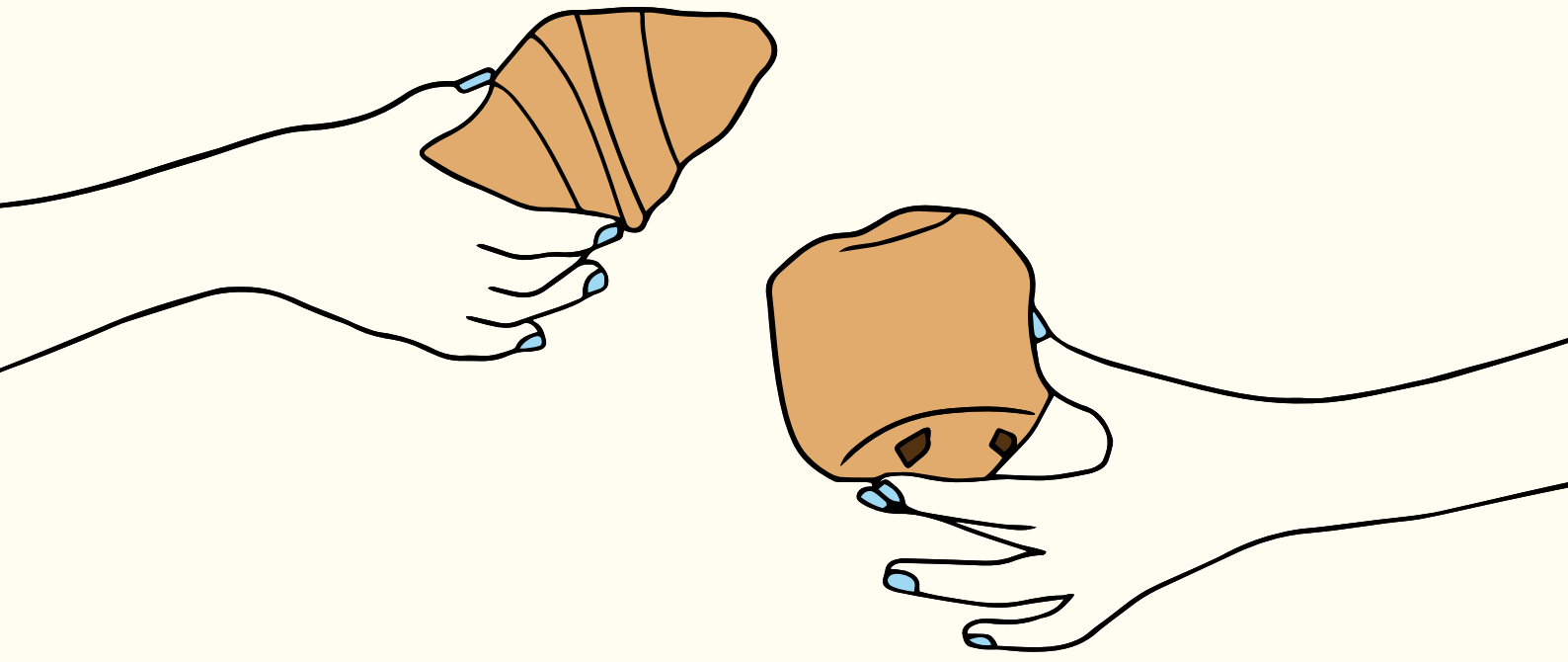
The development of BakeCraft was guided by Human–Computer Interaction (HCI) principles, emphasizing user-centered design, usability, accessibility, and continuous improvement based on user feedback. Understanding user behaviors, preferences, and pain points played an important role in shaping interface decisions and improving functionality to better align with real user needs.

Overall, BakeCraft highlights the importance of applying HCI approaches in designing digital solutions that are intuitive, inclusive, and capable of improving user experiences while simplifying the process of discovering, customizing, and ordering baked goods

References

- Figma. (2024). Design and prototyping tools. https://www.figma.com
- HCI Explained. (n.d.). How to Conduct Usability Testing [Video]. YouTube. https://www.youtube.com/watch?v=dHEx8QtCT74
- Nielsen Norman Group. (2024). Usability and user experience research. https://www.nngroup.com
- Usability.gov. (2024). User experience basics. https://www.usability.gov

Appendix



Consent form

Project Title: BakeCraft Mobile Application Design Project **institution**: [King Abdulaziz University]
Course: [MIS373]

Participation:
Your participation in this interview is completely voluntary. You have the choice to answer or skip any question and also you could stop the interview at any time.

Confidentiality:
All information you provide will be kept confidential and will only be used for academic purposes. Your name and personal information will not be included in the report. The transcript will be sent to you after the interview for confirmation.

Recording (Optional):
With your permission, the interview may be recorded only for note-taking purposes. The recording will not be shared and will be deleted after the project is completed.

By signing below, you confirm that:
You have read and understood the information above.
You voluntarily agree to participate in this interview.

Participant Name:
Signature:
Date:
Email :

Usability Testing Consent Form

Copy for the interviewer

Type of interview: Individual

Target participant:

General bakery consumers, including individuals with specific dietary preferences, food allergies (e.g., lactose/gluten intolerance), or health-conscious users seeking customized ingredients.

Context:

The session is conducted to evaluate the user experience of the "BakeCraft" application. BakeCraft enables users to select various types of bread and fully customize ingredients (such as removing lactose, adjusting sugar levels, or choosing specific fillings like custard and jam). The app connects users with nearby partner bakeries to ensure fresh preparation and delivery.

The participant's rights:

I am aware that I can leave the session at any moment without incurring any penalties by informing the researcher. Upon withdrawal, all my recorded information and data will be permanently removed from the study records.

Agreement:

This is to confirm that you are willing to participate in this usability study. I am aware of the project details, including the collection of data regarding my interaction with the app's customization features, ingredient selection, and navigation flow.

Confirmation:

I confirm that I am over the age of 18 and I grant permission for the data gathered (which may include voice/screen recording and observation notes) to be used for the study and subsequent academic publications. All data will be treated with strict confidentiality and used for research purposes only.

Name:

Signature:

Date:

Email:

Contact number (optional):

Information Sheet

copy for the participant

The purpose of the study: The objective of this research is to comprehend how consumers use mobile applications that enable them place orders for personalized baked goods from shops or bakeries. The insights will assist us in enhancing the application's usability and design in order to meet the needs and requirements of the users.

Interview Process: The interview will last roughly 10 - 15 minutes and be performed in a brief conversational approach. During the interview, participants will be questioned about how they often find or select bakeries or stores, their preferences when requesting personalized products, and their experiences ordering baked goods.

Context: The interview was held on [date] at [time] through [online meeting / in person]. Responses were recorded after aggrement.

Data Access: The information collected during this interview will only be accessed by the student research team conducting this study. The data will be used solely for academic purposes as part of a university project and will not be shared with external parties.

Contact information: If you have any questions about this study, you may contact the research team at:
Name: Heba Salah Al-deen
Phone number: 0532806206

Interview Script

Q1: Do you usually buy baked goods from bakeries?

- **Yes / No**

Q2: How do you usually find bakeries?

- **Social media**
- **Delivery apps**
- **Recommendations from friends**
- **Other**

Q3: How important are dietary options such as low sugar, gluten-free, or special ingredients to you?

Q4: Have you ever ordered customized baked goods before? If yes, can you describe your experience?

Q5: Do you face any difficulties when trying to find bakeries that offer customized baked goods?

Q6: What kind of customization options would you like to have when ordering baked goods?

Q7: When customizing your bakeries (like removing sugar or adding fillings), do you prefer to choose from a ready-made list or write your own notes to the baker?

Q8: If an ingredient you want to remove (like eggs) is essential to the bread's structure, would you like the app to suggest a "safe alternative" (like flaxseed)?

Q9: Since customized bread takes time to bake, how long are you willing to wait for a "made-from-scratch" order compared to a regular one?

Q10: For customized healthy bread that you might need daily, would you be interested in a "Weekly Subscription" where your specific loaf is delivered fresh at a set time?

Q11: If there was a mobile app that connects you with bakeries that prepare baked goods based on your preferences, would you use it? Why or why not?

Q12: What is the one must-have feature you need in the app to satisfy your bakery needs?

Interview Script

: السؤال الأول
هل تقوم عادةً بشراء المخبوزات من المخابز؟

:السؤال الثاني
كيف تجد المخابز عادةً؟

:السؤال الثالث
ما مدى أهمية الخيارات الغذائية مثل قليلة السكر، أو الخالية من الجلوتين، أو المكونات الخاصة بالنسبة لك؟

:السؤال الرابع
هل سبق لك طلب مخبوزات مخصصة من قبل؟ إذا كانت الإجابة نعم، هل يمكنك وصف تجربتك؟

:السؤال الخامس
هل تواجه أي صعوبات عند محاولة العثور على مخابز تقدم مخبوزات مخصصة؟

:السؤال السادس
ما نوع خيارات التخصيص التي تود توفرها عند طلب المخبوزات؟

:السؤال السابع
عند تخصيص طلبك، هل تفضل الاختيار من قائمة جاهزة أم كتابة ملاحظاتك الخاصة؟

:السؤال الثامن
إذا كان هناك مكون ترغب في إزالته ولكنه أساسي في المنتج، هل تود أن يقترح لك التطبيق بدائل مناسبة؟

:السؤال التاسع
نظرًا لأن إعداد المخبوزات المخصصة قد يستغرق وقتًا، كم من الوقت أنت مستعد للانتظار مقارنة بالطلب العادي؟

:السؤال العاشر
هل ستكون مهتمًا بفكرة اشتراك أسبوعي يتم من خلاله توصيل طلبك المخصص بشكل منتظم؟

:السؤال الحادي عشر
إذا توفر تطبيق يربطك بالمخابز بناءً على تفضيلاتك، هل ستستخدمه؟ ولماذا؟

:السؤال الثاني عشر
ما هي الميزة الأساسية التي يجب توفرها في التطبيق لتلبية احتياجاتك؟

Interview Transcript 1

Interviewer student Name: Maha Saleh

السؤال الأول :

هل تقوم عادةً بشراء المخبوزات من المخابز؟

"إيه أشتري، بس مو بشكل عشوائي أبد. يعني أنا صرت حريص أكثر على أكلي، خاصة بالفترة الأخيرة، فمو أي شيء يعجبني آخذه. لازم أتأكد إنه يناسب نظامي الغذائي، مثل يكون قليل سكر أو على الأقل مو مليان إضافات. أحيانًا أشتهي شيء وأروح أطلب، بس إذا ما لقيت معلومات واضحة عن المكونات، أتردد وأكنسل الفكرة كلها."

السؤال الثاني :

كيف تجد المخابز عادةً؟

"غالبًا أستخدم تطبيقات التوصيل زي هنقرستيشن أو جاهز، لأنها أسهل شيء بالنسبة لي. بس برضو أقضي وقت على إنستقرام أشوف حسابات مخابز، خصوصًا اللي يعرضون منتجات صحية. أحيانًا أشوف إعلان أو بوست ويشدني الشكل، فأدخل أشوف التفاصيل. وإذا أحد من أصحابي جرب مكان وقال لي إنه كويس وصحي، غالبًا أثق بكلامه وأجرب."

السؤال الثالث :

ما مدى أهمية الخيارات الغذائية مثل قليلة السكر، أو الخالية من الجلوتين، أو المكونات الخاصة بالنسبة لك؟

"مرة مهمة، يعني بالنسبة لي هذا العامل الأساسي قبل الطعم حتى. أنا أحاول أقلل السكر قد ما أقدر، وأفضل الأشياء اللي تكون حبوب كاملة أو فيها قيمة غذائية. ما أبغى آكل شيء أحس بعده إني غلطت أو خربت نظامي. فلو ما فيه خيارات صحية، غالبًا ما أشتري من الأساس."

السؤال الرابع :

هل سبق لك طلب مخبوزات مخصصة من قبل؟ إذا كانت الإجابة نعم، هل يمكنك وصف تجربتك؟

"إيه جربت أكثر من مرة، بس الصراحة التجربة مو مريحة. كنت أضطر أكتب ملاحظات طويلة، زي "قللوا السكر" أو "استخدموا نوع معين من الطحين"، بس ما كنت متأكد هل هم فهموا علي صح أو لا. أحيانًا يجي الطلب مختلف شوي عن اللي في بالي، وهذا يخليني أحس بعدم رضا."

السؤال الخامس :

هل تواجه أي صعوبات عند محاولة العثور على مخابز تقدم مخبوزات مخصصة؟

"كثير. أغلب المخابز إما ما عندهم تخصيص أصلًا، أو يكون بشكل بسيط جدًا. وبرضو المشكلة الكبيرة إنه ما فيه معلومات غذائية واضحة، يعني ما أدري كم السكر أو السعرات. هذا الشيء يخليني أتردد كثير قبل ما أطلب."

السؤال السادس :

ما نوع خيارات التخصيص التي تود توفرها عند طلب المخبوزات؟

أبغى تحكم واضح وبسيط. يعني أقدر أحدد كمية السكر، نوع الطحين (أبيض أو حبوب كاملة)، وأضيف أو أشيل مكونات" "بسهولة. مثلًا أبغى أشيل شيء معين أو أضيف بروتين، يكون الموضوع بضغطة زر، مو لازم أكتب واشرح.

السؤال السابع :

عند تخصيص طلبك، هل تفضل الاختيار من قائمة جاهزة أم كتابة ملاحظاتك الخاصة؟

"أكيد قائمة جاهزة، تكون مرتبة وواضحة. أحسها أسرع وأسهل، وتقلل احتمالية الخطأ. الكتابة متعبة وأحيانًا ما توصل الفكرة صح."

السؤال الثامن :

إذا كان هناك مكون ترغب في إزالته ولكنه أساسي في المنتج، هل تود أن يقترح لك التطبيق بدائل مناسبة؟

" طبعًا، هذا شيء مهم جدًا بالنسبة لي. أحيانًا أشيل مكون وما أعرف وش البديل المناسب، فلو التطبيق يقترح بدائل صحية يكون مرة مفيد ويوفر علي تفكير."

السؤال التاسع :

نظرًا لأن إعداد المخبوزات المخصصة قد يستغرق وقتًا، كم من الوقت أنت مستعد للانتظار مقارنة بالطلب العادي؟

"عادي أنتظر أكثر من الطلب العادي، لأن طبيعي بياخذ وقت. بس أهم شيء يكون واضح من البداية. يعني لو قالوا يوم أو يومين، ما عندي مشكلة، بس ما أبغى أتفاجأ بتأخير."

السؤال العاشر :

هل ستكون مهتمًا بفكرة اشتراك أسبوعي يتم من خلاله توصيل طلبك المخصص بشكل منتظم؟

" جدًا، خاصة إني غالبًا أكرر نفس الطلب. فلو فيه اشتراك يوصل لي طلبي بشكل أسبوعي، بيوفر علي وقت وجهد، وما أحتاج أفكر كل مرة."

السؤال الحادي عشر :

إذا توفر تطبيق يربطك بالمخابز بناءً على تفضيلاتك، هل ستستخدمه؟ ولماذا؟

"اكيد بستخدمه، لأنه بيختصر علي وقت كبير. بدل ما أدور في أكثر من تطبيق أو حساب، كل شيء يكون في مكان واحد ومناسب لي من البداية."

السؤال الثاني عشر :

ما هي الميزة الأساسية التي يجب توفرها في التطبيق لتلبية احتياجاتك؟

"أهم شيء عندي الشفافية. أبغى معلومات غذائية واضحة، مثل السكر والسعرات والمكونات. هذا الشيء يخليني أطلب وأنا مرتاح."

Interview Transcript 2

Interviewer student Name: Maha Saleh

السؤال الأول :
هل تقومين عادةً بشراء المخبوزات من المخابز؟
"أشتري، بس مو بشكل يومي. يعني إذا اشتهيت شيء حلى أو كان فيه مناسبة. بس دايم أحاول أختار شيء أخف أو صحي أكثر عشان ما أحس بالذنب بعدين."

السؤال الثاني :
كيف تجدين المخابز عادةً؟
"أكثر شيء من إنستقرام، أحب أشوف الصور والتقديم. إذا الشكل جذبني، أبدأ أقرأ التفاصيل. وأحيانًا أسأل صديقاتي، لأنهم يجربون أماكن كثير. التوصيات تساعدني أقرر بسرعة."

السؤال الثالث :
ما مدى أهمية الخيارات الغذائية مثل قليلة السكر، أو الخالية من الجلوتين، أو المكونات الخاصة بالنسبة لك؟
"مرة مهمة، خصوصًا إني أحاول أقلل السكر. إذا ما فيه خيارات صحية، أحس ما له داعي أطلب. أفضل أختار شيء يناسبني بدل ما أندم بعدين."

السؤال الرابع :
هل سبق لك طلب مخبوزات مخصصة من قبل؟ إذا كانت الإجابة نعم، هل يمكنك وصف تجربتك؟
" جربت، بس المشكلة إني ما أكون متأكدة إذا فعلاً طبقوا طلبي زي ما هو. أحس فيه نقص في الوضوح أو التواصل، وهذا يخليني أتردد أعيد التجربة."

السؤال الخامس :
هل تواجهين أي صعوبات عند محاولة العثور على مخابز تقدم مخبوزات مخصصة؟
"إيه، أحس الخيارات قليلة. وحتى لو فيه تخصيص، يكون محدود جدًا، يعني ما أقدر أتحكم بكل شيء زي ما أبغى."

السؤال السادس :
ما نوع خيارات التخصيص التي تودين توفرها عند طلب المخبوزات؟
"أبغى أتحكم في السكر، ونوع الدقيق، وأضيف أشياء صحية زي الشوفان أو البروتين. يعني أبغى أحس إني أقدر أضبط الطلب على حسب احتياجي."

السؤال السابع :
عند تخصيص طلبك، هل تفضلين الاختيار من قائمة جاهزة أم كتابة ملاحظاتك الخاصة؟
"أفضل قائمة جاهزة، تكون واضحة وسهلة. الكتابة أحيانًا تكون مربكة، وأخاف ما يوصل اللي أبغاه بشكل صحيح."

السؤال الثامن :
إذا كان هناك مكون ترغب في إزالته ولكنه أساسي في المنتج، هل تودين أن يقترح لك التطبيق بدائل مناسبة؟
"إيه أكيد، وصراحة اشوفها فكرة جداا ممتزة واتمنى لو في شي كذا لان اوقات ما احبذ بعض المكونات و خصوصًا لو فيه مكون أساسي. هذا يساعدني أختار بدون ما أخرب الطعم أو النتيجة."

السؤال التاسع :
نظرًا لأن إعداد المخبوزات المخصصة قد يستغرق وقتًا، كم من الوقت أنتِ مستعدة للانتظار مقارنة بالطلب العادي؟
"عادي أنتظر، بس مو فترة طويلة مرة. يوم أو يومين يعتبر مناسب بالنسبة لي وبرضو كمان يعتمد على هل الطلبيه كبيرة او صغيرة اذا كبيره صراحه ماعندي مشكله انتظر اكثر خصوصا وقت المناسبات."

السؤال العاشر :
هل ستكونين مهتمة بفكرة اشتراك أسبوعي يتم من خلاله توصيل طلبك المخصص بشكل منتظم؟
"ايه ممكن، خصوصًا لو فيه خيارات تناسب روتيني اليومي. هذا بيسهّل علي كثير."

السؤال الحادي عشر :
إذا توفر تطبيق يربطك بالمخابز بناءً على تفضيلاتك، هل ستستخدمينه؟ ولماذا؟
"أكيد، لأنه بيوفر علي وقت ويخليني أوصل لخيارات مناسبة بسرعة بدل ما أحتار وأقعد أجرب."

السؤال الثاني عشر :
ما هي الميزة الأساسية التي يجب توفرها في التطبيق لتلبية احتياجاتك؟
"أبغى وضوح كامل، خاصة في المكونات والسعرات. لما تكون المعلومات واضحة، أحس بثقة أكثر وأنا أطلب ويااليت الاسعار تكون متناسبه والتوصيل كذلك ."

Interview Transcript 3

Interviewer student Name: Heba Salah

السؤال الأول :
هل تقوم عادةً بشراء المخبوزات من المخابز؟
"أيوة و بشكل معتاد نوعًا ما. تقريبًا كل أسبوع"

السؤال الثاني :
كيف تجد المخابز عادةً؟
"أغلب الأحيان من خلال السوشال ميديا، خصوصًا الانستقرام، وأحيانا برضو من خلال اقتراحات بعض الاصدقاء."

السؤال الثالث :
ما مدى أهمية الخيارات الغذائية مثل قليلة السكر، أو الخالية من الجلوتين، أو المكونات الخاصة بالنسبة لك؟
"مهمة جدًا. أنا أعاني من حساسية الغلوتين و لازم قبل كل طلب أتأكد من المكونات."

السؤال الرابع :
هل سبق لك طلب مخبوزات مخصصة من قبل؟ إذا كانت الإجابة نعم، هل يمكنك وصف تجربتك؟
"جربت كم مرة. أحد المرات اتواصلت مع مخبز عن طريق الانستقرام و طلبت منهم يستبدلوا مكون بمكون آخر و وافقوا، بس لما وصل الطلب ما كان نفس طلبي بالضبط; اعتقد انهم ما فهموا طلبي كويس. المرة الثانية كانت تجربة كويسة و ناجحة."

السؤال الخامس :
هل تواجه أي صعوبات عند محاولة العثور على مخابز تقدم مخبوزات مخصصة؟
"نعم، بعض المخابز ما توضح اذا عندهم بدائل صحية فأقضي وقت وأنا أبحث في مخابز مختلفة عشان أشوف خيار مناسب، أحيانا الموضوع يكون صعب و مرهق."

السؤال السادس :
ما نوع خيارات التخصيص التي تود توفرها عند طلب المخبوزات؟
"امكانية تبديل المكونات بسهولة، اني مثلا احذف أي شيء يحتوي على غلوتين و استبدله ببديل آمن و مناسب. كمان ممكن تكون ميزة جيدة انه يكون في معلومات واضحة عن مكونات كل منتج عشان ما اتسائل اذا ممكن يحتوي على مسببات حساسية او لا."

السؤال السابع :
عند تخصيص طلبك، هل تفضل الاختيار من قائمة جاهزة أم كتابة ملاحظاتك الخاصة؟
"بالنسبة لي الأفضل اني اختار من قائمة جاهزة. الكتابة ممكن الأسلوب ما يكون واضح بالنسبة للخباز فممكن يكون في احتمالية انه ما يوضح طلبي كويس."

السؤال الثامن:
إذا كان هناك مكون ترغب في إزالته ولكنه أساسي في المنتج، هل تود أن يقترح لك التطبيق بدائل مناسبة؟
"أيوة أكيد، فكرة جيدة لأن احيانا حتى أنا ما عندي فكرة اذا في بديل مناسب، اهم شيء البدائل تكون آمنة."

السؤال التاسع

نظرًا لأن إعداد المخبوزات المخصصة قد يستغرق وقتًا، كم من الوقت أنت مستعد للانتظار مقارنة بالطلب العادي؟

"ممكن من عدة ساعات إلى يوم كامل، بس يُفضل انه يكون عندي علم مسبق بمدة الانتظار."

لسؤال العاشر

هل ستكون مهتمًا بفكرة اشتراك أسبوعي يتم من خلاله توصيل طلبك المخصص بشكل منتظم؟

"أشوفها فكرة مثيرة للاهتمام. اذا لقيت مخبز مناسب وأثق فيه و في جودة المنتجات أو المخبوزات، أكيد بفكر في الموضوع، لأنه غالبا بيختصر علي وقت."

السؤال الحادي عشر :

إذا توفر تطبيق يربطك بالمخابز بناءً على تفضيلاتك، هل ستستخدمه؟ ولماذا؟

"أكيد بدون تردد، حاليا أواجه صعوبة اني ألقى خيارات متعددة تكون آمنة و تناسب حالتي الصحية. فكرة انه يكون كل شيء في مكان واحد و يكون عندي امكانية البحث و التعديل أو الفلترة بناء على تفضيلاتي هي فكرة ممتازة بنظري. لأنه عندي رغبة اني اطلب بسهولة بدون ما أقضي وقت طويل في البحث عن المكونات."

السؤال الثاني عشر :

ما هي الميزة الأساسية التي يجب توفرها في التطبيق لتلبية احتياجاتك؟

"غالبًا أهم حاجة هو فلتر الحساسية، أو اني اقدر أدخل نوع مشكلتي الصحية و التطبيق يعرض لي الخيارات الآمنة بالنسبة لي."

Interview Transcript 4

Interviewer student Name: Heba Salah

السؤال الأول :
هل تقوم عادةً بشراء المخبوزات من المخابز؟
"أيوة، بين كل فترة و فترة بمشي المخابز، بس ما بشكل دائم."

السؤال الثاني:
كيف تجد المخابز عادةً؟
"غالبًا من تطبيقات التوصيل، أو أحيانا بسمع عنها في السوشال ميديا عن طريق الصدفة وأنا أتصفح مثلا يطلع لي بوست عن مخبز معين أو شخص يصور تجربته."

السؤال الثالث:
ما مدى أهمية الخيارات الغذائية مثل قليلة السكر، أو الخالية من الجلوتين، أو المكونات الخاصة بالنسبة لك؟
"نوعًا ما مهم بالنسبة لي، لكن بصراحة ما أحرص كثير على الموضوع. أنا عندي حساسية لاكتوز و ما أشدد دائمًا على اني أطلب منتجات خالية من اللاكتوز، مع انه أحس بالأعراض المصاحبة للحساسية حتى لو بشكل خفيف، لكن أحاول اني أبدأ أهتم أكتر."

السؤال الرابع:
هل سبق لك طلب مخبوزات مخصصة من قبل؟ إذا كانت الإجابة نعم، هل يمكنك وصف تجربتك؟
"حصل جربت، مرة طلبت قهوة في أحد الكافيهات و سألت اذا ممكن يسووها بحليب خالي اللاكتوز. وافقوا يسووها بس ما كان مكتوب عندهم في المنيو انه في بدائل للحليب العادي متوفرة، اتوقع لو ما سألت ما كان ممكن أعرف و غالبا كنت طلبت بحليب عادي."

السؤال الخامس:
هل تواجه أي صعوبات عند محاولة العثور على مخابز تقدم مخبوزات مخصصة؟
"بعض الأحيان ايوة، اذا ما كان واضح في المنيو او قائمة المكونات اذا تحتوي على أي منتجات ألبان ف اضطر اني أسأل."

السؤال السادس:
ما نوع خيارات التخصيص التي تود توفرها عند طلب المخبوزات؟
"اني استبدل منتج الألبان ببدائل أخرى بسهولة و تكون الخيارات متوفرة قدامي بشكل واضح بدون اسئلة كتيرة."

السؤال السابع:
عند تخصيص طلبك، هل تفضل الاختيار من قائمة جاهزة أم كتابة ملاحظاتك الخاصة؟
"القائمة أفضل لأنها أسهل و أسرع في الاستخدام و حتى من ناحية الوضوح بالنسبة للمخبز و الشخص اللي بينفذ الطلب."

السؤال الثامن:
إذا كان هناك مكون ترغب في إزالته ولكنه أساسي في المنتج، هل تود أن يقترح لك التطبيق بدائل مناسبة؟
"في رأيي بتكون حاجة مفيدة، بالذات لمنتجات الألبان المستخدمة في المخبوزات، خصوصًا انه ما عندي فكرة كافية عنها."

السؤال التاسع:
نظرًا لأن إعداد المخبوزات المخصصة قد يستغرق وقتًا، كم من الوقت أنت مستعد للانتظار مقارنة بالطلب العادي؟
"على حسب المنتج، اذا مخبوزات بكميات كبيرة مثلا ممكن انتظر يوم. لكن اذا حاجة بسيطة و كميتها قليلة غالبا ما بنتظر اكتر من ساعة أو ساعة و نص."

السؤال العاشر:
هل ستكون مهتمًا بفكرة اشتراك أسبوعي يتم من خلاله توصيل طلبك المخصص بشكل منتظم؟
"للأمانة على حسب السعر. بس بشكل عام ممكن أجرب بالذات اذا حاجة ممكن استهلكها بشكل دائم و الجودة ممتازة."

السؤال الحادي عشر :
إذا توفر تطبيق يربطك بالمخابز بناءً على تفضيلاتك، هل ستستخدمه؟ ولماذا؟
"ما عندي مشكلة أستخدمه، أهم حاجة بالنسبة لي انه المكونات تكون واضحة تمامًا."

السؤال الثاني عشر :
ما هي الميزة الأساسية التي يجب توفرها في التطبيق لتلبية احتياجاتك؟
"مؤشر يوضّح لي بشكل مباشر اذا المنتج يحتوي على منتجات ألبان أو لا بدون ما أجتهد في البحث."

Interview Transcript 5

Interviewer student Name: Aisha Alkaff

السؤال الأول :

هل تقوم عادةً بشراء المخبوزات من المخابز؟

ايه، أشتري من المخابز بشكل شبه يومي أو كم مرة في الأسبوع، خصوصًا لما أكون طالع أو بين مشاوير. بالنسبة لي المخبوزات خيار سريع وسهل، ما يحتاج تفكير ولا تجهيز، بس أوقف وآخذ شي وخلاص

السؤال الثاني:

كيف تجد المخابز عادةً؟

غالبًا ما أبحث بشكل متعمق، أروح للي قريب مني أو اللي متعود عليه. وإذا كنت في مكان جديد، أفتح تطبيق توصيل وأختار أول شي يطلع لي ويكون تقييمه كويس. ما أحب أضيع وقت في المقارنة أو البحث الطويل.

السؤال الثالث:

ما مدى أهمية الخيارات الغذائية مثل قليلة السكر، أو الخالية من الجلوتين، أو المكونات الخاصة بالنسبة لك؟

بصراحة ما أعطيها اهتمام كبير، إلا إذا كان عندي مزاج معين أو أبغى أخفف سكر لفترة. لكن بشكل عام أنا أركز أكثر على السرعة والطعم، مو مرة على التفاصيل الغذائية، إلا لو كانت واضحة وسهلة الاختيار بدون تعقيد.

السؤال الرابع:

هل سبق لك طلب مخبوزات مخصصة من قبل؟ إذا كانت الإجابة نعم، هل يمكنك وصف تجربتك؟

جربت مرة أطلب شي مخصص، زي إني أبغى أقلل السكر أو أغير مكون، بس حسّيت الموضوع معقد شوي. احتجت أكتب ملاحظات وأتأكد إذا الطلب ينفذ أو لا، وأخذ وقت أطول من الطلب العادي. فبصراحة ما تحمست أعيد التجربة كثير

السؤال الخامس:

هل تواجه أي صعوبات عند محاولة العثور على مخابز تقدم مخبوزات مخصصة؟

ايه، أحس صعب أعرف أي مخبز يوفر تخصيص بسهولة. أحيانًا ما يكون واضح في التطبيق، وأضطر أتواصل أو أسأل، وهذا ياخذ وقت وما يناسبني. أفضل يكون كل شي واضح وسريع بدون ما أرجع وأسأل.

السؤال السادس:

ما نوع خيارات التخصيص التي تود توفرها عند طلب المخبوزات؟

أفضل تكون الخيارات بسيطة وواضحة، زي إني أختار الحجم، أو أقلل سكر، أو أشيل مكون معين بضغطة زر. ما أبغى خيارات كثيرة ومعقدة، أبغى شي سريع أقدر أخلصه خلال ثواني

السؤال السابع:

عند تخصيص طلبك، هل تفضل الاختيار من قائمة جاهزة أم كتابة ملاحظاتك الخاصة؟

أكيد أفضل قائمة جاهزة، لأن الكتابة تاخذ وقت وتخليني أفكر أكثر. أنا أبغى أخلص الطلب بسرعة، فكل ما كانت الخيارات جاهزة وواضحة يكون أفضل بالنسبة لي.

السؤال الثامن:

إذا كان هناك مكون ترغب في إزالته ولكنه أساسي في المنتج، هل تود أن يقترح لك التطبيق بدائل مناسبة؟

ايه أكيد، هذا بيسهل علي كثير. بدل ما أحتار أو أكنسل الطلب، لو التطبيق يقترح بدائل مباشرة بيكون أسرع وأسهل، خصوصًا إذا كانت الاقتراحات مناسبة لنفس الشي اللي أبغاه

السؤال التاسع:

نظرًا لأن إعداد المخبوزات المخصصة قد يستغرق وقتًا، كم من الوقت أنت مستعد للانتظار مقارنة بالطلب العادي؟

أنا افهم إن الطلب المخصص ممكن ياخذ وقت أطول، بس مو مرة طويل. ممكن أستنى زيادة بسيطة عن الطلب العادي، بس إذا حسّيت الموضوع بيطول كثير غالبًا بغير رأيي وأطلب شي جاهز

السؤال العاشر:

هل ستكون مهتمًا بفكرة اشتراك أسبوعي يتم من خلاله توصيل طلبك المخصص بشكل منتظم؟

ممكن أهتم بالفكرة إذا كانت فعلاً بتوفر علي وقت وما تحتاج مني مجهود. يعني لو أقدر أحدد مرة وحدة والباقي يتم تلقائي، ممكن تكون فكرة حلوة. بس إذا فيها تعقيد أو متابعة مستمرة، غالبًا ما راح أستخدمها

السؤال الحادي عشر:

إذا توفر تطبيق يربطك بالمخابز بناءً على تفضيلاتك، هل ستستخدمه؟ ولماذا؟

ايه، ممكن أستخدمه إذا كان بيسهل علي الموضوع ويخليني أوصل للي أبغاه بسرعة. أهم شي يكون واضح وسريع وما فيه خطوات كثيرة، لأن هذا الشي اللي أبحث عنه أصلًا.

السؤال الثاني عشر:

ما هي الميزة الأساسية التي يجب توفرها في التطبيق لتلبية احتياجاتك؟

أهم ميزة بالنسبة لي هي السرعة وسهولة الاستخدام. أبغى تطبيق أفتحُه وأطلب خلال أقل من دقيقة بدون ما أضيع وقت أو أتوه بين الخيارات. كل ما كان أبسط وأسرع، كل ما استخدمته أكثر.

Interview Transcript 6

Interviewer student Name: Aisha Alkaff

السؤال الأول :
هل تقوم عادةً بشراء المخبوزات من المخابز؟
ايه، أشتري من المخابز بين فترة وفترة، مو يوميًا بس إذا اشتهيت شي أو كان عندي مناسبة صغيرة أو حتى كقهوة مع شي حلو

السؤال الثاني:
كيف تجد المخابز عادةً؟
غالبًا أشوف اللي حولي أو من خلال تطبيقات التوصيل، وأحيانًا أبحث في السوشال ميديا إذا كنت أبغى شي معين أو شكل حلو

السؤال الثالث:
ما مدى أهمية الخيارات الغذائية مثل قليلة السكر، أو الخالية من الجلوتين، أو المكونات الخاصة بالنسبة لك؟
تفرق معاي إلى حد ما، خصوصًا موضوع السكر. أحاول أختار أشياء أخف شوي، بس مو دايم ألتزم، يعني حسب المزاج

السؤال الرابع:
هل سبق لك طلب مخبوزات مخصصة من قبل؟ إذا كانت الإجابة نعم، هل يمكنك وصف تجربتك؟
ايه جربت أطلب شي مخصص قبل، كان كيك بسيط. التجربة كانت حلوة بس احتاجت وقت وترتيب، يعني مو شي أسويه بشكل متكرر.

السؤال الخامس:
هل تواجه أي صعوبات عند محاولة العثور على مخابز تقدم مخبوزات مخصصة؟
أحيانًا، مو كل المخابز توضح إذا عندها تخصيص أو لا، وأحيانًا أحس الموضوع يبغاله تواصل أكثر من اللازم.

السؤال السادس:
ما نوع خيارات التخصيص التي تود توفرها عند طلب المخبوزات؟
أحب يكون فيه خيارات زي اختيار النكهة، مستوى السكر، وبعض الإضافات البسيطة. ما أبغى شي معقد بس يكون فيه مرونة.

السؤال السابع:
عند تخصيص طلبك، هل تفضل الاختيار من قائمة جاهزة أم كتابة ملاحظاتك الخاصة؟
أفضل قائمة جاهزة، بس يكون فيه خيار أضيف ملاحظة بسيطة إذا احتجت، يعني الاثنين مع بعض.

السؤال الثامن:
إذا كان هناك مكون ترغب في إزالته ولكنه أساسي في المنتج، هل تود أن يقترح لك التطبيق بدائل مناسبة؟
ايه أكيد، هذا بيسهل علي القرار، خصوصًا لو ما كنت عارف إيش البديل المناسب.

السؤال التاسع:
نظرًا لأن إعداد المخبوزات المخصصة قد يستغرق وقتًا، كم من الوقت أنت مستعد للانتظار مقارنة بالطلب العادي؟
أنا افهم إن الطلب المخصص ممكن ياخذ وقت أطول، بس مو مرة طويل. ممكن أستنى زيادة بسيطة عن الطلب العادي، بس إذا حسّيت الموضوع بيطول كثير غالبًا بغير رأيي وأطلب شي جاهز

السؤال العاشر:
هل ستكون مهتمًا بفكرة اشتراك أسبوعي يتم من خلاله توصيل طلبك المخصص بشكل منتظم؟
ممكن، خصوصًا لو كان عندي روتين معين، زي كل أسبوع آخذ نفس الشي. بس لازم يكون سهل وما يقيدني.

السؤال الحادي عشر:
إذا توفر تطبيق يربطك بالمخابز بناءً على تفضيلاتك، هل ستستخدمه؟ ولماذا؟
ايه ممكن أستخدمه، خصوصًا لو كان يجمع الخيارات في مكان واحد ويوفر علي وقت البحث.

السؤال الثاني عشر:
ما هي الميزة الأساسية التي يجب توفرها في التطبيق لتلبية احتياجاتك؟
أهم شي يكون واضح وسهل الاستخدام، وأقدر أوصل للي أبغاه بدون تعقيد، مع شوية خيارات حلوة بدون ما أحس إني بضياع.

Interview Transcript 7

Interviewer student Name: Heba Salah

السؤال الأول :

هل تقومين عادةً بشراء المخبوزات من المخابز؟

"ايوة بشتري من المخابز بس مو بشكل منتظم، وأغلب الأحيان ببحث عن المخبز قبل أزوره وأجربه."

السؤال الثاني:

كيف تجدين المخابز عادةً؟

"ممكن توصيات من الاصدقاء أو أبحث أنا بنفسي في السوشال ميديا او الانترنت عن مخابز بحس انها تناسبني."

السؤال الثالث:

ما مدى أهمية الخيارات الغذائية مثل قليلة السكر، أو الخالية من الجلوتين، أو المكونات الخاصة بالنسبة لك؟

"مهمة جدًا بالنسبة لي صراحة، بالذات مع حساسية الغلوتين، مهم عندي أقرأ كل حاجة قبل ما أجرب أي منتج جديد، خصوصًا انه سبق وحصلت لي allergic reactions فدائمًا بحرص اني ما اتهاون."

السؤال الرابع:

هل سبق لك طلب مخبوزات مخصصة من قبل؟ إذا كانت الإجابة نعم، هل يمكنك وصف تجربتك؟

"جربت وكان الموضوع بالنسبة لي ممل و مرهق، لأني احتجت أشرح أكتر من مرة انه عندي حساسية عشان أأكد لهم، مع ذلك كنت نوعًا ما حاسة بتردد أو قلق طول فترة الانتظار انه ممكن يكون في مكون مسبب للحساسية، بس لما وصل الطلب و اتأكدت منه كان تمام بس حسيت عملية الطلب شوية مرهقة يعني احس ممكن تكون أسهل."

السؤال الخامس:

هل تواجهين أي صعوبات عند محاولة العثور على مخابز تقدم مخبوزات مخصصة؟

"صعوبات كتيرة صراحة، أغلب الأماكن ما بيكون عندهم معلومات كافية أو مفصلة عن المكونات أو تكون مكتوبة بشكل عام جدًا، وللأسف اذا ما عرفت مكونات المنتج كويس غالبًا بغير رأيي وما أشتري أو أشوف مكان غير."

السؤال السادس:

ما نوع خيارات التخصيص التي تودين توفرها عند طلب المخبوزات؟

"اني أقدر أشوف كل المكونات بالتفصيل وأقدر أحذف أي مكون قد يسبب حساسية و أشوف البدائل كمان بالتفصيل و تكون واضحة. كمان انه يكون عندي ملف غذائي محفوظ في التطبيق بكل معلوماتي الصحية لأن عملية ادخالها كل مرة من جديد مملة و تاخد وقت."

السؤال السابع:

عند تخصيص طلبك، هل تفضلين الاختيار من قائمة جاهزة أم كتابة ملاحظاتك الخاصة؟

"أكيد قائمة جاهزة لأنها بتكون جزء من النظام و اختياراتي بتكون أوضح، ما أمانع لو في خانة صغيرة مثلا لو انا حابة أضيف حاجة غير موجودة لكن زي ما قلت القائمة أضمن."

السؤال الثامن:

إذا كان هناك مكون ترغبين في إزالته ولكنه أساسي في المنتج، هل تودين أن يقترح لك التطبيق بدائل مناسبة؟

"أيوه فكرة كويسة، بس لازم أكون عارفة كل التفاصيل عن البديل قبل أوافق اني اختاره."

السؤال التاسع:
نظرًا لأن إعداد المخبوزات المخصصة قد يستغرق وقتًا، كم من الوقت أنت مستعدة للانتظار مقارنة بالطلب العادي؟
"ما عندي مانع أنتظر يومين أو ثلاثة بمقابل اني أحصل على منتجات آمنة ١٠٠٪ و صحية و تساهم في صحة و راحة جهازي الهضمي."

السؤال العاشر :
هل ستكونين مهتمة بفكرة اشتراك أسبوعي يتم من خلاله توصيل طلبك المخصص بشكل منتظم؟
"لو لقيت المخبز المناسب، الجدير بالثقة فأكيد ايوة."

السؤال الحادي عشر :
إذا توفر تطبيق يربطك بالمخابز بناءً على تفضيلاتك، هل ستستخدمينه؟ ولماذا؟
"بستخدمه ما عندي مشكلة بس أهم حاجة انه المعلومات دائمًا تكون دقيقة و محدثة بشكل دائم. بعض التطبيقات للأسف تكون المعلومات قديمة أو غلط و بالتالي احتمالية حدوث أي خطأ واردة."

السؤال الثاني عشر:
ما هي الميزة الأساسية التي يجب توفرها في التطبيق لتلبية احتياجاتك؟
"قائمة المكونات تكون تفصيلية و دقيقة ما تكون مكتوبة بشكل عام أو مبهم، لازم أشوف مكونات فعلية عشان أحس بالأمان اثناء استخدام التطبيق."

Interview Transcript 8

Interviewer student Name: Hala Alghamdi

السؤال الأول :
هل تقوم عادةً بشراء المخبوزات من المخابز؟
"ايوه كثير، خاصة للمناسبات زي العزايم أو لما يكون فيه جمعة مع الاهل او مع صحباتي لان الكل يحبها."

السؤال الثاني:
كيف تجد المخابز عادةً؟
"غالبًا من الإنستقرام او التيك توك واحيانا أسال صحباتي."

السؤال الثالث:
ما مدى أهمية الخيارات الغذائية مثل قليلة السكر، أو الخالية من الجلوتين، أو المكونات الخاصة بالنسبة لك؟
"شخصيا مو مهمه، لكن بعض صحباتي عندهم سكر وبعضهم يسوون دايت ف احاول اجيب مخبوزات تناسب نظامهم الصحي اذا تجمعنا."

السؤال الرابع:
هل سبق لك طلب مخبوزات مخصصة من قبل؟ إذا كانت الإجابة نعم، هل يمكنك وصف تجربتك؟
"ايوه، طلبت كيكة بتصميم معين، كان سعرها عالي بس ما طلع نفس الصورة ونفس ماتوقعت."

السؤال الخامس:
هل تواجه أي صعوبات عند محاولة العثور على مخابز تقدم مخبوزات مخصصة؟
"ايوه، صعب جدا أوصل فكرتي بالضبط، واحس كل مرة لازم أشرح كثير واكرر كلامي خاصة اذا كان تصميم خاص مافي تنفيذ واضح له ويكون اسوأ اذا كنت اشرح وجه لوجه."

السؤال السادس:
ما نوع خيارات التخصيص التي تود توفرها عند طلب المخبوزات؟
"بيكون شيء حلو لو قدرت اختار الألوان، الكتابة، اغير الحشوة، او شكل الكيك وحتى نسبة السكر عشان صحبتي."

السؤال السابع:
عند تخصيص طلبك، هل تفضل الاختيار من قائمة جاهزة أم كتابة ملاحظاتك الخاصة؟
"أفضل أشوف خيارات جاهزة وأعدل عليها لان الشرح صعب وياخذ وقت وجهد وممكن مايكون واضح ويسوون لي طلب خطأ بسبب سوء فهم."

السؤال الثامن:
إذا كان هناك مكون ترغب في إزالته ولكنه أساسي في المنتج، هل تود أن يقترح لك التطبيق بدائل مناسبة؟
"ايوه أكيد، بدل ما أحتار وقت طويل وانا افكر بيساعدني اتخيل الشكل النهائي للطلب."

السؤال التاسع:
نظرًا لأن إعداد المخبوزات المخصصة قد يستغرق وقتًا، كم من الوقت أنت مستعد للانتظار مقارنة بالطلب العادي؟
"عادي أنتظر فترة طويلة إذا النتيجة تستاهل بس طبعا كل ماكان اسرع كان افضل."

:السؤال العاشر
هل ستكون مهتمًا بفكرة اشتراك أسبوعي يتم من خلاله توصيل طلبك المخصص بشكل منتظم؟
"لا مو مهم بالنسبة لي."

:السؤال الحادي عشر
إذا توفر تطبيق يربطك بالمخابز بناءً على تفضيلاتك، هل ستستخدمه؟ ولماذا؟
"ايه أكيد بيريحني بدل ما أدور بكل مكان عن متجر يلبي طلباتي."

:السؤال الثاني عشر
ما هي الميزة الأساسية التي يجب توفرها في التطبيق لتلبية احتياجاتك؟
"أشوف شكل الطلب قبل ما أطلب بيخليني اكون مرتاحة واتأكد انه نفس الي ابغاه وبرضو ميزة الطلب وتحديد وقت التوصيل بيخليني اقدر اطلب من بدري وافضى اجهز للعزيمة."

Interview Transcript 9

Interviewer student Name: Hala Alghamdi

السؤال الأول :
هل تقوم عادةً بشراء المخبوزات من المخابز؟
"ايوه لكن مو كثير."

السؤال الثاني:
كيف تجد المخابز عادةً؟
"ابحث عنها في الانستقرام او Google maps."

السؤال الثالث:
ما مدى أهمية الخيارات الغذائية مثل قليلة السكر، أو الخالية من الجلوتين، أو المكونات الخاصة بالنسبة لك؟
"مهمة مره، عندنا بالعائلة حساسية حادة من اللاكتوز عشان كذا مانشتري مخبوزات كثير لازم نكون حذرين في كل شيء ناكله."

السؤال الرابع:
هل سبق لك طلب مخبوزات مخصصة من قبل؟ إذا كانت الإجابة نعم، هل يمكنك وصف تجربتك؟
"ايوه، طلبت اكثر من نوع مخبوزات زي الكروسان وكب كيك بدون لاكتوز وسكر قليل واغلبهم كانو متعاونين بس فيه مره كان تعاملهم سيء ومهملين اعطوني الطلب غلط ومرضت بسبب هذا الشيء ف اشتكيت على المخبز."

السؤال الخامس:
هل تواجه أي صعوبات عند محاولة العثور على مخابز تقدم مخبوزات مخصصة؟
"ما أثق 100% إنهم ملتزمين، خاصة انهم احيانا مايأخذون الحساسية بجدية وما يعطوني تفاصيل واضحة ف اتوتر من انهم نسوها."

السؤال السادس:
ما نوع خيارات التخصيص التي تود توفرها عند طلب المخبوزات؟
"لو اقدر ازيل بعض المكونات زي الحليب واغير حشوة الكروسان مثلا واقلل السكر فيه بيكون شيء حلو مره وبيقلل توتري من اني اتحسس."

السؤال السابع:
عند تخصيص طلبك، هل تفضل الاختيار من قائمة جاهزة أم كتابة ملاحظاتك الخاصة؟
"أفضل يكون مكس بين الاثنين أكتب ملاحظاتي بالتفصيل عشان يكون واضح اني اعاني من حساسية حادة وينتبهون واقدر برضو اعدل من القائمة واختار."

السؤال الثامن:
إذا كان هناك مكون ترغب في إزالته ولكنه أساسي في المنتج، هل تود أن يقترح لك التطبيق بدائل مناسبة؟
"ضروري جداً، بيساعدني اعرف ايش الموجود من بدائل بدون ما احتاج لاني ابحث بنفسي عنها."

السؤال التاسع:
نظرًا لأن إعداد المخبوزات المخصصة قد يستغرق وقتًا، كم من الوقت أنت مستعد للانتظار مقارنة بالطلب العادي؟
"مو كثير لأني ما احس طلبي صعب هو نفس المخبوزات العادية بس باختلاف بسيط يناسبني ويناسب اهلي."

:السؤال العاشر
هل ستكون مهتمًا بفكرة اشتراك أسبوعي يتم من خلاله توصيل طلبك المخصص بشكل منتظم؟
"بيكون شيء حلو لو فيه منتجات صحية جاهزة توصلنا بشكل مستمر وتناسب حالتنا الصحية."

:السؤال الحادي عشر
إذا توفر تطبيق يربطك بالمخابز بناءً على تفضيلاتك، هل ستستخدمه؟ ولماذا؟
"ايوه طبعا خاصة لو يعطيني تفاصيل المكونات واضحة بيساعدني اوفر جهد البحث."

:السؤال الثاني عشر
ما هي الميزة الأساسية التي يجب توفرها في التطبيق لتلبية احتياجاتك؟
"شفافية المكونات في التطبيق لان وضوحها بيساعدني اخفف قلق وتوتر من ان الطلب ممكن يكون غلط ويسبب لنا حساسية."

Interview Transcript 10

Interviewer student Name: Hala Alghamdi

السؤال الأول :
هل تقوم عادةً بشراء المخبوزات من المخابز؟
"ايوه وبشكل يومي تقريبا خاصة زي العيش والصامولي واحيانا الحلويات زي الدونات."

السؤال الثاني:
كيف تجد المخابز عادةً؟
"في المناسبات دايما ابحث عن مخابز في التيك توك او الإنستقرام لاني احتاج انواع محددة بس في الايام العادية عادة اخذ المخبوزات من مخابز قريبة من بيتنا لاننا متعودين عليها."

السؤال الثالث:
ما مدى أهمية الخيارات الغذائية مثل قليلة السكر، أو الخالية من الجلوتين، أو المكونات الخاصة بالنسبة لك؟
"مهمة احيانا عندي افراد عائلة يعانون من مرض السكر لازم ننتبه ايش نجيب لهم."

السؤال الرابع:
هل سبق لك طلب مخبوزات مخصصة من قبل؟ إذا كانت الإجابة نعم، هل يمكنك وصف تجربتك؟
"ايوه قد طلبت اكثر من مره بس دايما ما الاقي الشيء الي ابغاه جاهز ولا ابغى اطلبهم يسوون لي فاضطر اكنسل وما اخذ شيء."

السؤال الخامس:
هل تواجه أي صعوبات عند محاولة العثور على مخابز تقدم مخبوزات مخصصة؟
"ايوه جدا، صعب الاقي مخبز يوفر المخبوزات الي احتاجها جاهزة لازم اطلبهم يخبزونها من جديد وذا الشيء ياخذ وقت واحيانا حتى مايوافقون."

السؤال السادس:
ما نوع خيارات التخصيص التي تود توفرها عند طلب المخبوزات؟
"اهم شيء بالنسبة لي ولعائلتي هو خيارات تعديل نسبة السكر في الحلويات وتغيير نوع الدقيق لاننا نحب الدقيق البُر في المخبوزات اكثر وتغيير الحجم عشان الاطفال في العائلة."

السؤال السابع:
عند تخصيص طلبك، هل تفضل الاختيار من قائمة جاهزة أم كتابة ملاحظاتك الخاصة؟
"الخيارات الجاهزة اسهل واقدر أعدل عليها بسرعة وبيكون كويس لو كانت تنحفظ واقدر ارجع اطلبها كل مره."

السؤال الثامن:
إذا كان هناك مكون ترغب في إزالته ولكنه أساسي في المنتج، هل تود أن يقترح لك التطبيق بدائل مناسبة؟
"ايوه خصوصًا لو الشيء مو متوفر لان البحث عن بديل للمكون بياخذ مني وقت، افضل ان يكون كل شيء واضح."

السؤال التاسع:

نظرًا لأن إعداد المخبوزات المخصصة قد يستغرق وقتًا، كم من الوقت أنت مستعد للانتظار مقارنة بالطلب العادي؟

"أفضل يكون سريع لأني من النوع الي يطلب باخر لحظة خاصة لما اتناقش مع اهلي ويقررون فجأة يطلبون شيء عشان بنروح مكان مثلا، فيفضلون يوصل الاكل بسرعة."

:السؤال العاشر

هل ستكون مهتمًا بفكرة اشتراك أسبوعي يتم من خلاله توصيل طلبك المخصص بشكل منتظم؟

"اكيد بيكون مفيد جدا لأن استهلاك الخبز في عائلتنا كبير."

:السؤال الحادي عشر

إذا توفر تطبيق يربطك بالمخابز بناءً على تفضيلاتك، هل ستستخدمه؟ ولماذا؟

"ايوه بيوفر وقت ويسهل عملية البحث عن مخبوزات تناسبني انا واهلي."

:السؤال الثاني عشر

ما هي الميزة الأساسية التي يجب توفرها في التطبيق لتلبية احتياجاتك؟

"سرعة الطلب وسهولة التخصيص وبيكون شيء كويس لو اقدر احفظ الطلب واطلبه المره الجاية وتكون الاسعار مناسبة للكل."

Questionnaire form:

قسم 1 من 5

Survey for a Mobile Application for Customizing and Ordering Baked Goods

نحن نصمم تطبيق جوال لمساعدتك على تخصيص وطلب المخبوزات بسهولة! 🍞
شارك في هذا الاستبيان السريع الذي يستغرق 3 دقائق فقط لتخبرنا بتفضيلاتك، وساعدنا على إنشاء تجربة طلب سلسة وممتعة.
جميع إجاباتك ستكون مجهولة ولن تُستخدم إلا لأغراض المشروع الأكاديمي.

We're designing a mobile app to help users customize and order baked goods easily! 🥐
Take this quick 3-minute survey to share your preferences and help us create a smooth and fun ordering experience. Your answers are anonymous and only for our project.

1. Do you agree to participate in this survey for academic purposes ? *
١. هل توافق على المشاركة في هذا الاستبيان لأغراض أكاديمية؟

- Yes, I agree / نعم، أوافق
- No, I do not agree / لا، لا أوافق

بعد القسم 1 المتابعة إلى القسم التالي

قسم 2 من 5

Demographics – المعلومات العامة

الوصف (اختياري)

2. What is your age? *
٢. كم عمرك؟

- Under 18 / أقل من 18
- 18–24
- 25-34
- 34-49
- +50

3. What is your gender? *
٣. ما جنسك؟

- Male/ذكر
- Female/أنثى

4. How often do you buy baked goods? *
٤. كم مرة تشتري مخبوزات؟

- Daily/يوميا
- Several times a week/عدة مرات في الأسبوع
- Once a week/مرة في الأسبوع
- Once a month/مرة في الشهر
- Rarely/نادرا

5. Where do you usually buy baked goods? *
٥. من أين تشتري المخبوزات غالباً؟

- Nearby bakery/مخبز قريب
- Cafe / Coffee shop/ مقهى / محل حلويات
- Delivery app/ تطبيق توصيل

قسم 3 من 5

User Experience & Customization – تجربة المستخدم وتخصيص الطلب

الوصف (اختياري)

6. Have you ever ordered customized baked goods (e.g., cake with a name, special design)?
٦. هل سبق أن طلبت مخبوزات مخصصة (مثل كيك باسم أو تصميم خاص)؟

- Yes / نعم
- No / لا

7. If yes, how did you place your order?
٧. إذا نعم، كيف قمت بالطلب؟

- Visiting bakery / زيارة المخبز
- Using a delivery app / عبر تطبيق توصيل
- Through Instagram / WhatsApp / عبر إنستغرام / واتساب

8. What is the biggest difficulty you face when ordering customized baked goods? *
٨. ما أكبر صعوبة تواجهها عند طلب مخبوزات مخصصة؟

- Explaining the order clearly / صعوبة شرح الطلب
- Limited options / قلة الخيارات
- Finding a suitable bakery / صعوبة إيجاد مخبز مناسب

9. What matters most to you when choosing a bakery? *
٩. ما أكثر شيء يهمك عند اختيار المخبز؟

- Quality / الجودة
- Price / السعر
- Location / قرب الموقع
- Reviews / التقييمات

10. What type of customization do you prefer when ordering baked goods? (You can select more than one) *
١٠. ما نوع التخصيص الذي تفضله عند طلب المخبوزات؟ (يمكن اختيار أكثر من خيار)

- Choosing the flavor / اختيار النكهة
- Choosing shape/design / اختيار الشكل / التصميم
- Choosing size / اختيار الحجم
- Choosing ingredients / اختيار المكونات
- Adding a name or message / كتابة اسم أو رسالة

11. Do you want to see a preview of the product before ordering? *
١١. هل تفضل رؤية المنتج قبل الطلب؟

- Yes / نعم
- No / لا

12. How would you like to customize your order? *
١٢. ما الطريقة المفضلة لتخصيص المنتج؟

- Designing using a 3D preview with predefined choices / تصميم المنتج على شاشة ثلاثية الأبعاد باستخدام خيارات مح...
- Uploading an image / رفع صورة للتصميم المطلوب
- Writing a description / كتابة وصف للتصميم

بعد القسم 3 المتابعة إلى القسم التالي

قسم 4 من 5

App Features & Preferences – ميزات التطبيق والتفضيلات

الوصف (اختياري)

13. Which of the following features would you find most useful in a mobile app for ordering customized baked goods? *
١٣. ما الميزات التالية التي ستجدها الأكثر فائدة في تطبيق لطلب المخبوزات المخصصة؟

- [ ] Filtering bakeries based on dietary needs/تصفية المخابز حسب الاحتياجات الغذائية
- [ ] Ingredient customization options/إمكانية تخصيص المكونات
- [ ] Order tracking/تتبع الطلب
- [ ] User reviews and ratings/تقييمات وآراء المستخدمين
- [ ] Price comparison between bakeries/مقارنة الأسعار بين المخابز
- [ ] أخرى:

14. Would you like the app to save your previous customized orders for easier reordering in the future? *
١٤. هل تفضل أن يقوم التطبيق بحفظ طلباتك المخصصة السابقة لتسهيل إعادة الطلب لاحقًا؟

- ○ Yes/نعم
- ○ No/لا
- ○ Maybe/ربما

15. How useful would it be if the app suggested bakeries based on your dietary restrictions or preferences? *
١٥. ما مدى فائدة أن يقترح التطبيق مخابز بناءً على قيودك أو تفضيلاتك الغذائية؟

- ○ Very useful/مفيد جدا
- ○ Useful/مفيد
- ○ Neutral/محايد
- ○ Not useful/غير مفيد

16. Would you prefer the app to include a feature that allows you to communicate directly with the bakery for special customization requests? *
١٦. هل تفضل أن يتضمن التطبيق ميزة تتيح لك التواصل مباشرة مع المخبز لطلبات التخصيص الخاصة؟

- ○ yes/نعم
- ○ No/لا
- ○ Maybe/ربما

17. Would it be helpful if the app allowed you to view pictures of customized baked goods previously made by the bakery? *
١٧. هل سيكون من المفيد أن يتيح لك التطبيق مشاهدة صور لمخبوزات مخصصة قام المخبز بإعدادها سابقًا؟

- ○ Yes/نعم
- ○ No/لا
- ○ Maybe/ربما

18. How important is it for you to see bakery ratings before ordering? *
١٨. ما مدى أهمية رؤية تقييم المخبز قبل الطلب؟

- ○ Very important / مهم جدا
- ○ Important / مهم
- ○ Not important / غير مهم

بعد القسم 4 المتابعة إلى القسم التالي

قسم 5 من 5

Delivery – التوصيل

الوصف (اختياري)

19. What is your preferred method for receiving your. order? *
١٩. ما الطريقة المفضلة لاستلام الطلب؟

- ○ Home delivery / توصيل للمنزل
- ○ Pickup from the bakery / الاستلام من المخبز

20. Would you like to track your order in real-time? *
٢٠. هل ترغب في تتبع طلبك بشكل مباشر؟

- ○ Yes / نعم
- ○ No / لا

21. Would you prefer notifications about your order status? *
٢١. هل تفضل تلقي إشعارات حول حالة طلبك؟

- ○ Yes / نعم
- ○ No / لا

22. Do you have any suggestions or ideas to improve the app? (Optional)
٢٢. هل لديك أي اقتراحات أو أفكار لتحسين التطبيق؟ (اختياري)

نص الإجابة الطويلة

Questionnaire data analysis:

1. 100% of respondents agreed to participate.

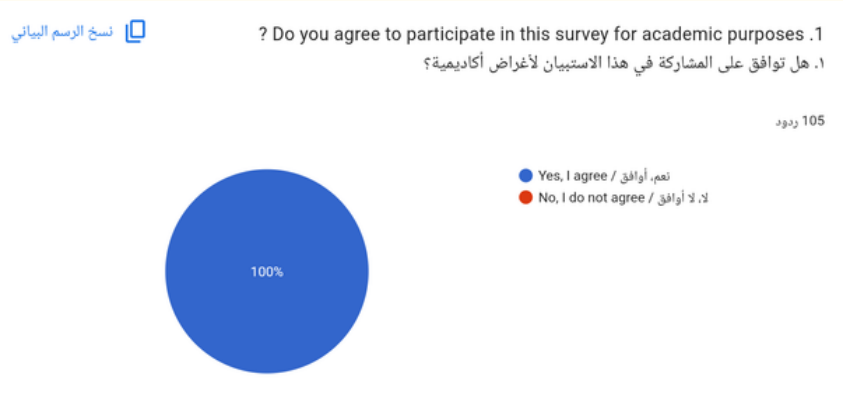


Figure 73 - data Analysis

2.The dominant age group is18-24 (68.6%), indicating that the primary target users are young to mid-career adults. This demographic is likely comfortable using mobile applications.

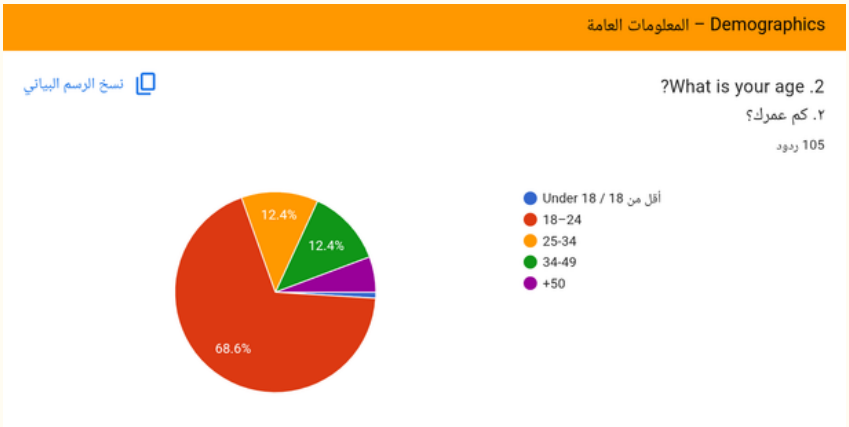


Figure 74 - data Analysis

3.The survey respondents are predominantly female (82.9%). This suggests the app's design and marketing should consider the preferences of this user group. Of course, we must also ensure the design is suitable for males too.

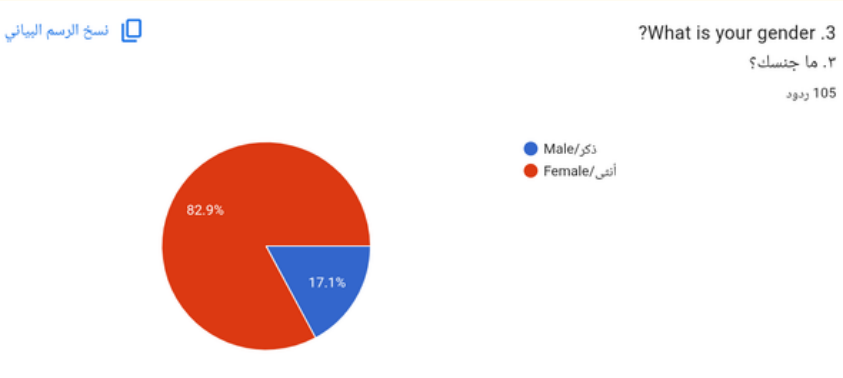


Figure 75 - data Analysis

4.The largest group (42.9%) buys baked goods Several times a week, followed by 24.8% who buy once a week. This shows a mix of regular and occasional customers, suggesting the app should cater to both spontaneous and planned orders.

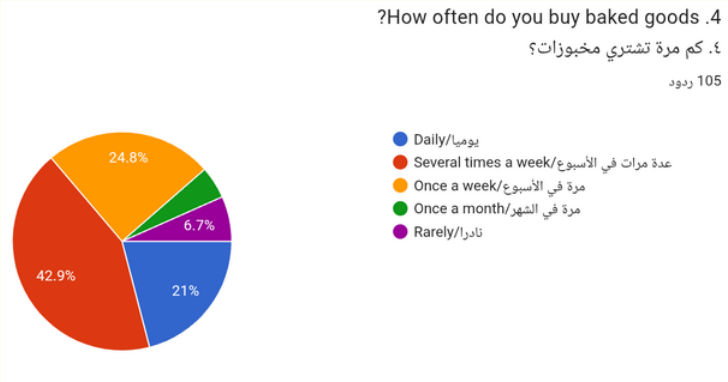


Figure 76 - data Analysis

5. An overwhelming 87.6% of users buy from a nearby bakery. This confirms that proximity is a key factor in their current habits.

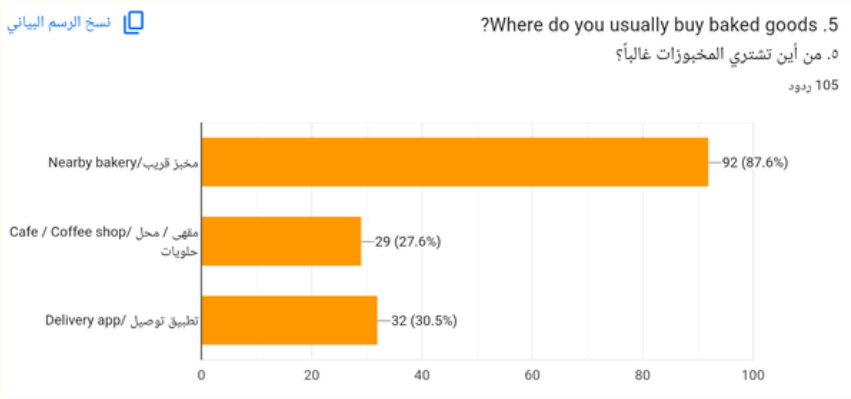


Figure 77 - data Analysis

6.A large majority (84.8%) have ordered customized baked goods before. This validates the core concept of the project, showing a strong existing demand for personalization.

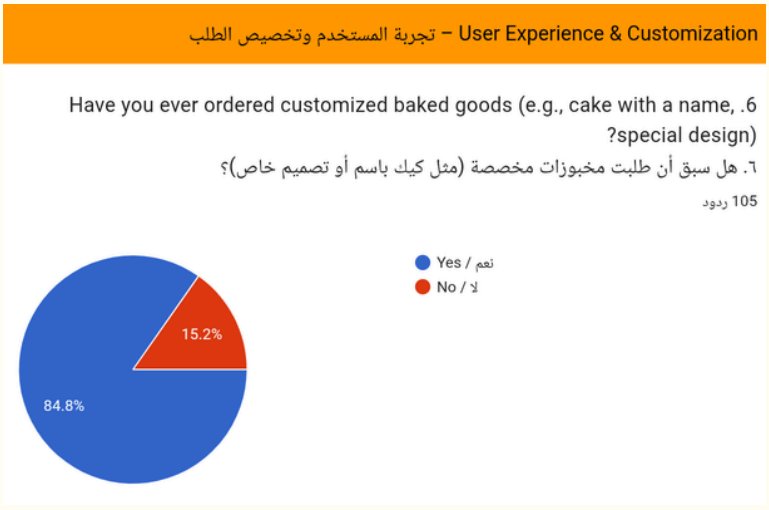


Figure 78 - data Analysis

7.The majority of users (55.9%) prefer visiting the bakery in person. However, a significant portion (52%) use Instagram or WhatsApp, highlighting the importance of social media and messaging apps in the current ordering process.

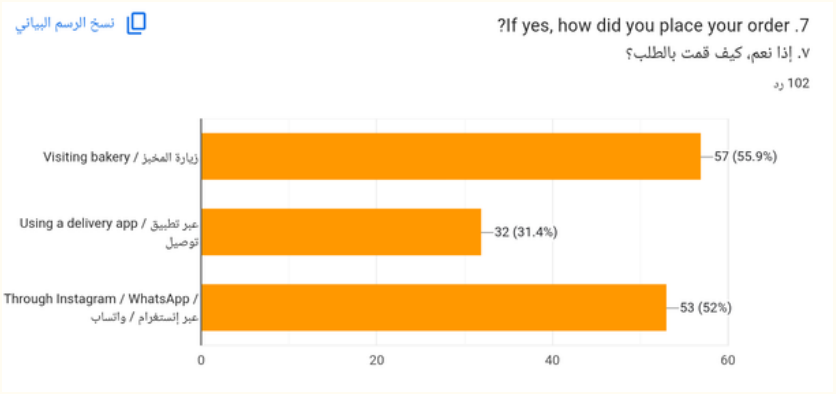


Figure 79 - data Analysis

8.The biggest challenge is "Finding a suitable bakery" (60%), followed closely by "Limited options" (48.6%). This strongly indicates a user need for a platform that helps discover bakeries that can fulfill specific requests.

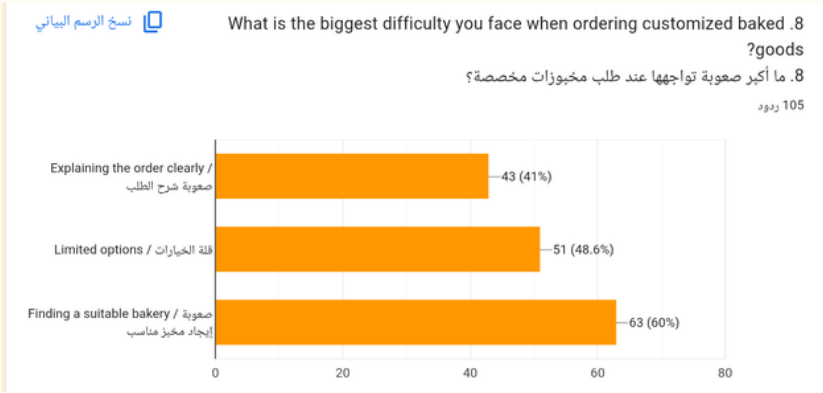


Figure 80 - data Analysis

9. Quality" is the most important factor for nearly 90% of users. "Price" is also a major consideration (68.6%), while location and reviews are less prioritized but still relevant.

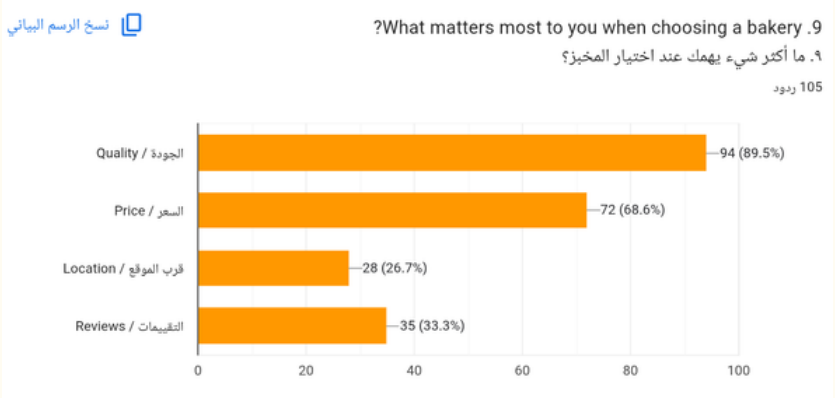


Figure 81 - data Analysis

10. The most popular choice was flavor (76.2%), followed closely by size and shape/design. Emotional personalization is just as important to consumers as functional customization, as seen by the fact that tailored message (47.6%) marginally outranked ingredient customization (44.8%).

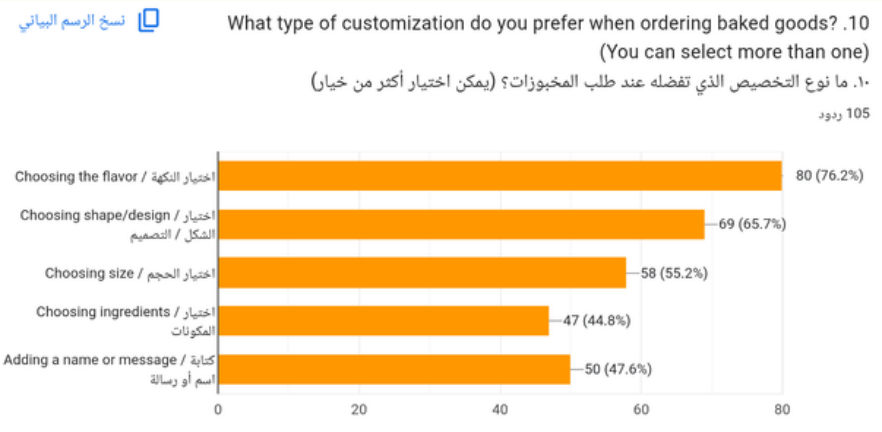


Figure 82 - data Analysis

11. All respondents expressed a desire to preview products before placing an order, demonstrating complete agreement, establishing this as an essential platform feature.

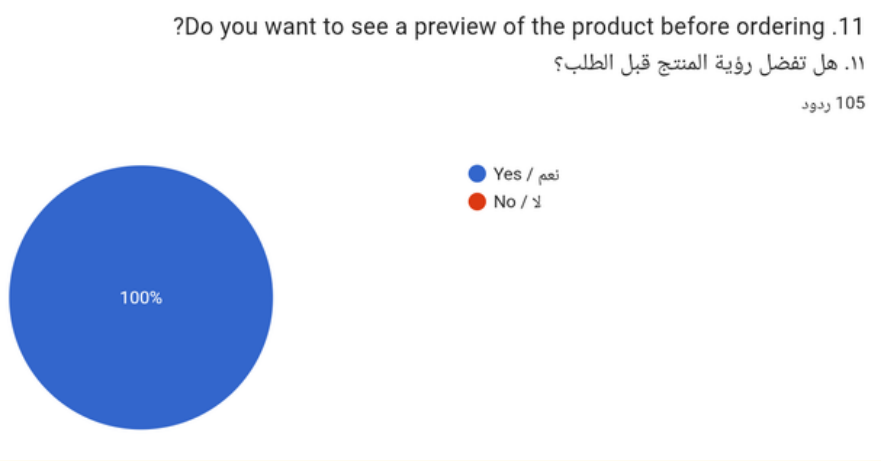


Figure 83 - data Analysis

12. Visual customizing techniques were highly preferred; 61.9% of respondents chose 3D design tools and image uploads, while only 39% chose written descriptions. This implies that intuitive, image-driven interfaces are preferred by users than text-based input.

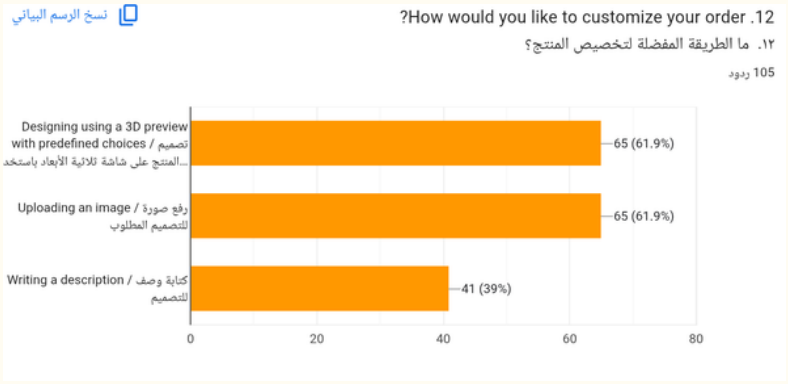


Figure 84 - data Analysis

13. Price comparison amongst bakeries came up first (59.6%), followed by nutritional filtering (54.8%) and ingredient modification (58.7%). These results are indicative of a user base that values openness and well-informed choices.

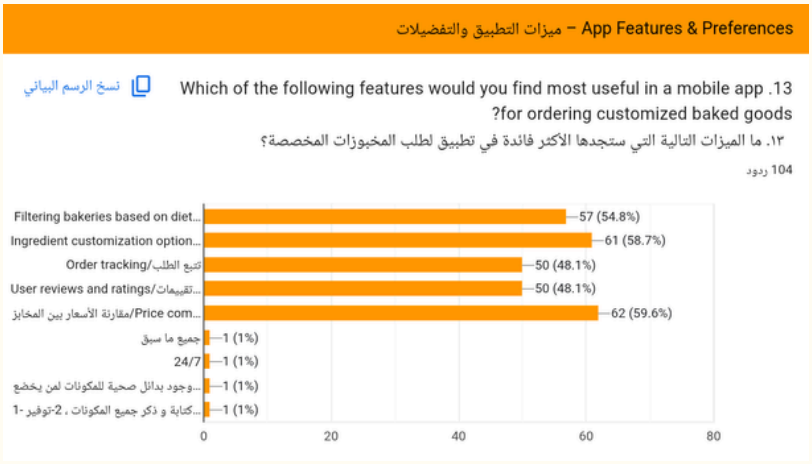


Figure 85 - data Analysis

14. The option to save previous customized orders was selected by a vast majority (92.9%), highlighting the importance of efficient reordering as a fundamental functional need.

Figure 86 - data Analysis

15. Dietary-based recommendations were regarded as helpful or very helpful by more than 93% of respondents, indicating that this feature is a major user expectation rather than an optional improvement.

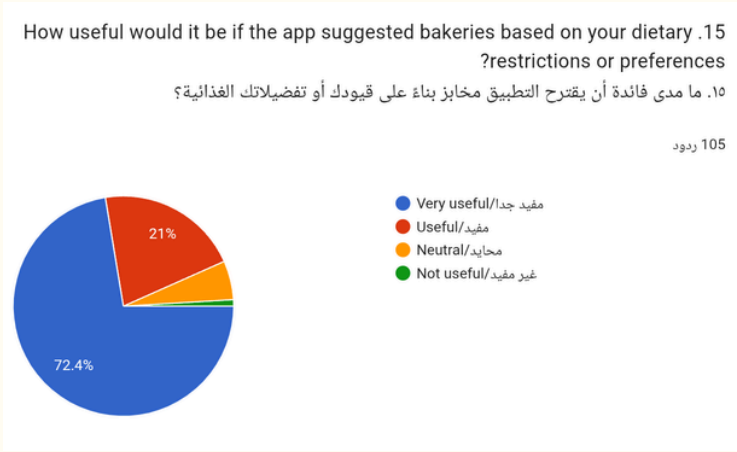


Figure 87 - data Analysis

16. 91.4% of respondents said they would rather speak with bakers directly, indicating that they are aware of the limitations of automatic customizing options when dealing with complex or subtle requests.

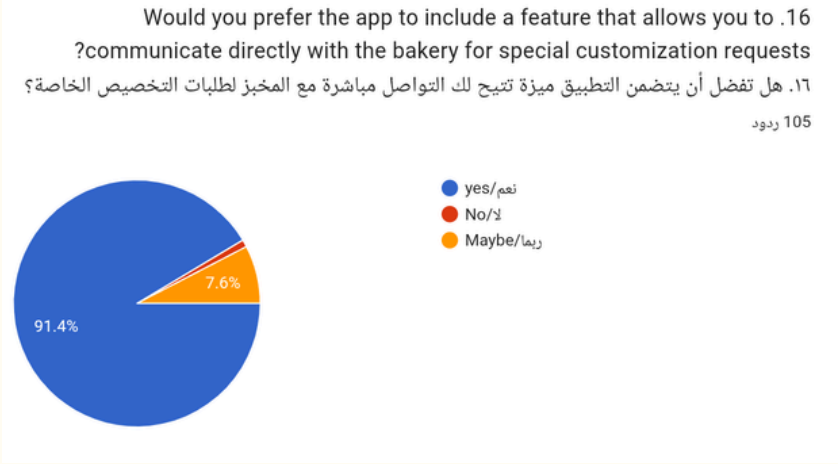


Figure 88 - data Analysis

17. Access to bakeries' prior customization portfolios was supported by 93.3% of respondents, underscoring the importance of visual credibility and skillful work in influencing user confidence and buy intent.

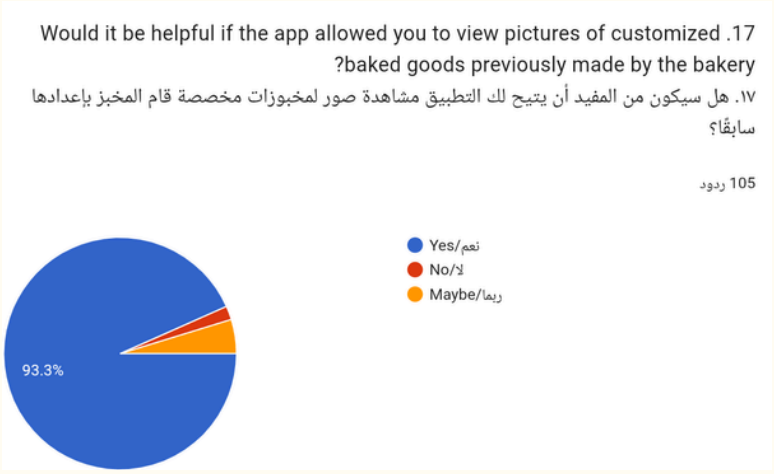


Figure 89 - data Analysis

18. 73.3% of respondents classified bakery ratings as very important, and 23.6% rated them as important, indicating a very high level of importance. This confirms that user confidence in platform-based transactions is largely dependent on peer rating processes.

Figure 90 - data Analysis

19. The overwhelming majority (82.9%) favored home delivery versus bakery pickup, suggesting that smooth delivery logistics ought to be the top operational priority.

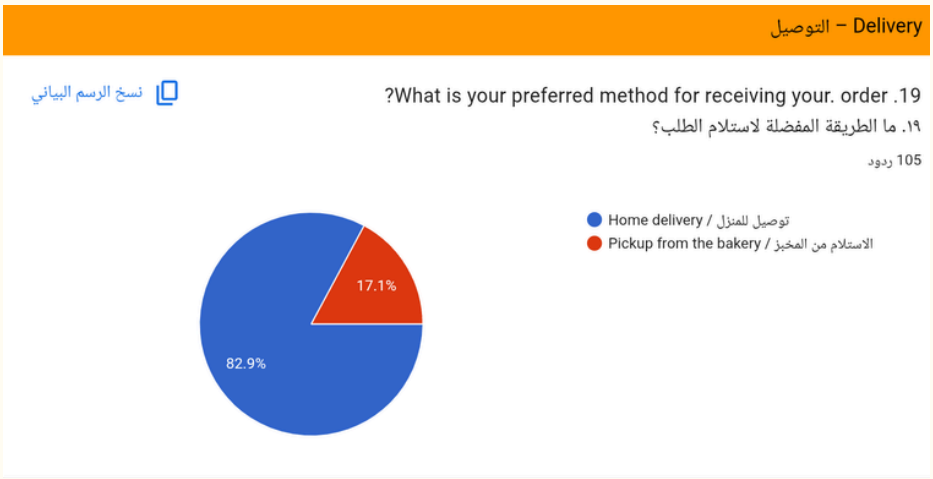


Figure 91 - data Analysis

20. Real-time tracking was shown to have common support, making it a standard user expectation rather than a differentiator.

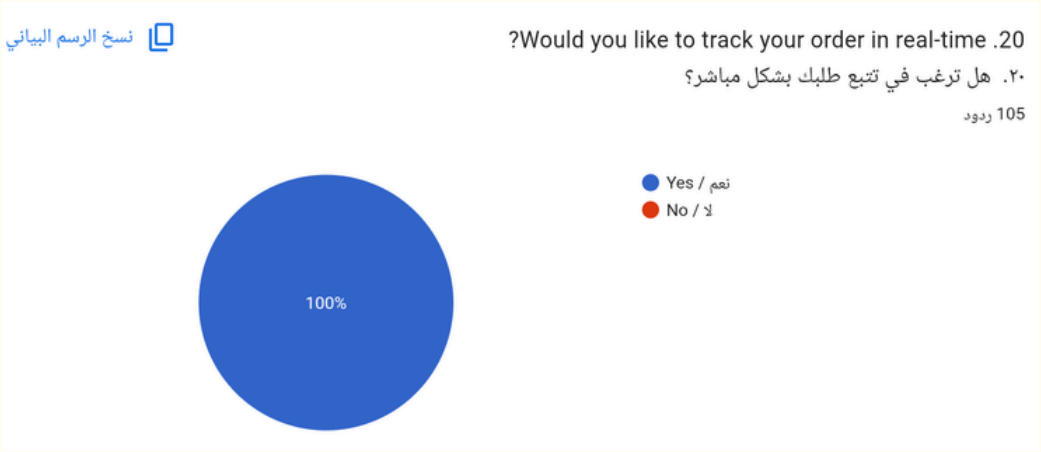


Figure 92 - data Analysis

21. Proactive order status updates were preferred by 96.2% of respondents. When combined with Q20, these results show that ongoing, open communication is expected during the fulfillment process.

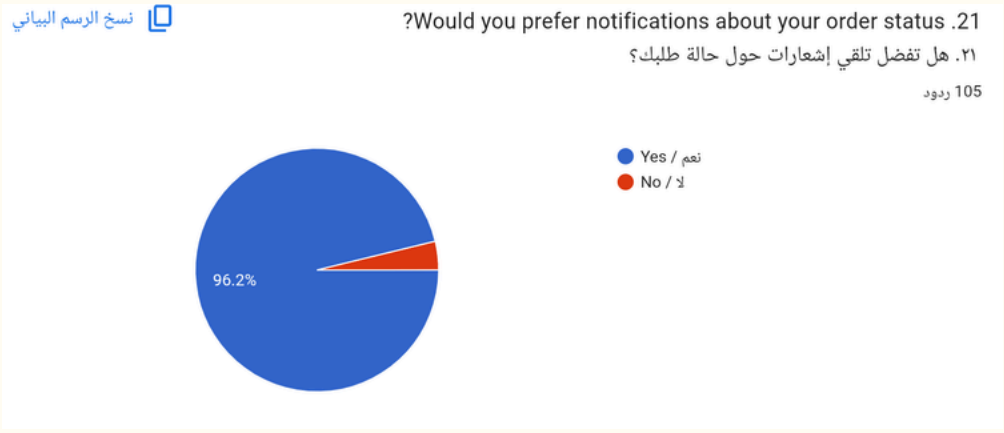


Figure 93 - data Analysis

Table of Tasks

Tasks involved in	Student Name
Contribution was made to the completion of the index and the entire project was reviewed, and ideas and guidance were given for all aspects of the project in a collective and collaborative manner. **1. Project goals and objectives** 1.3 Goals and Objectives **2. Empathize** 2.1.1 Questionnaire pilot study 2.1.3 Observation pilot Study 2.2.1 The questionnaire 2.2.2 The interview 2.2.3 The Observation 2.4.1 Persona 1 **3. Define** 3.1 User Problems and Challenges 3.2 The 5 W's Framework 3.3 Problem Statements **4. Ideate** 4.1.1 Direct Competitors (Overview, Aspects, Research) 4.1.2 Indirect competitors (Overview, Aspects, Research) 4.2.1 How Might We (Persona 1) crazy 8 (persona 1) **5. Prototype** 5.1 Storyboard Sketching and Wire-framing 5.1.1 Storyboard: Big-picture and close-up 5.1.2 Low-fidelity wireframes and sketches 5.2 Prototyping 5.2.1 Mock-up prototypes **6. Test and feedback** 6.2 Usability Testing 6.2.1 usability testing sessions with users 6.2.2 pain points, usability issues, and areas for improvement **Appendix** consent form two interviews The Survey Results	**Maha Saleh 2314771**
Contribution was made to the completion of the index and the entire project was reviewed, and ideas and guidance were given for all aspects of the project in a collective and collaborative manner. **1. Project goals and objectives** 1.1 The purpose of the mobile application **2. Empathize** 2.2.1 The questionnaire 2.2.2 The interview 2.2.3 The Observation 2.4.2 Persona 2 **3. Define** 3.1 User Problems and Challenges 3.2 The 5 W's Framework 3.3 Problem Statements 2.1.3 Observation pilot Study 2.1.2 Interview pilot study **4. Ideate** 4.2.1 How Might We (Persona 2) crazy 8 (persona 1) 4.2.4 Goal statements **5. Prototype** 5.1 Storyboard Sketching and Wire-framing 5.1.1 Storyboard: Big-picture and close-up (persona2) 5.2 Prototyping 5.2.1 Mock-up prototypes **6. Test and feedback** 6.3.1 Analyze the feedback and observations **Appendix** Information sheet	**Heba Salah 2314683**

Table of Tasks

Tasks involved in	Student Name
Contribution was made to the completion of the index and the entire project was reviewed, and ideas and guidance were given for all aspects of the project in a collective and collaborative manner. **1. Project goals and objectives** 1.2 The target audience and their needs 1.3 Project goals and objectives **2. Empathize** 2.1.3 Observation pilot Study 2.2.3 The Observation 2.2.2 The interview 2.4.2 Persona 4 **3. Define** 2.3.1 Aggregate Empathy Maps 2.5 Pain points and opportunities 3.5 Value Proposition 3.5.1 Features and Benefits 3.5.2 Application Value 3.5.3 Meeting Users' Needs 3.5.4 Official Value Proposition 3.6 Problem statement to guide the development process **4. Ideate** 4.1.4 Summarize Findings 4.2.1 How Might We (HMW) and Crazy 8 (Persona 4) 4.2.2 Potential solutions that align with the defined problem statement **5. Prototype** 5.1.1 Storyboard: Big-picture and close-up (Persona 4) 5.1.2 Lo - fi prototype Wireframes **6. Test and feedback** 6.3.2 Make informed design and functionality decisions **Conclusion** **Appendix** Usability testing consent form Three interviews	**Hala Alghamdi 2317918**
Contribution was made to the completion of the index and the entire project was reviewed, and ideas and guidance were given for all aspects of the project in a collective and collaborative manner. **Introduction** **2. Empathize** 2.1.3 Observation pilot Study 2.2.1 The questionnaire 2.2.3 The Observation 2.2.2 The interview 2.4.2 Persona 3 **3. Define** 3.4 Hypothesis Statements If Then 3.6 Problem statement to guide the development process **4. Ideate** 4.2.1 How Might We (Persona 3) crazy 8 (persona 3) 4.2.3 Evaluating ideas 4.3 Mapping design features to each persona in a table to fulfill their needs, wants, and desires **5. Prototype** 5.1 Storyboard Sketching and Wire-framing 5.1.1 Storyboard: Big-picture and close-up (persona3) 5.1.2 Low-fidelity wireframes and sketches **6. Test and feedback** 6.2 Usability Testing 6.2.1 usability testing sessions with users 6.2.2 pain points, usability issues, and areas for improvement **Appendix** Usability testing consent form Two interviews Questionnaire form	**Aisha Alkaff 2314692**